CITY BUILDERS

Fire Brigate



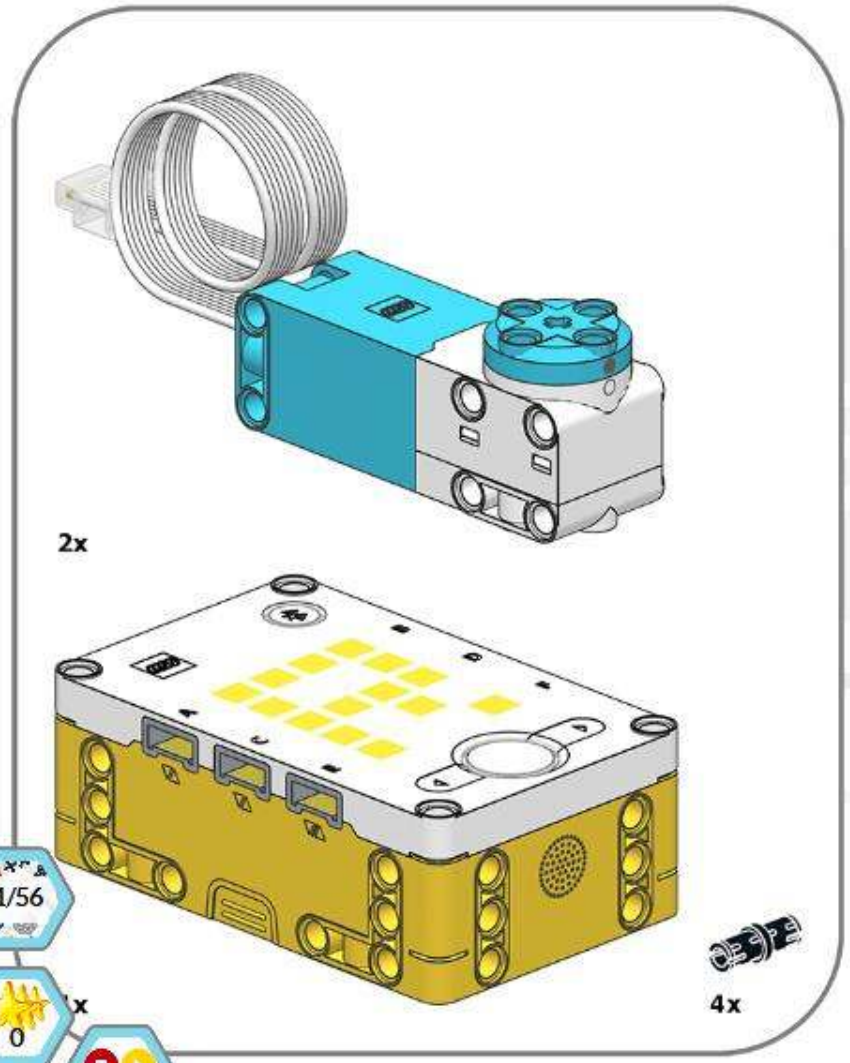
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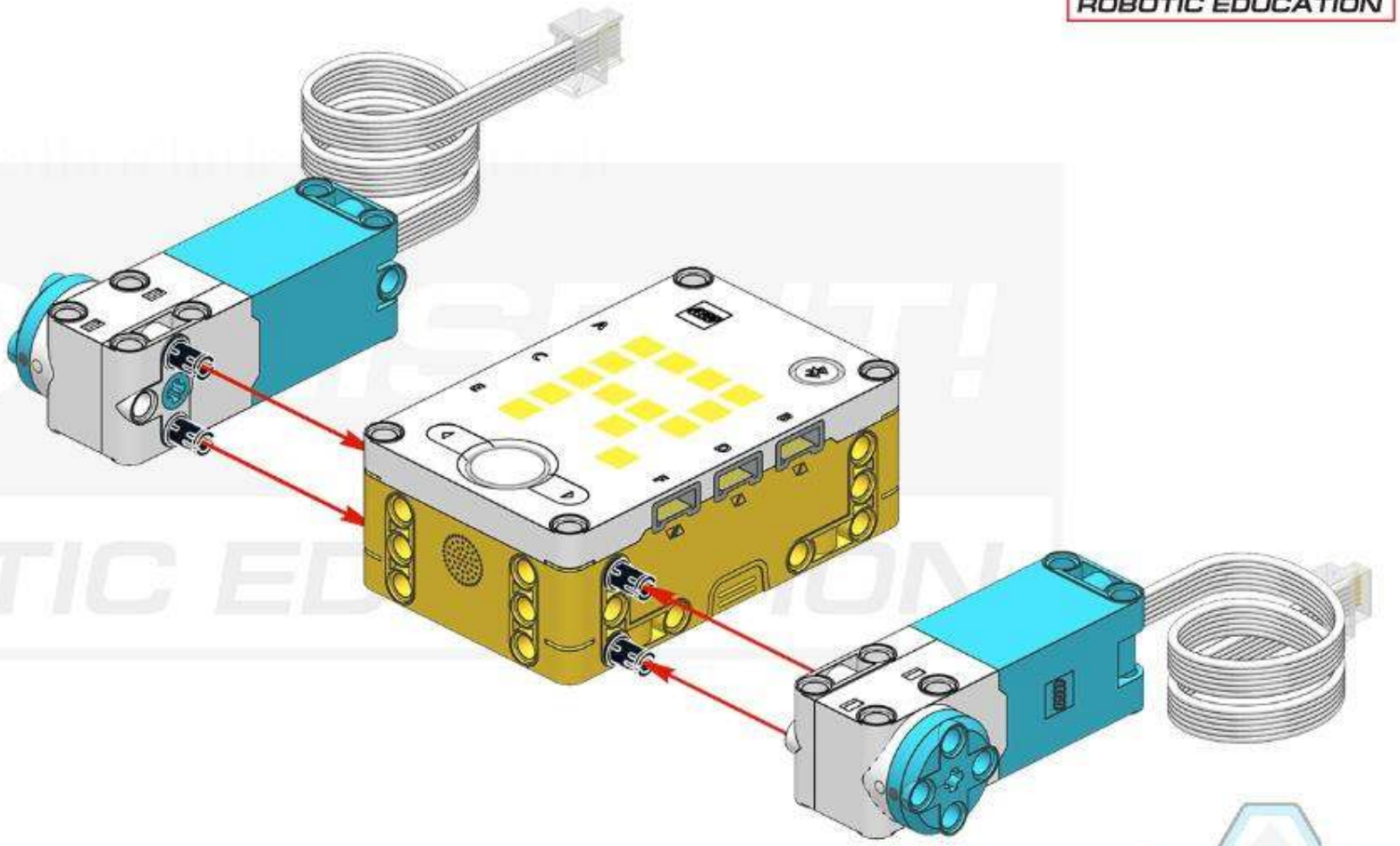
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BUILDING

LITTLE SCIENTISTS ACADEMY







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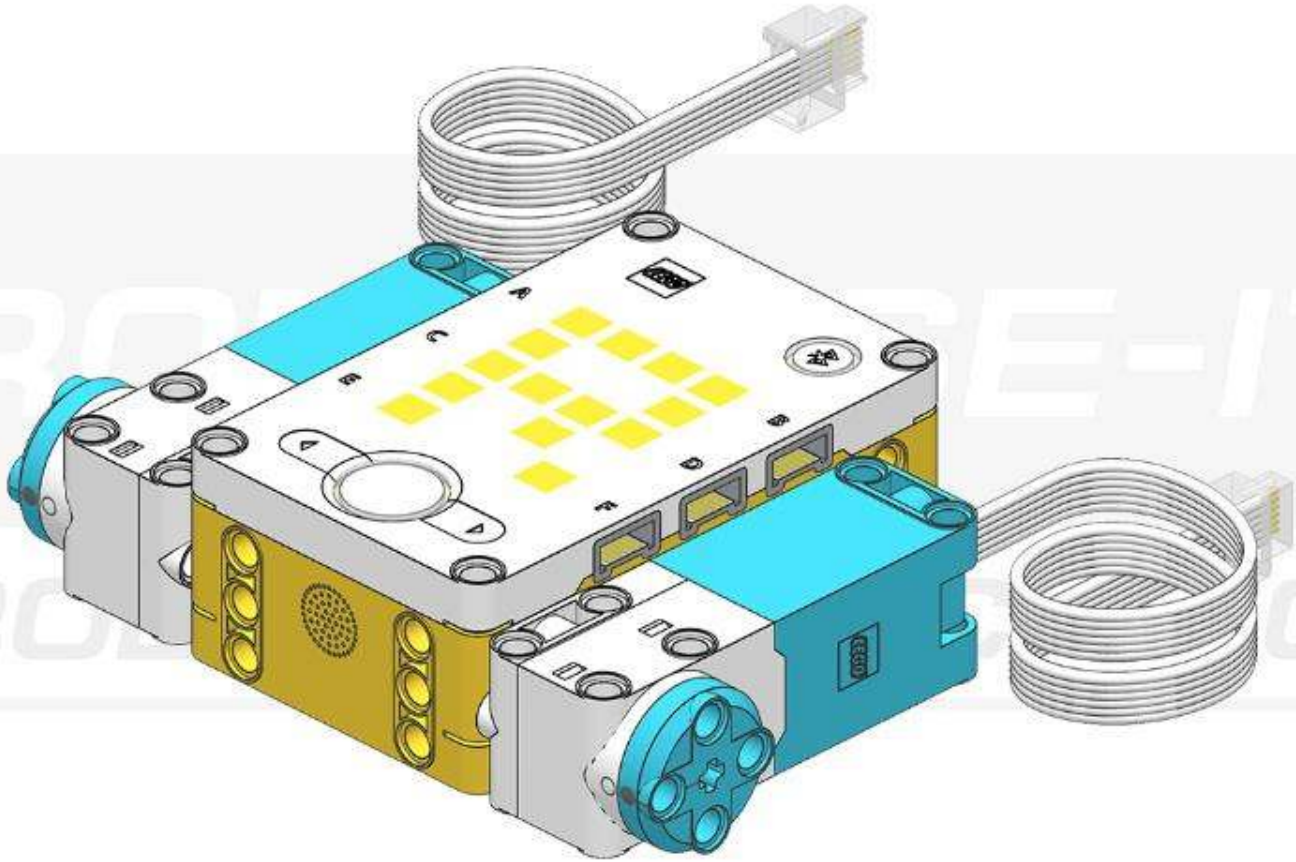
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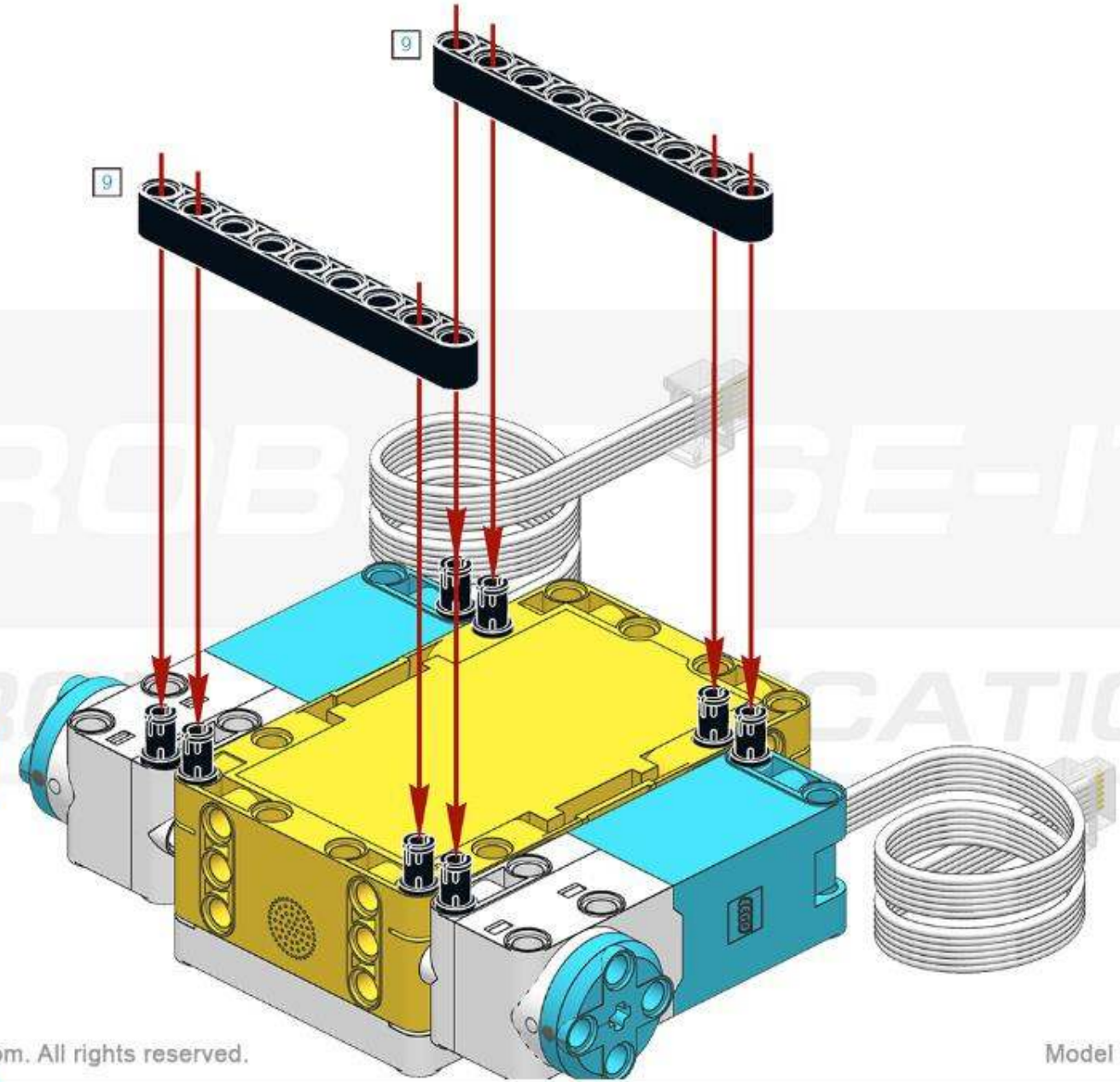
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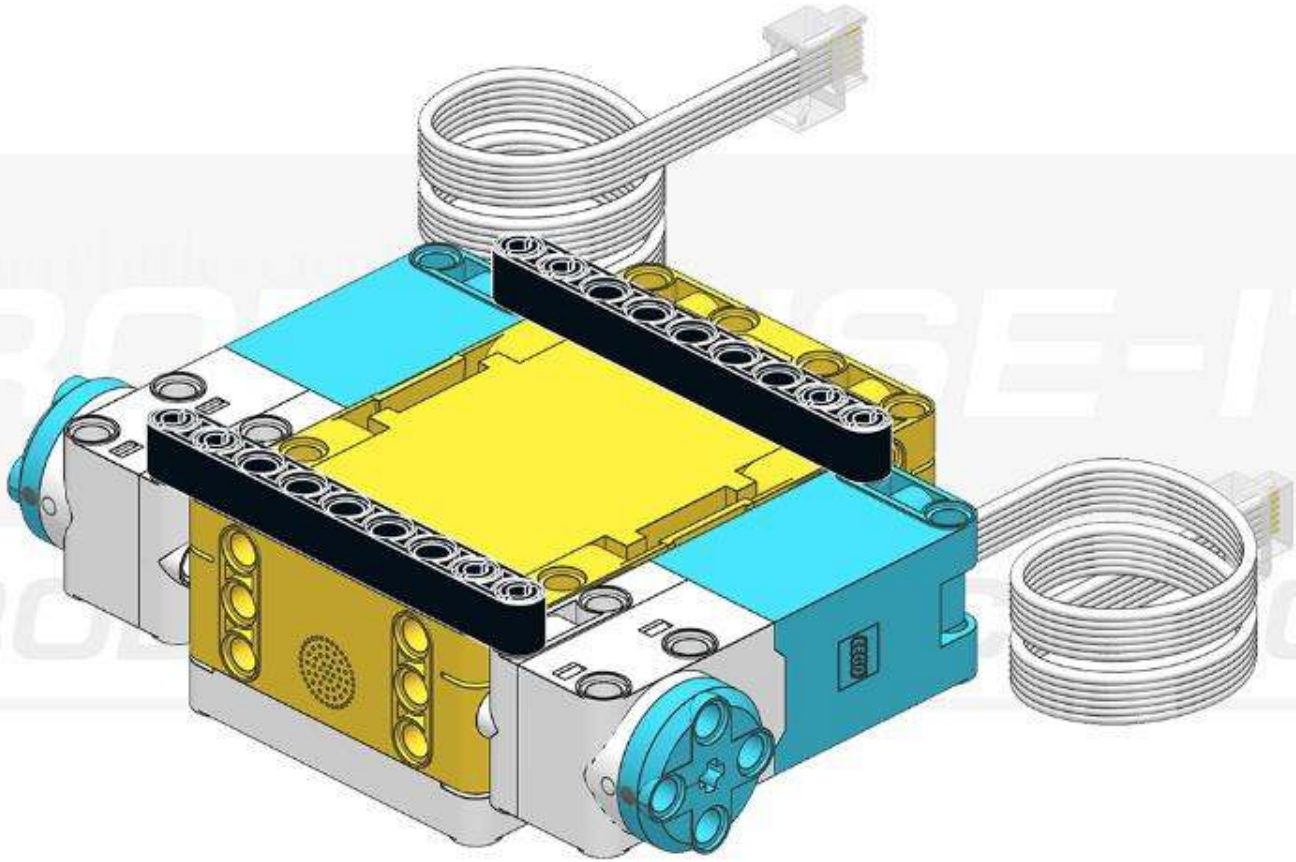
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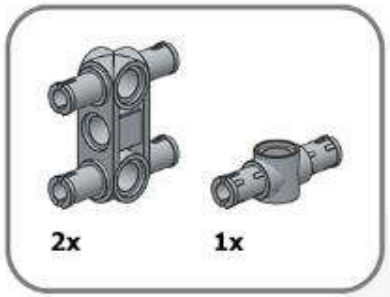
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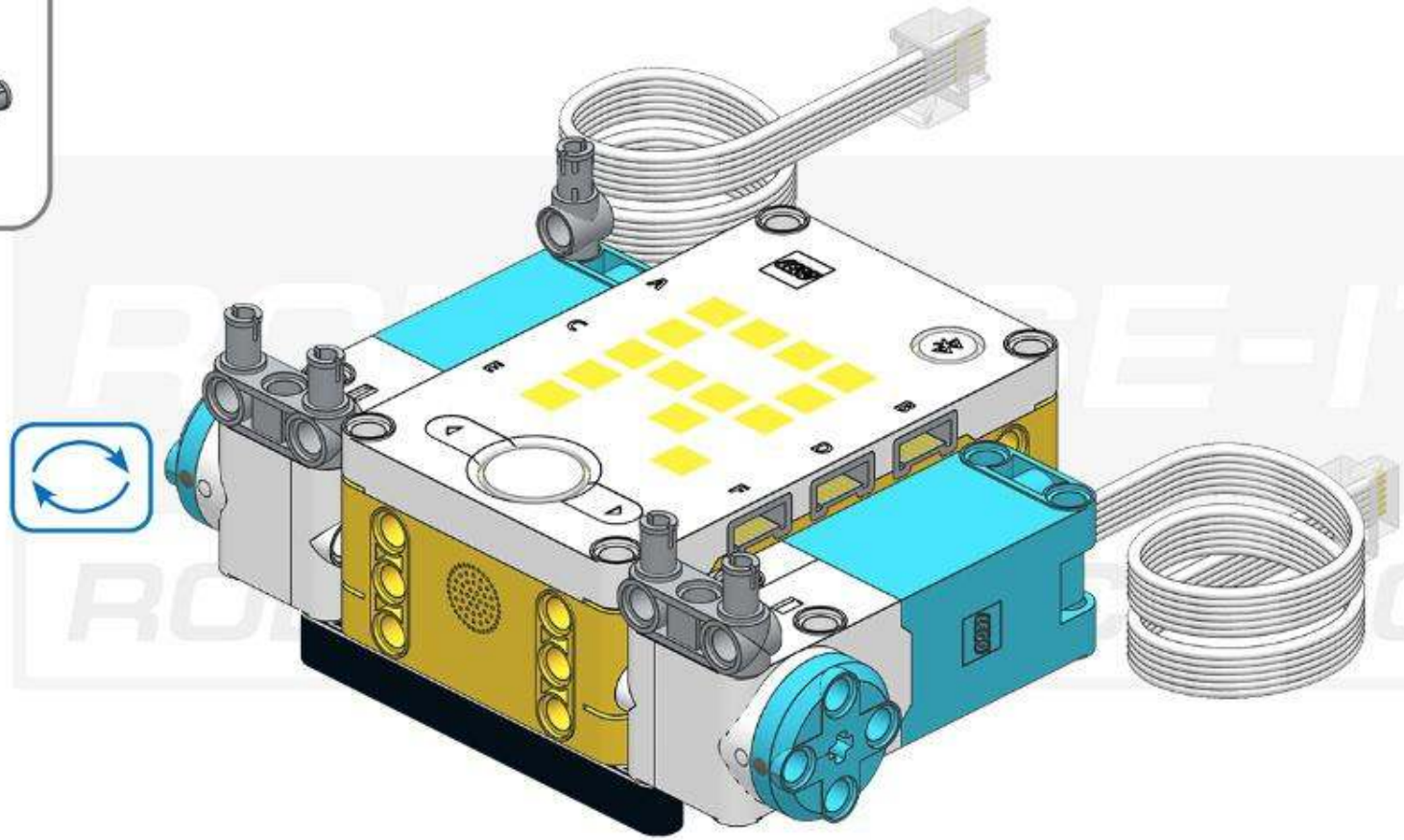
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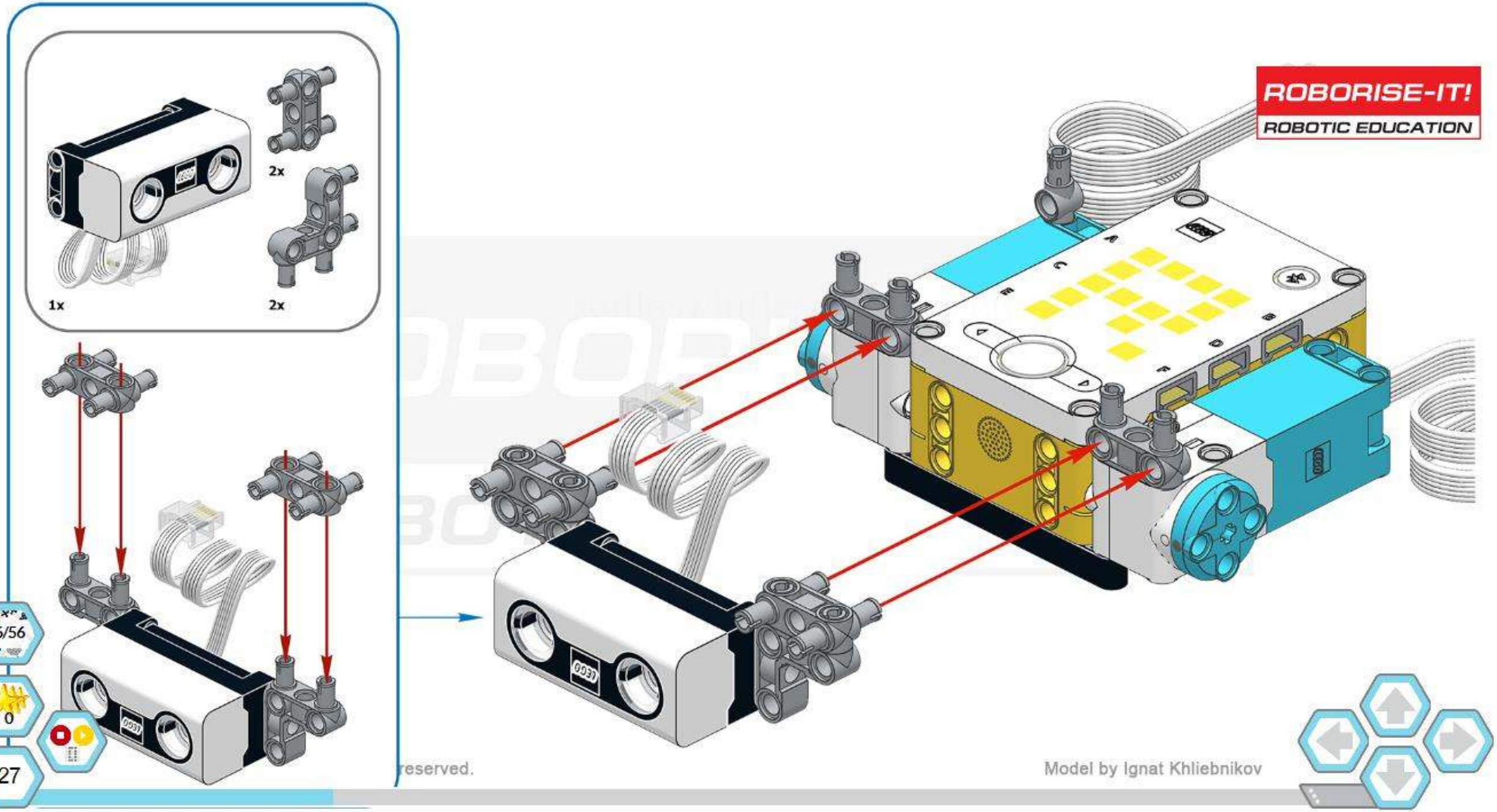


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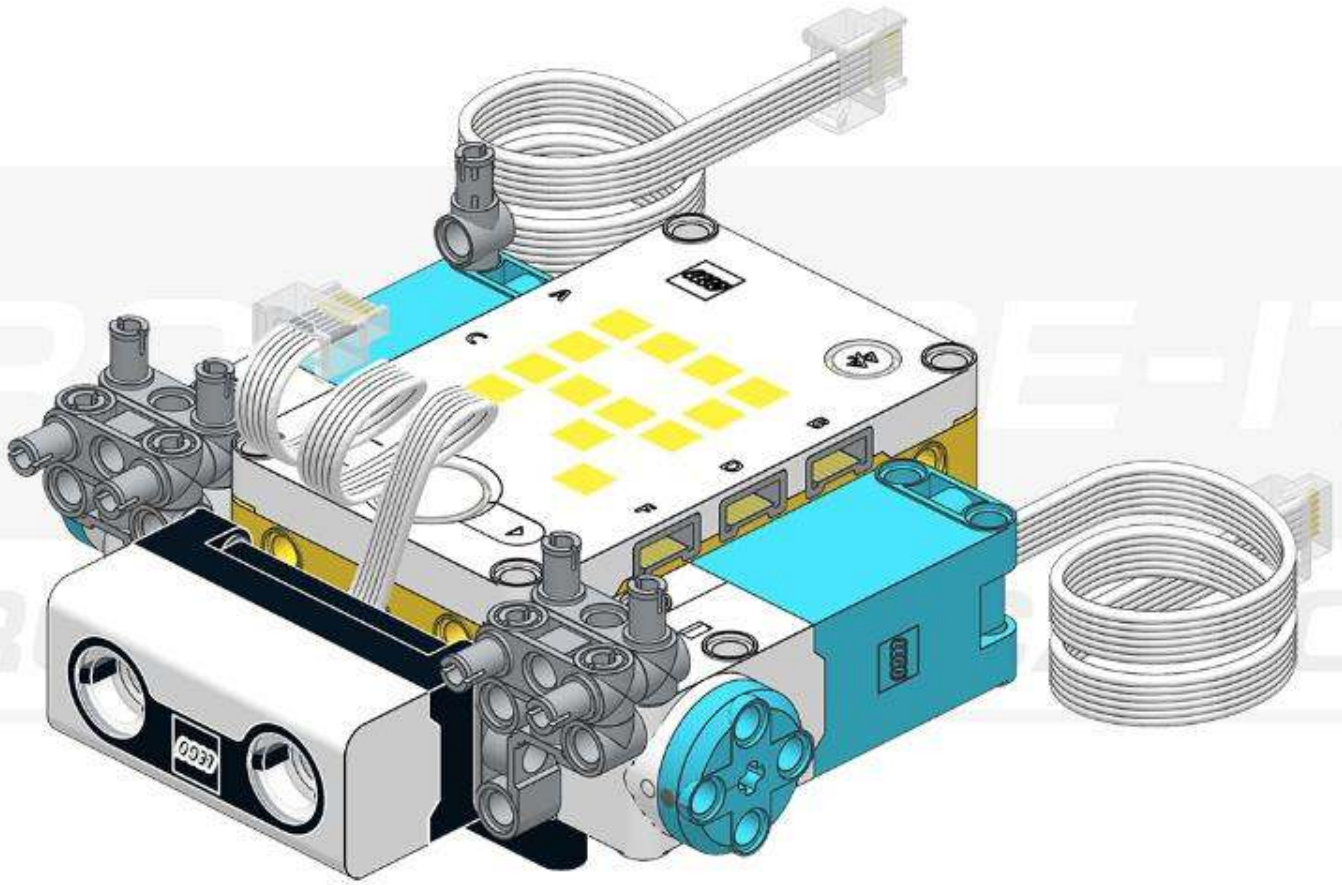
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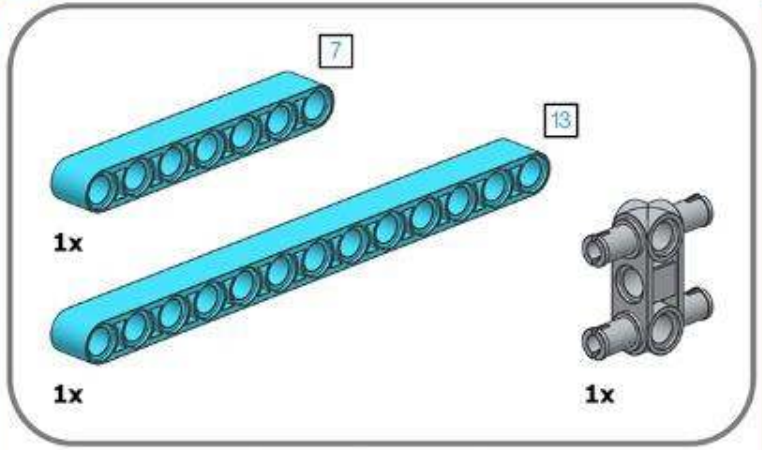
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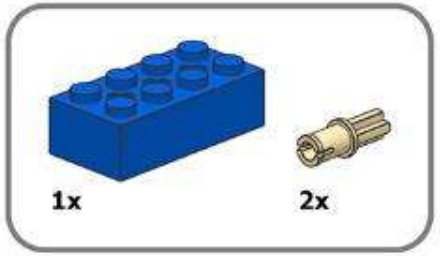
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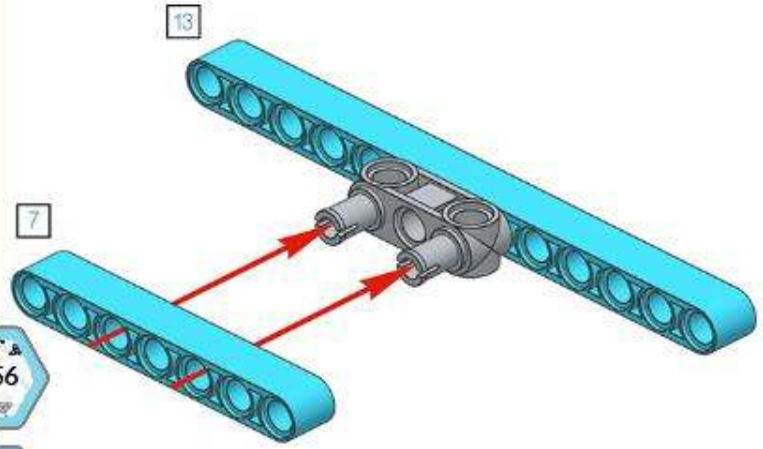
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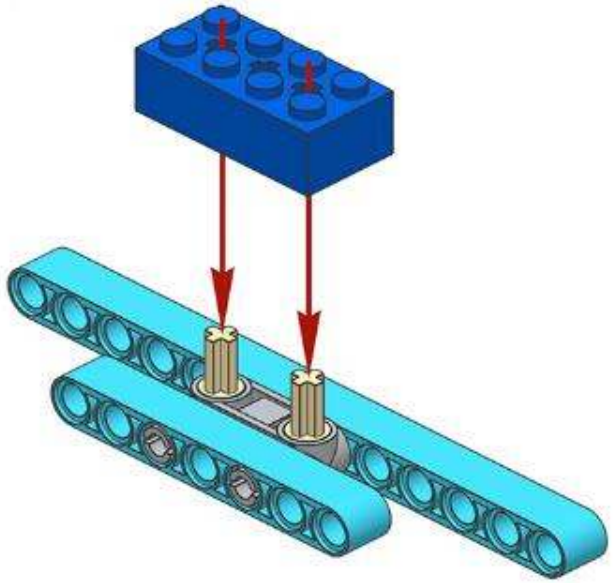


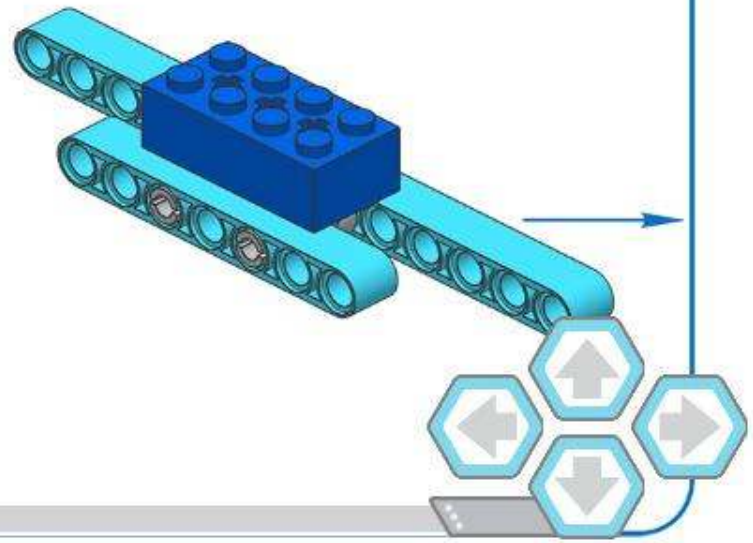
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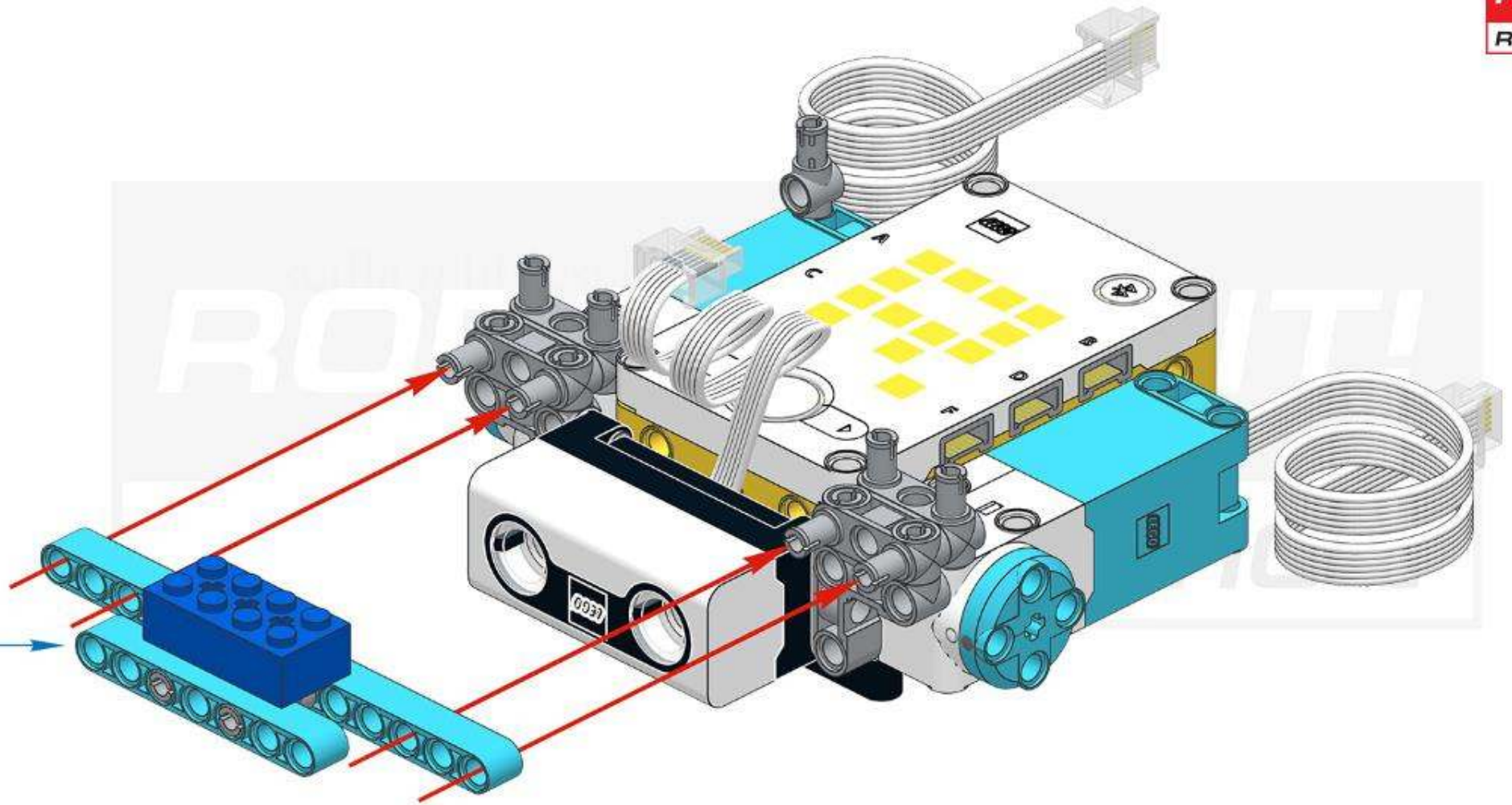
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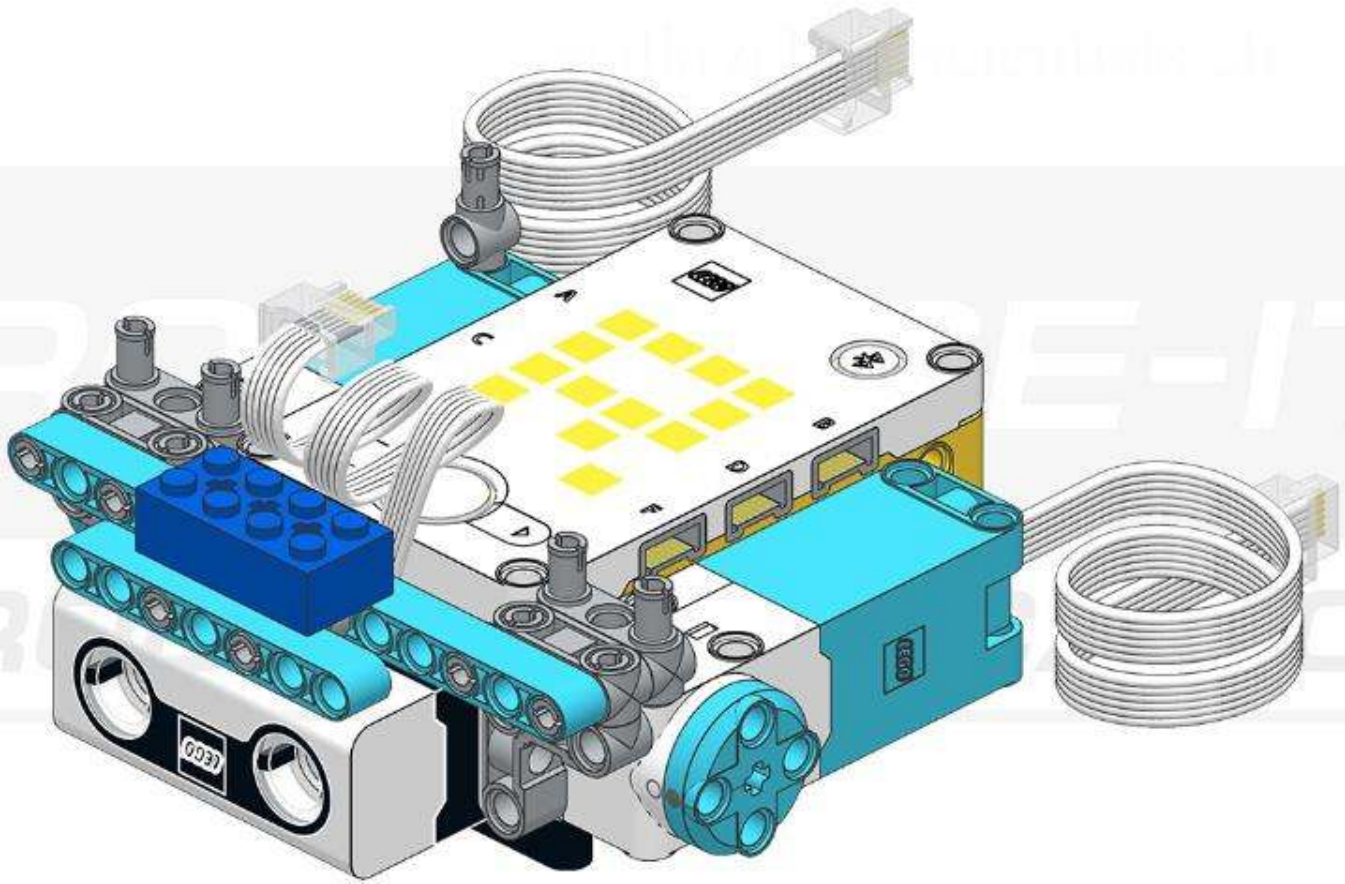
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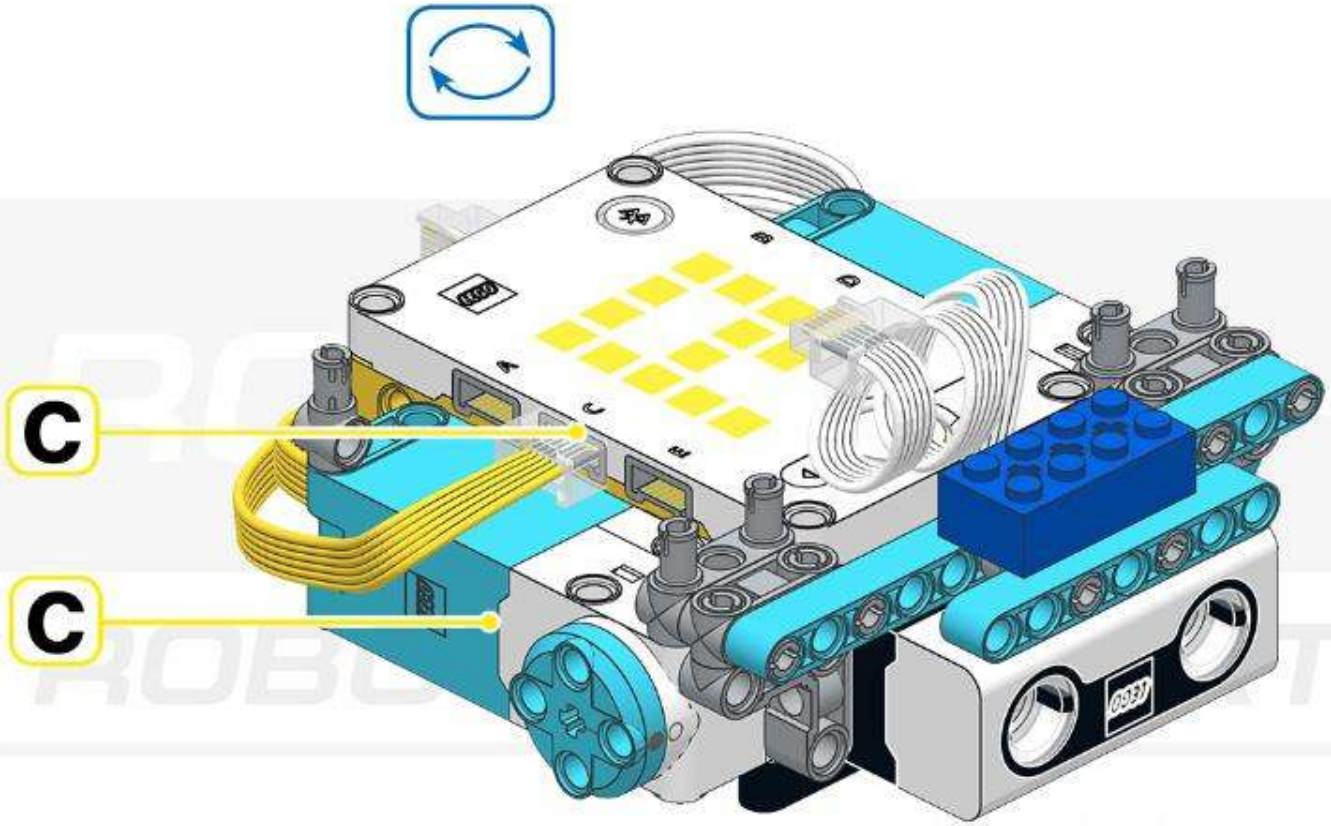


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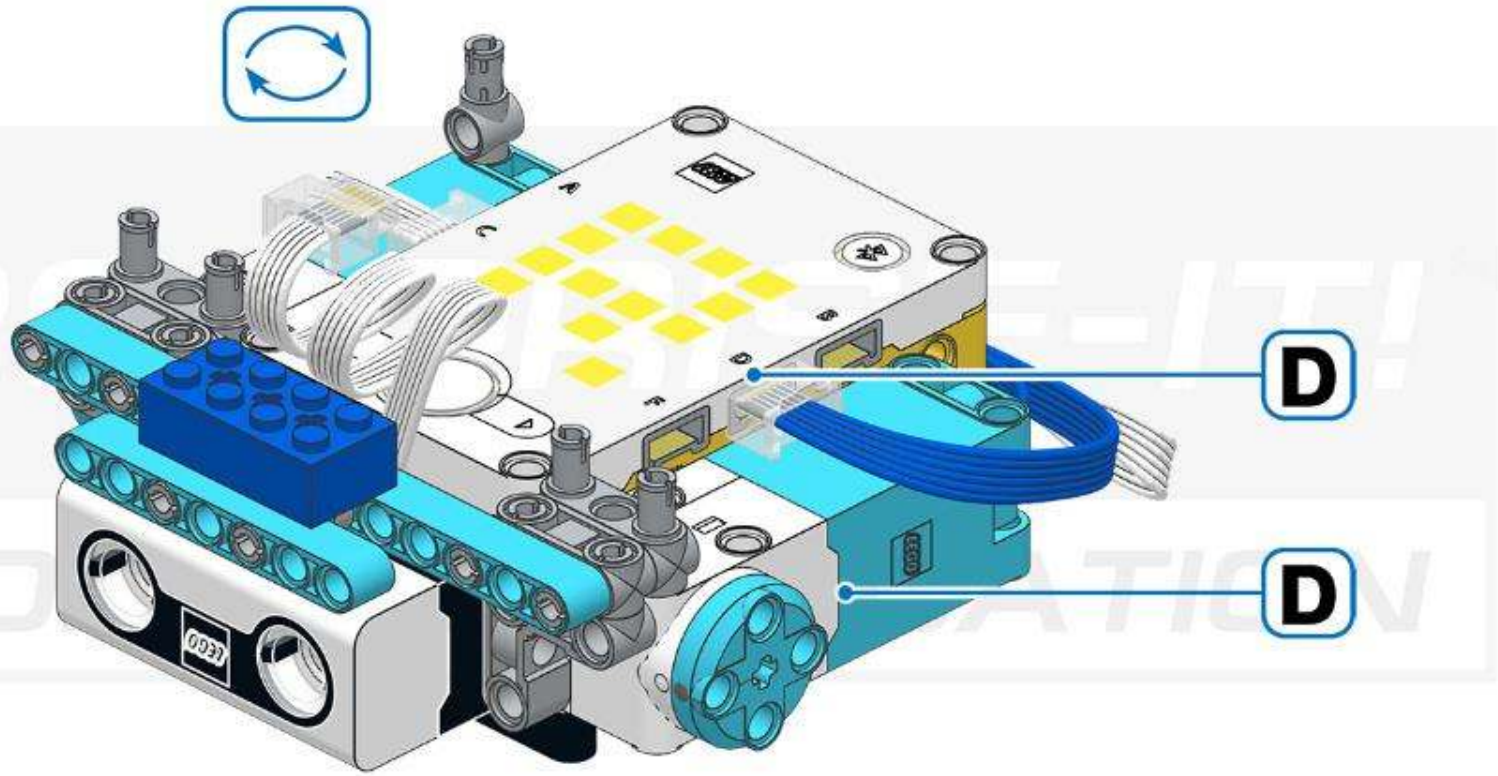
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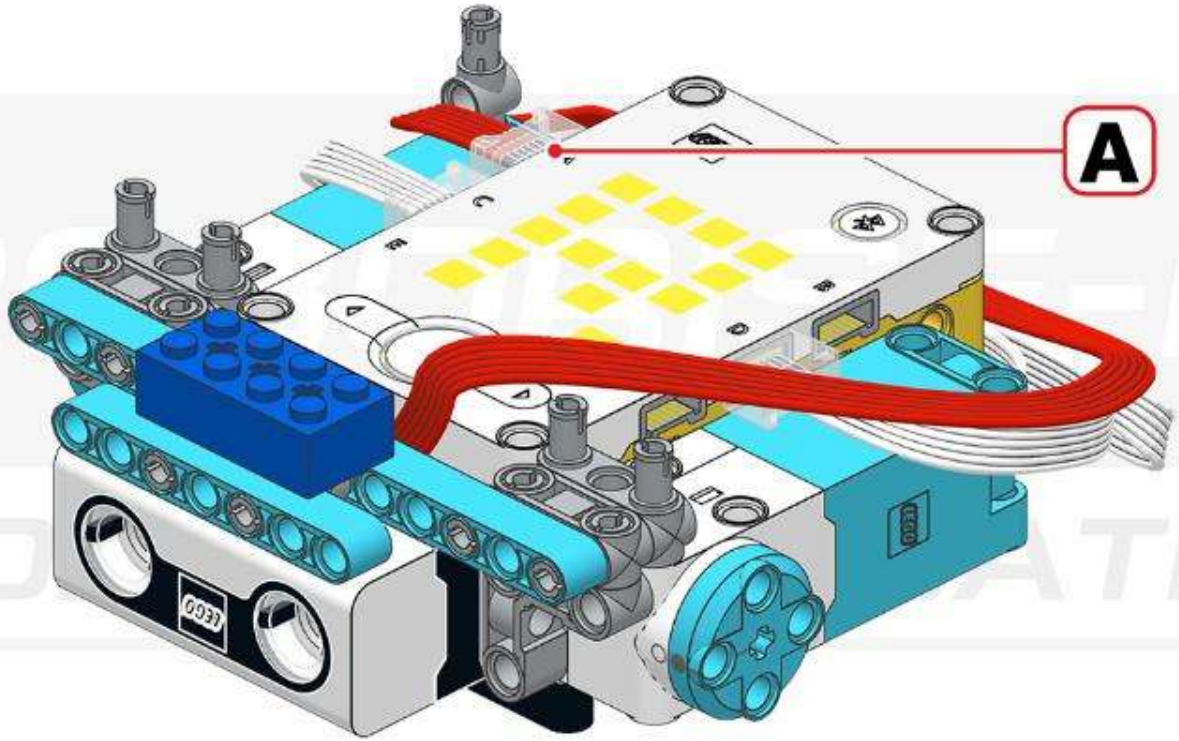
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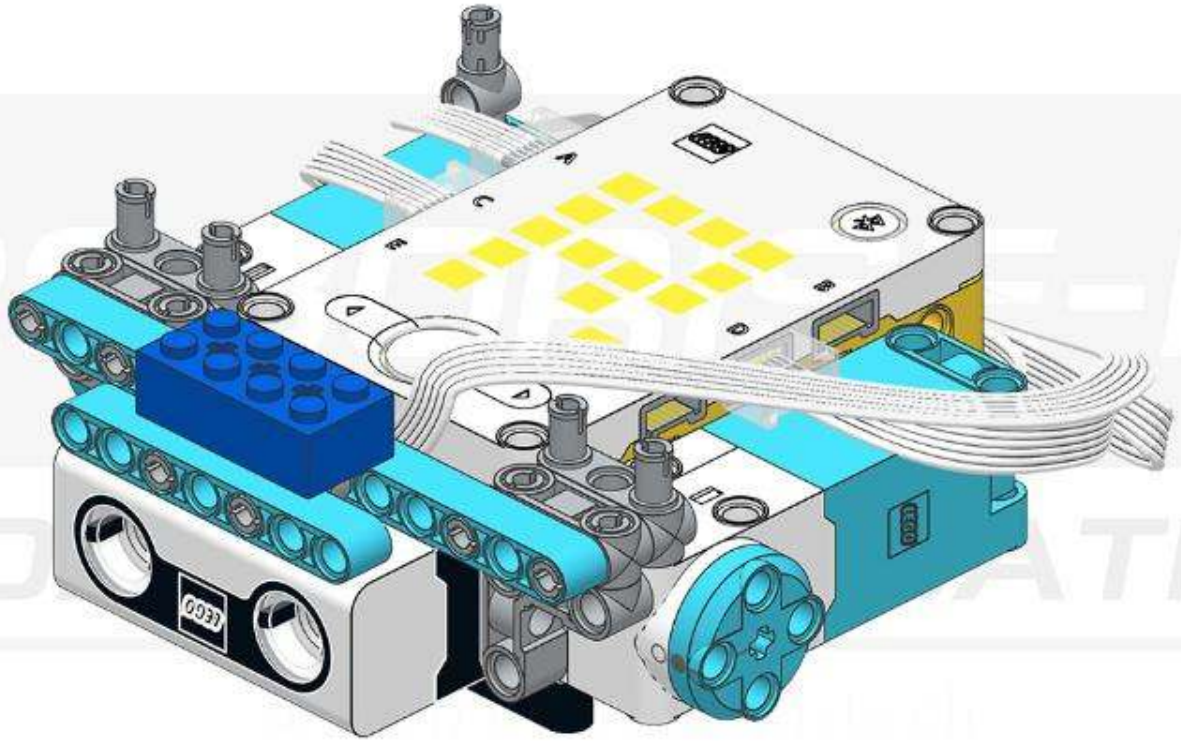
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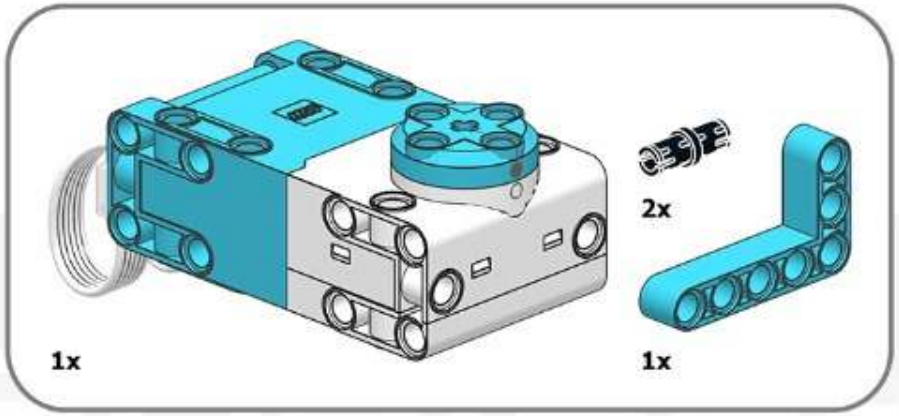
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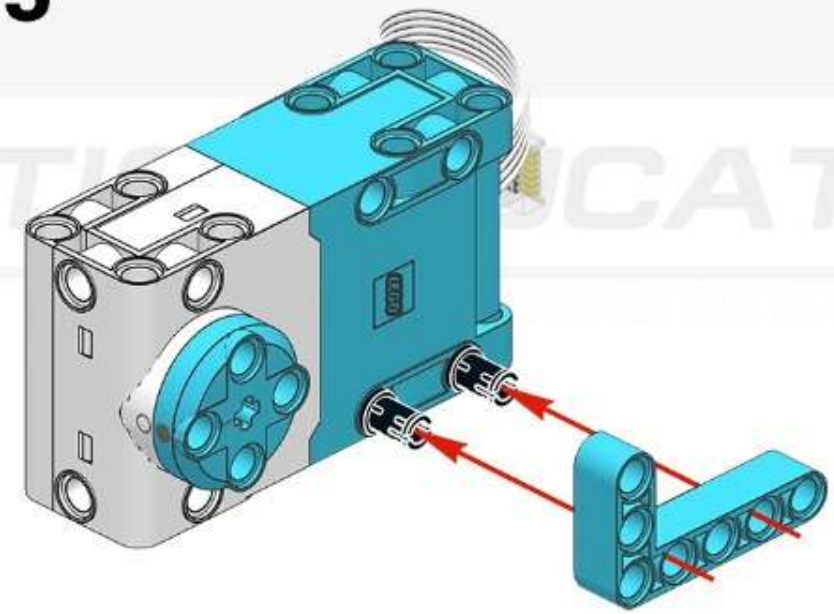
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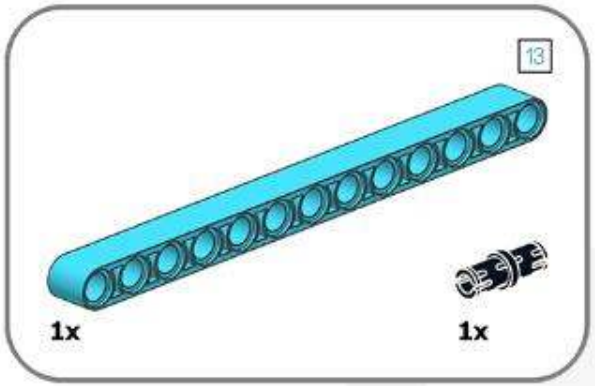
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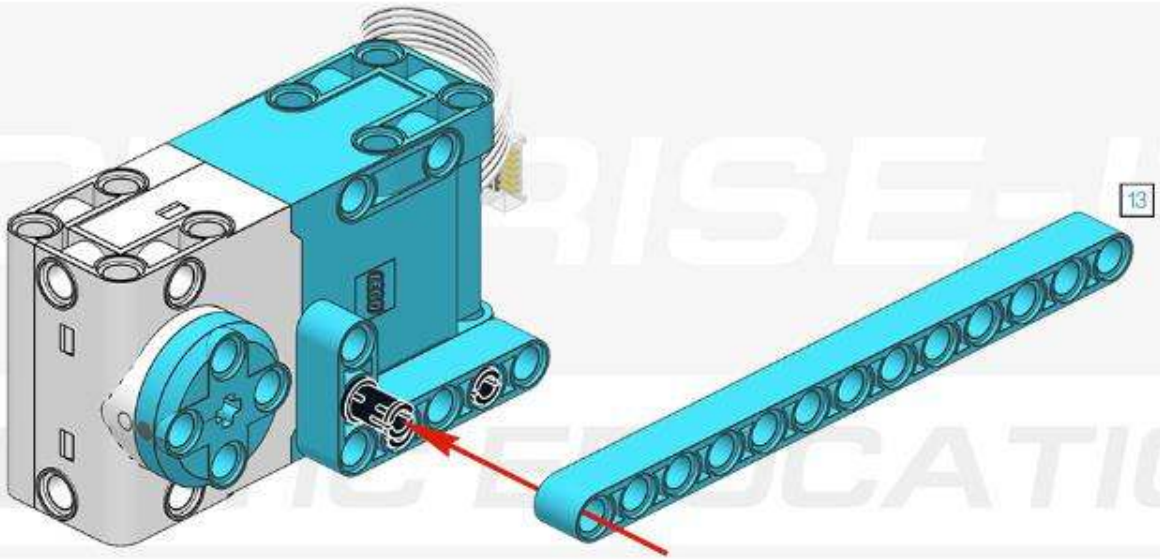
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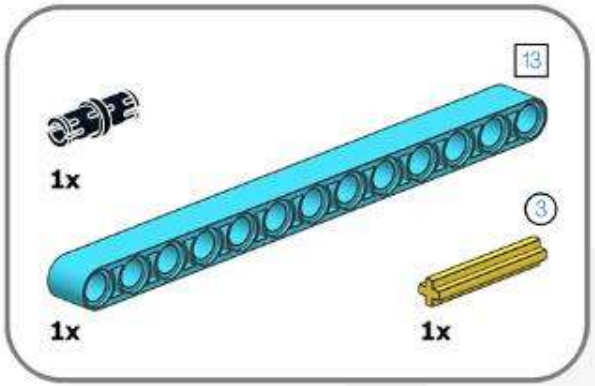


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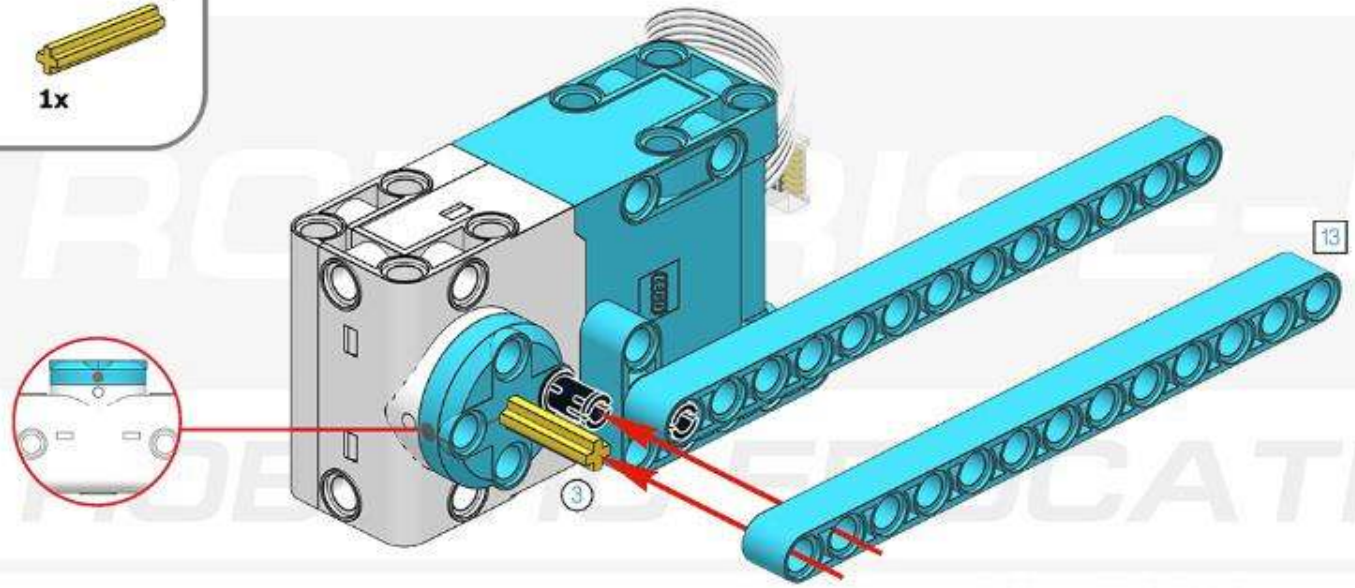


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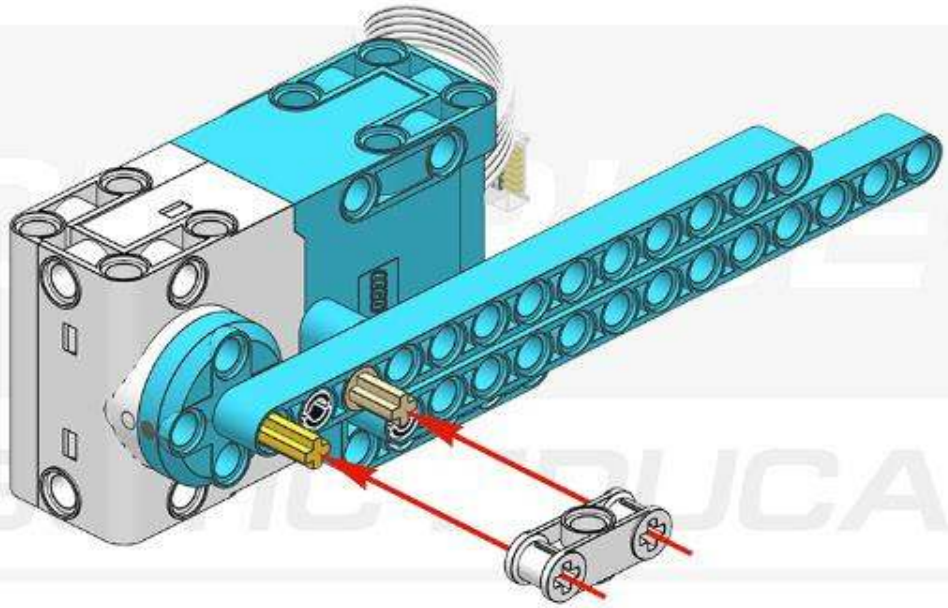
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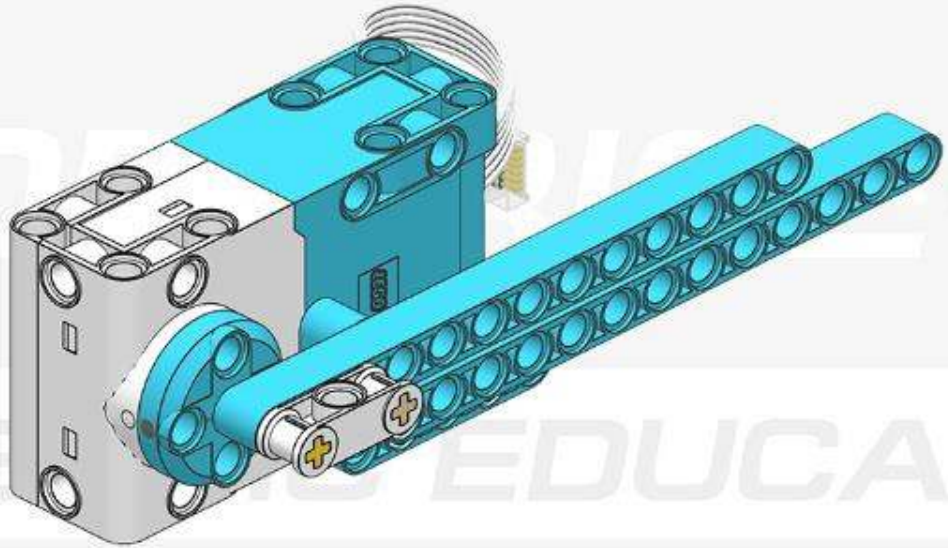
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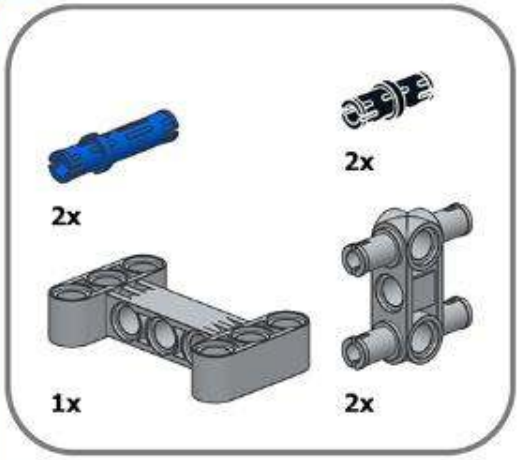
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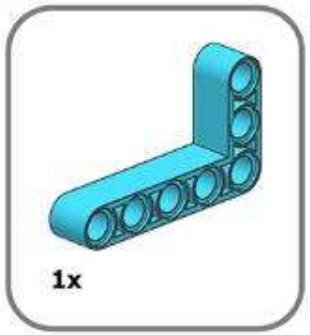
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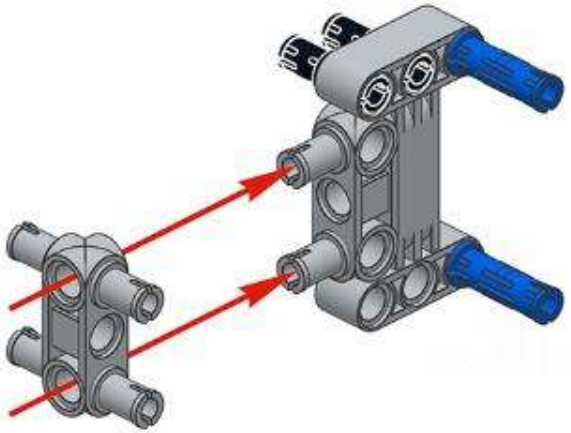
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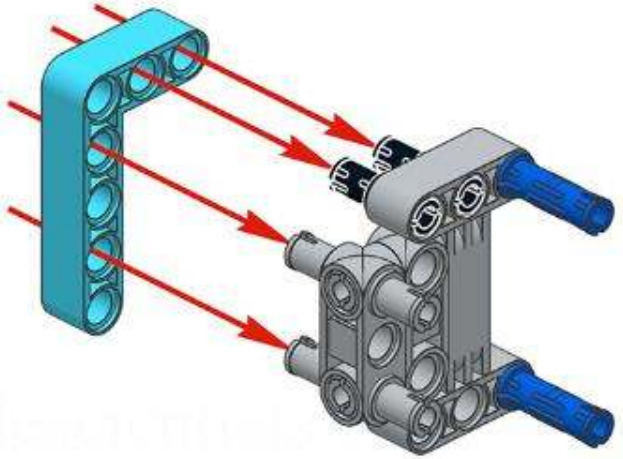


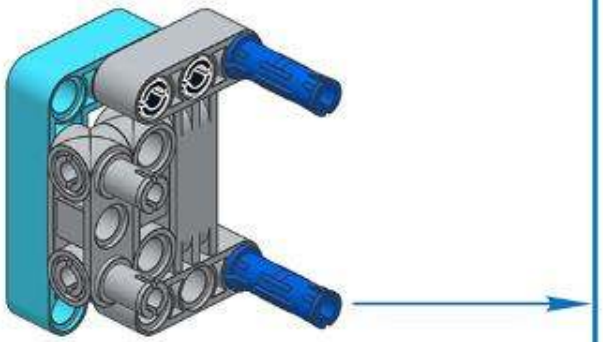
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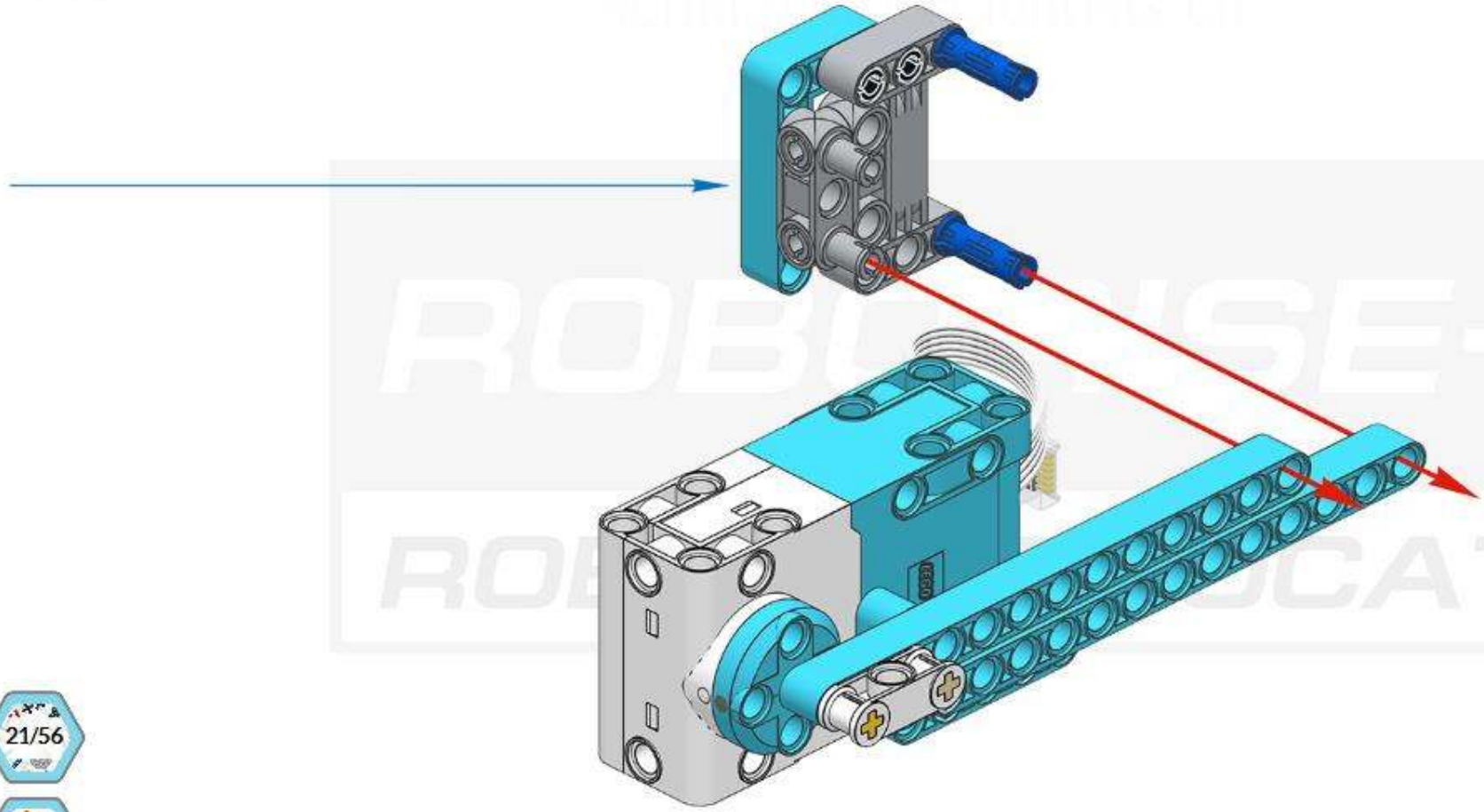
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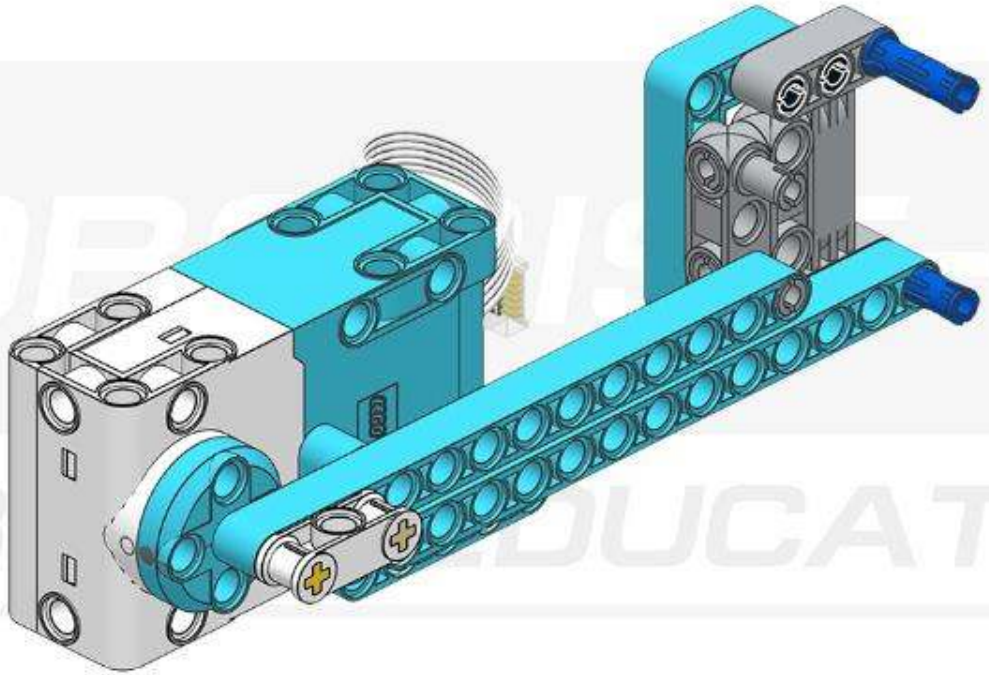
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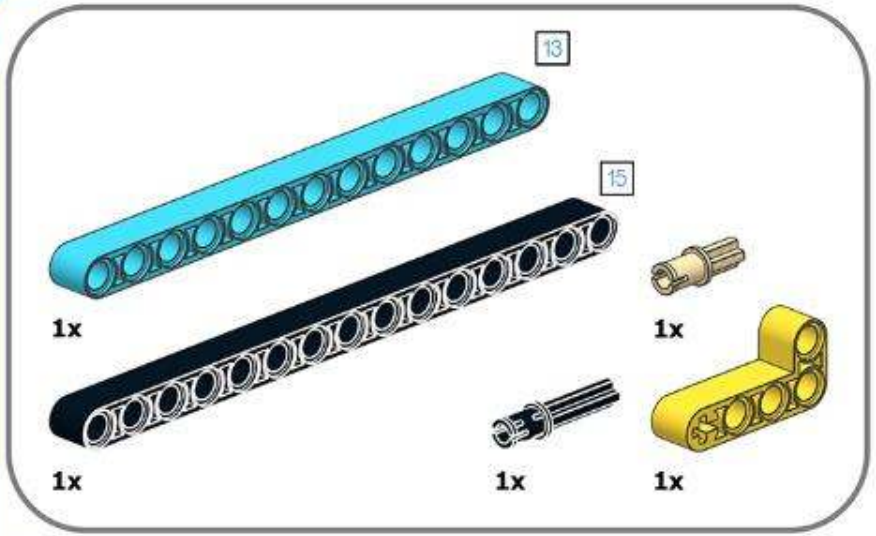
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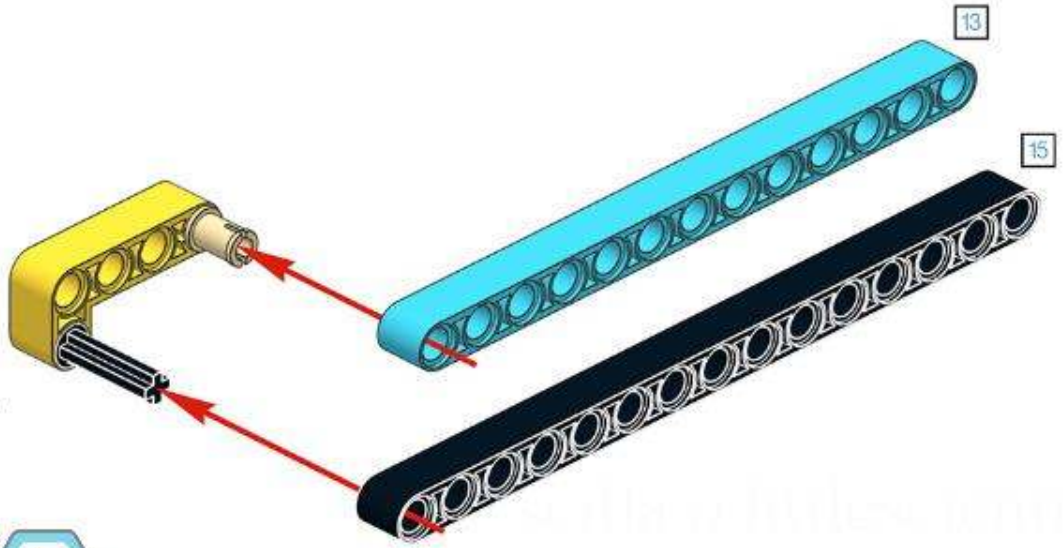


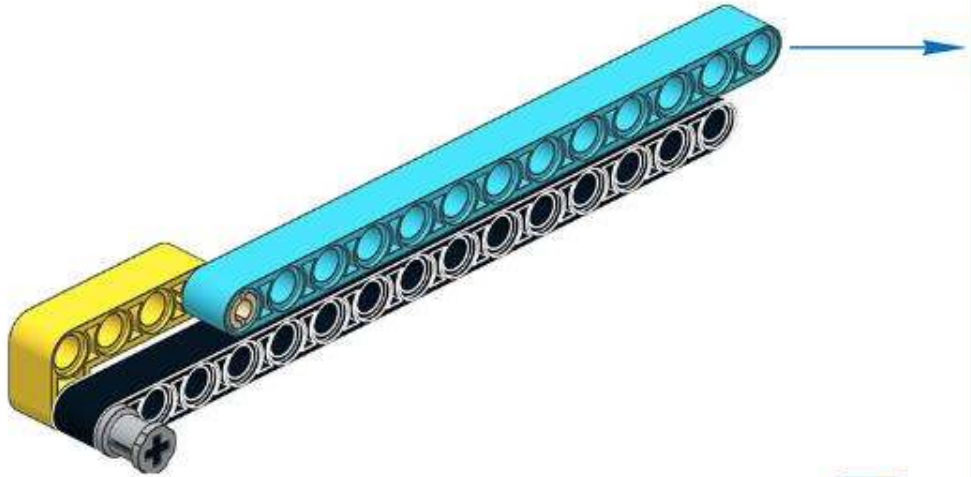
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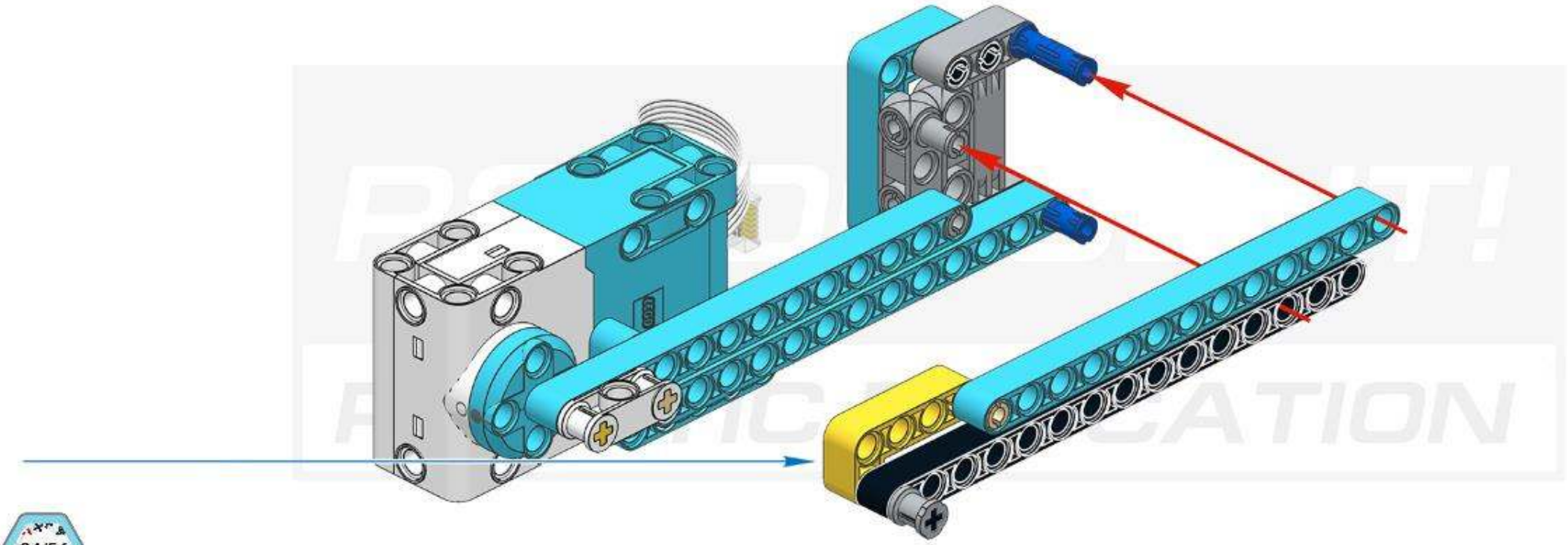
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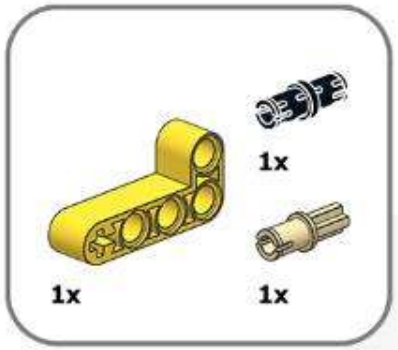
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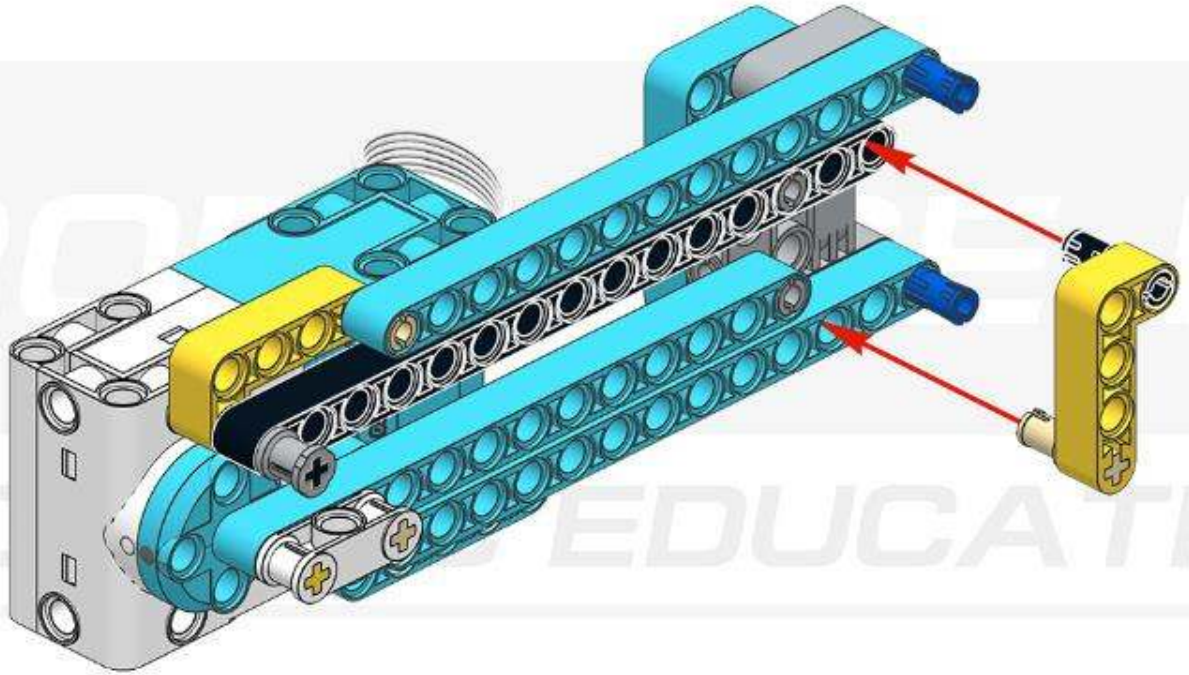
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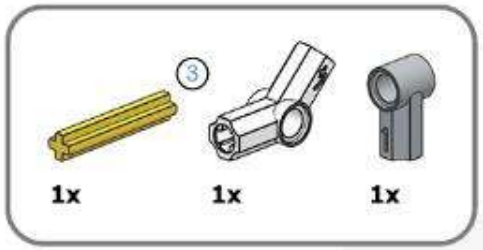


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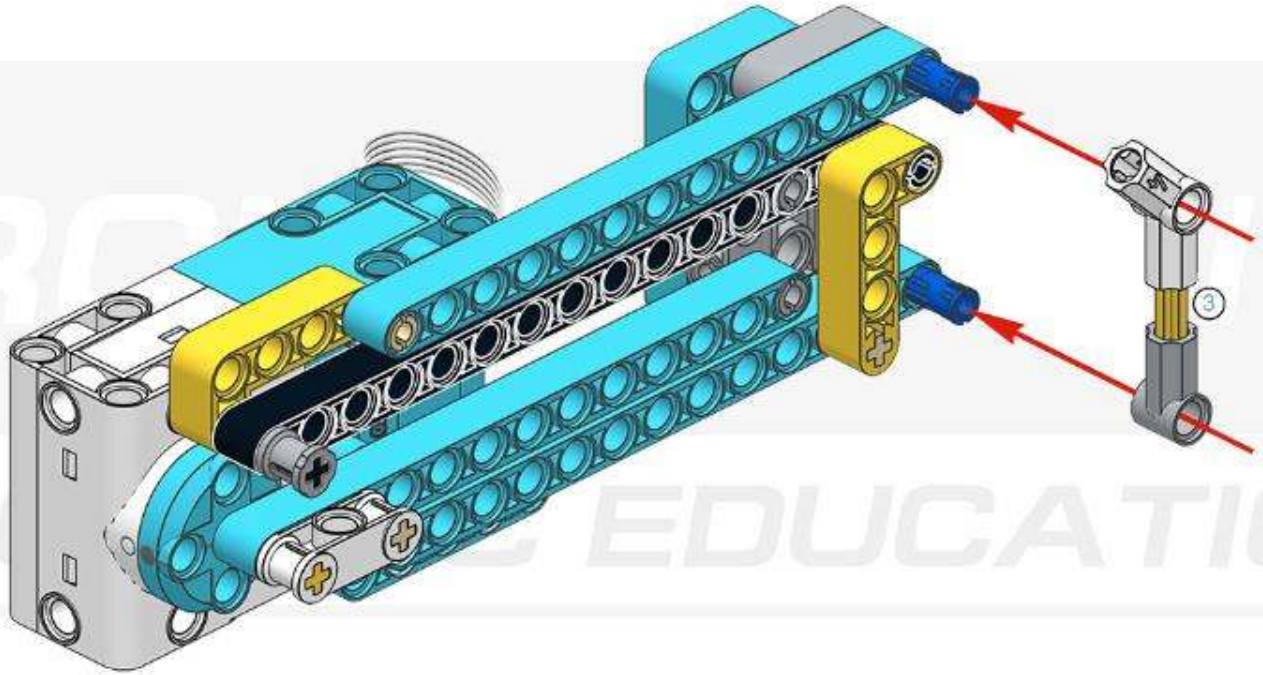


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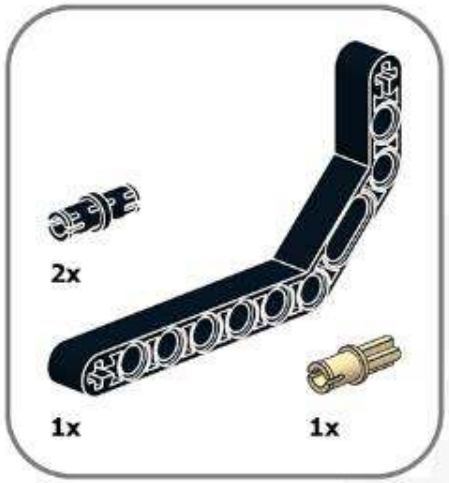


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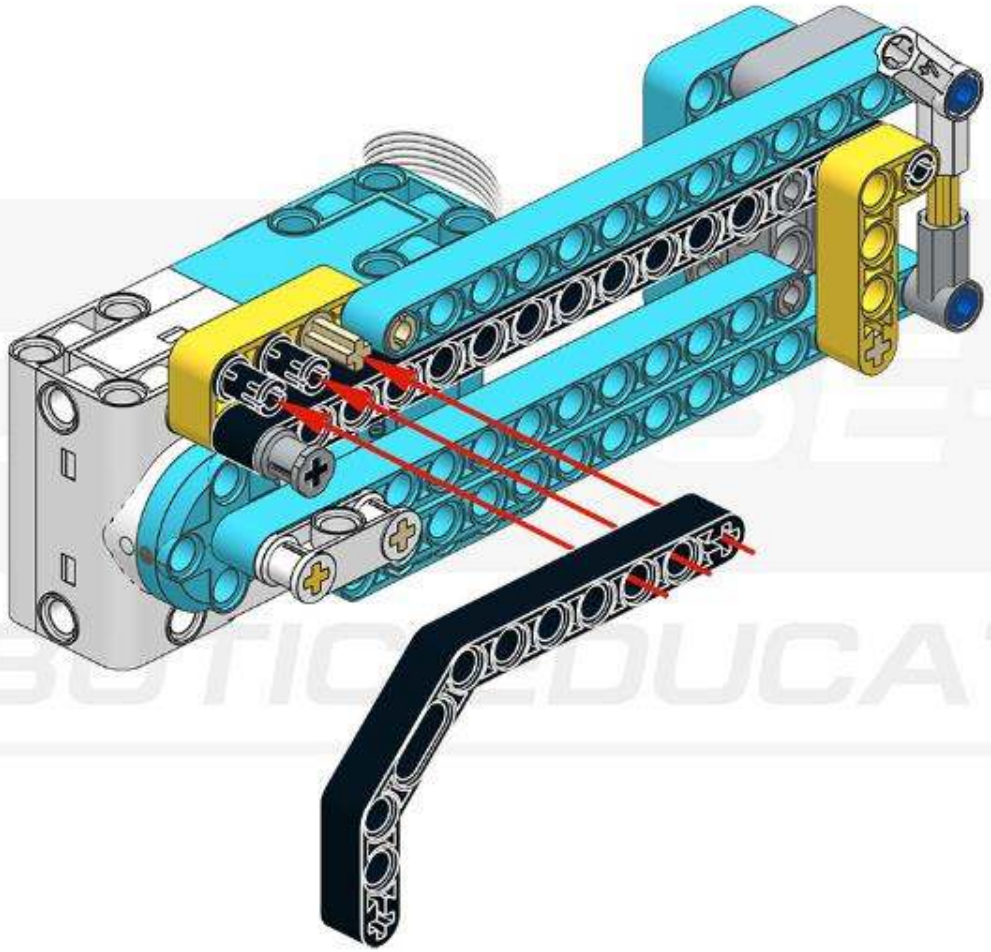


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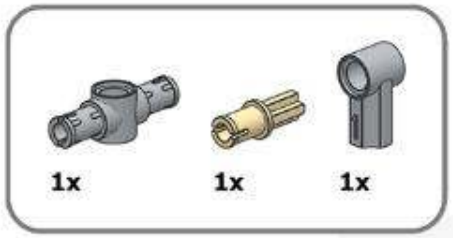
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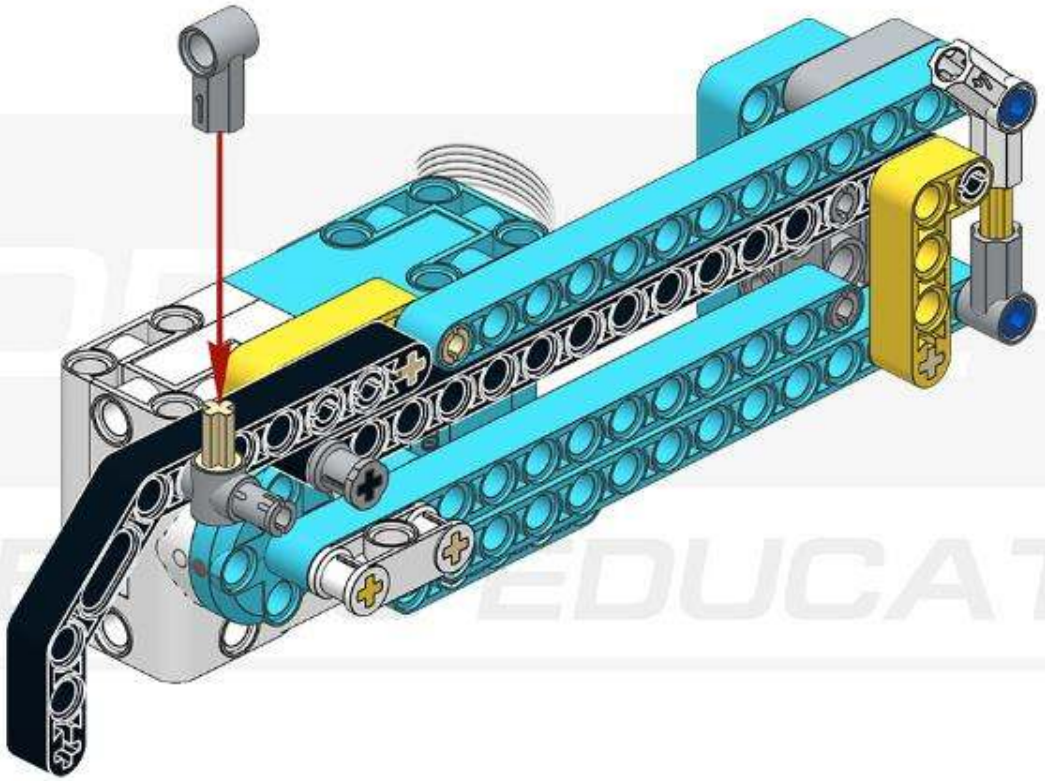
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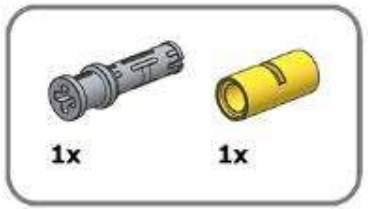
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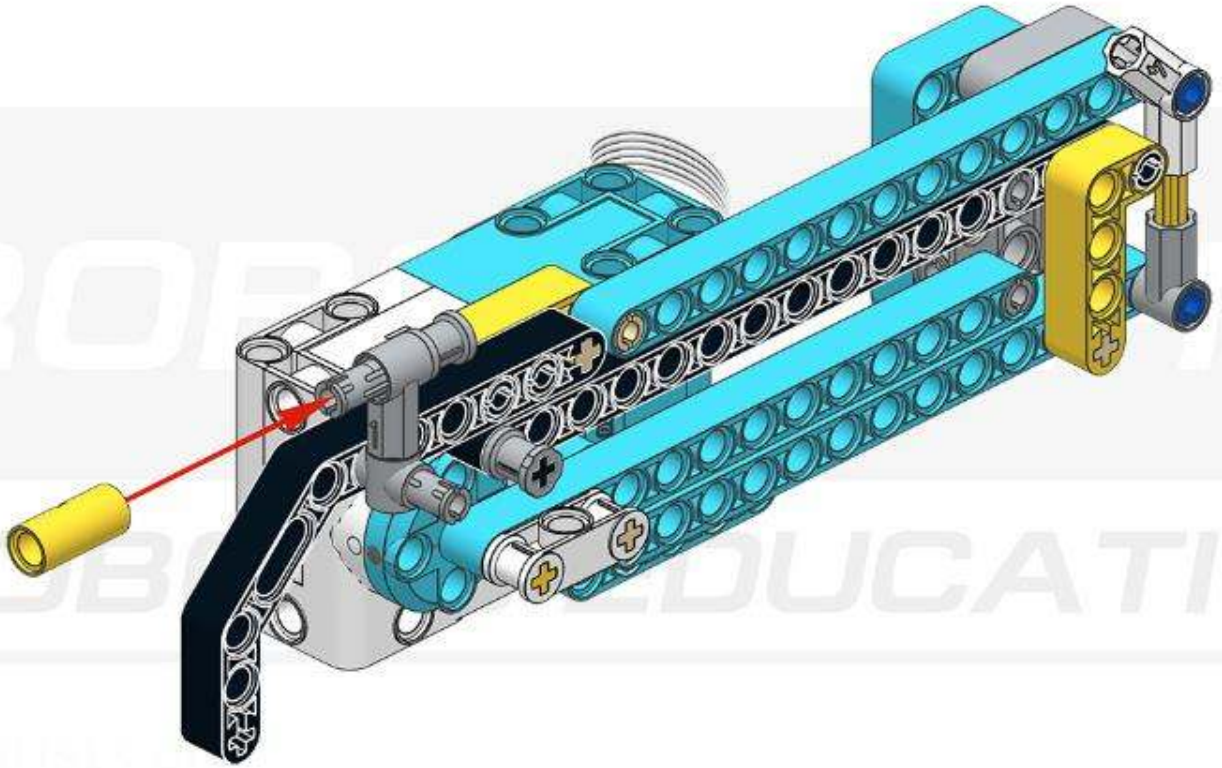
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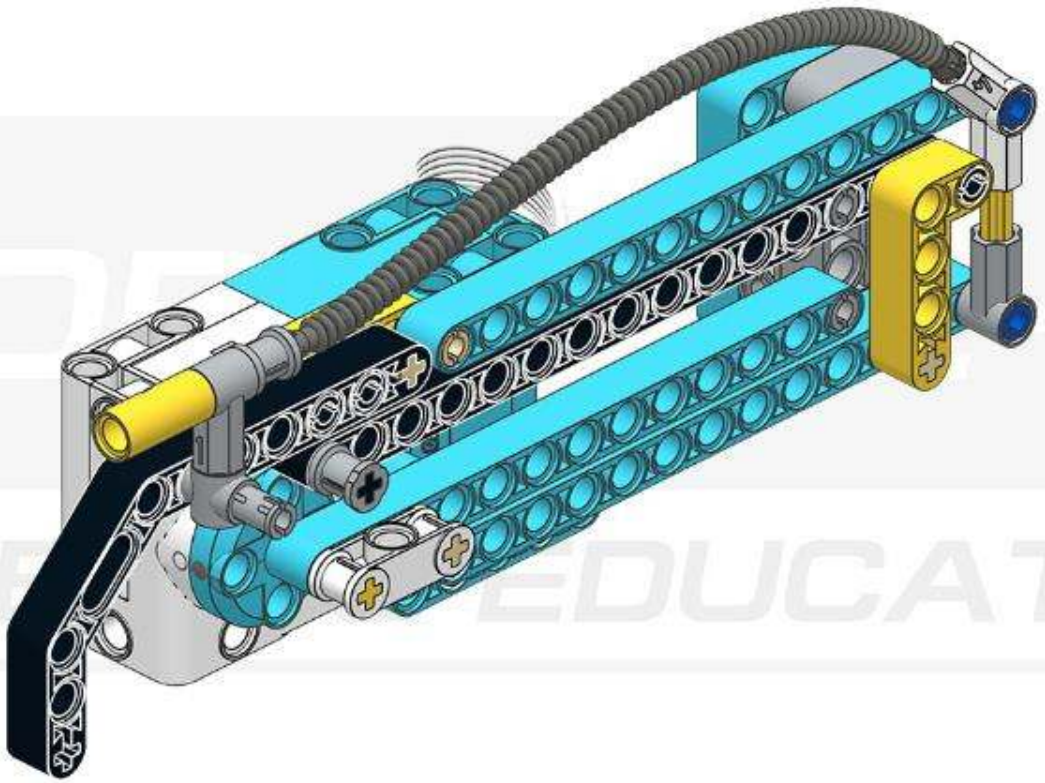
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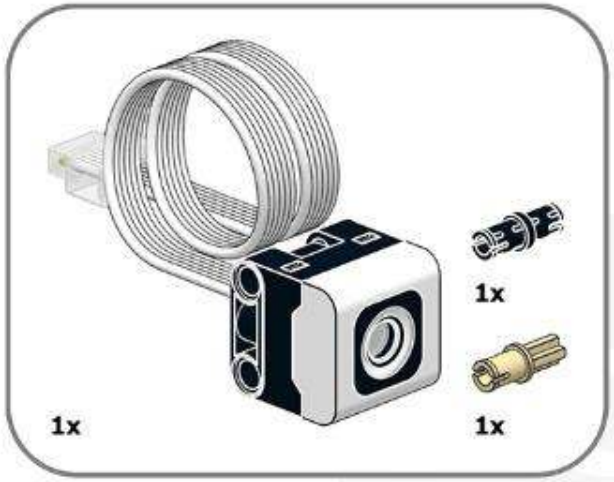
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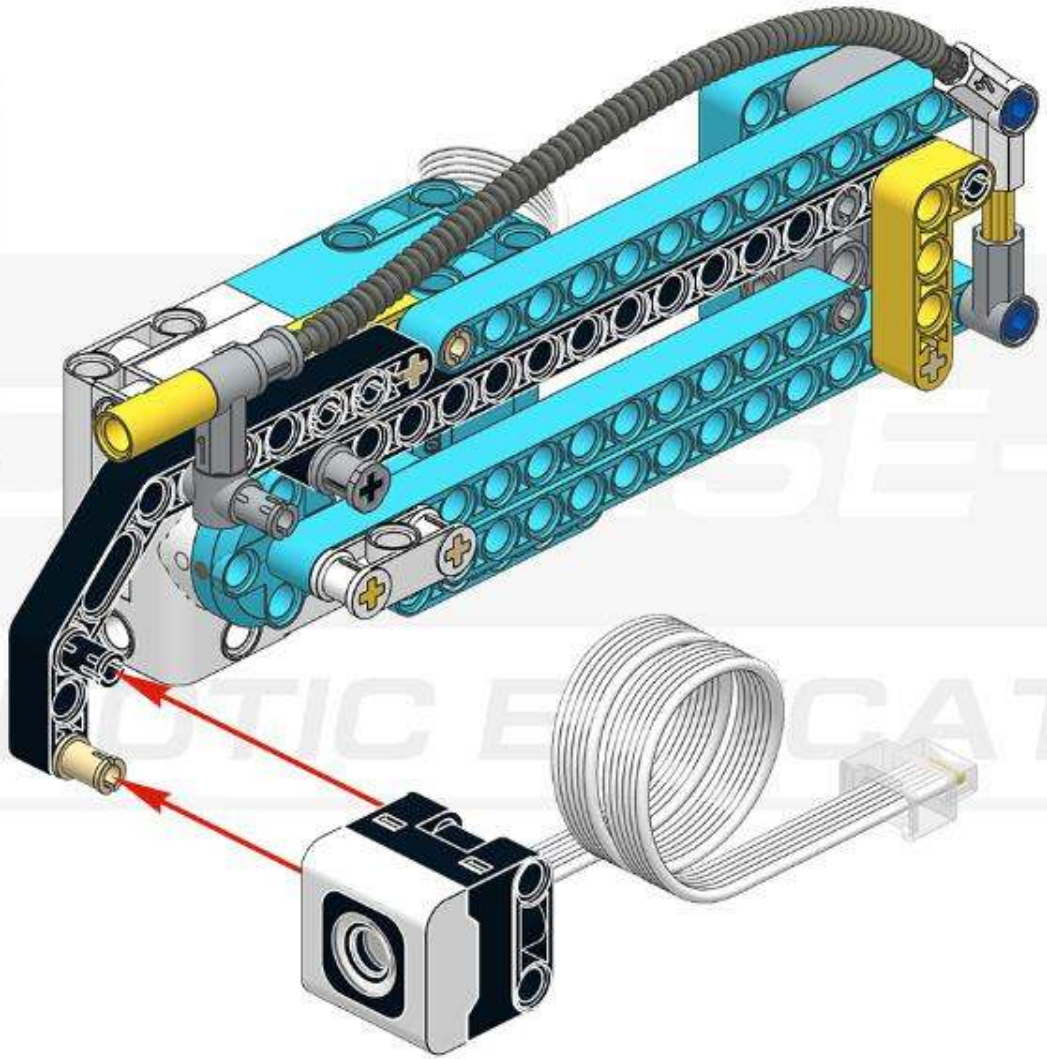
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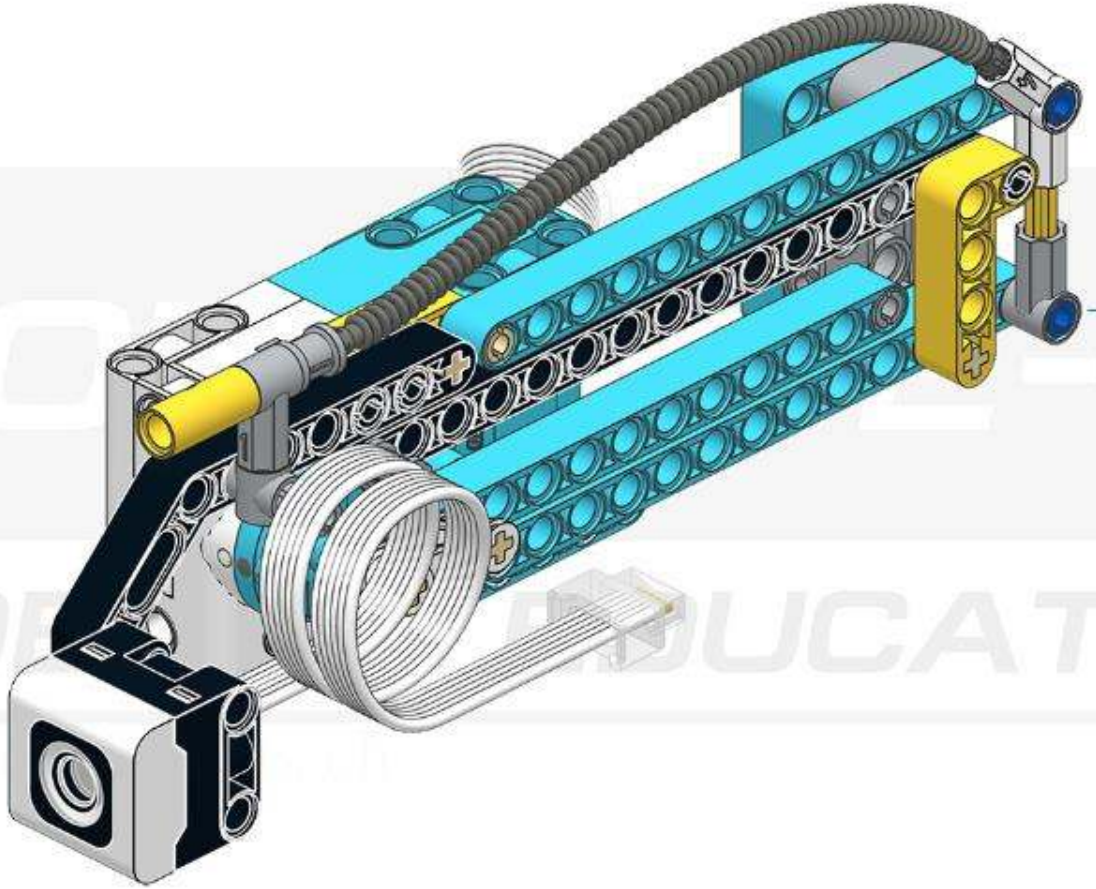
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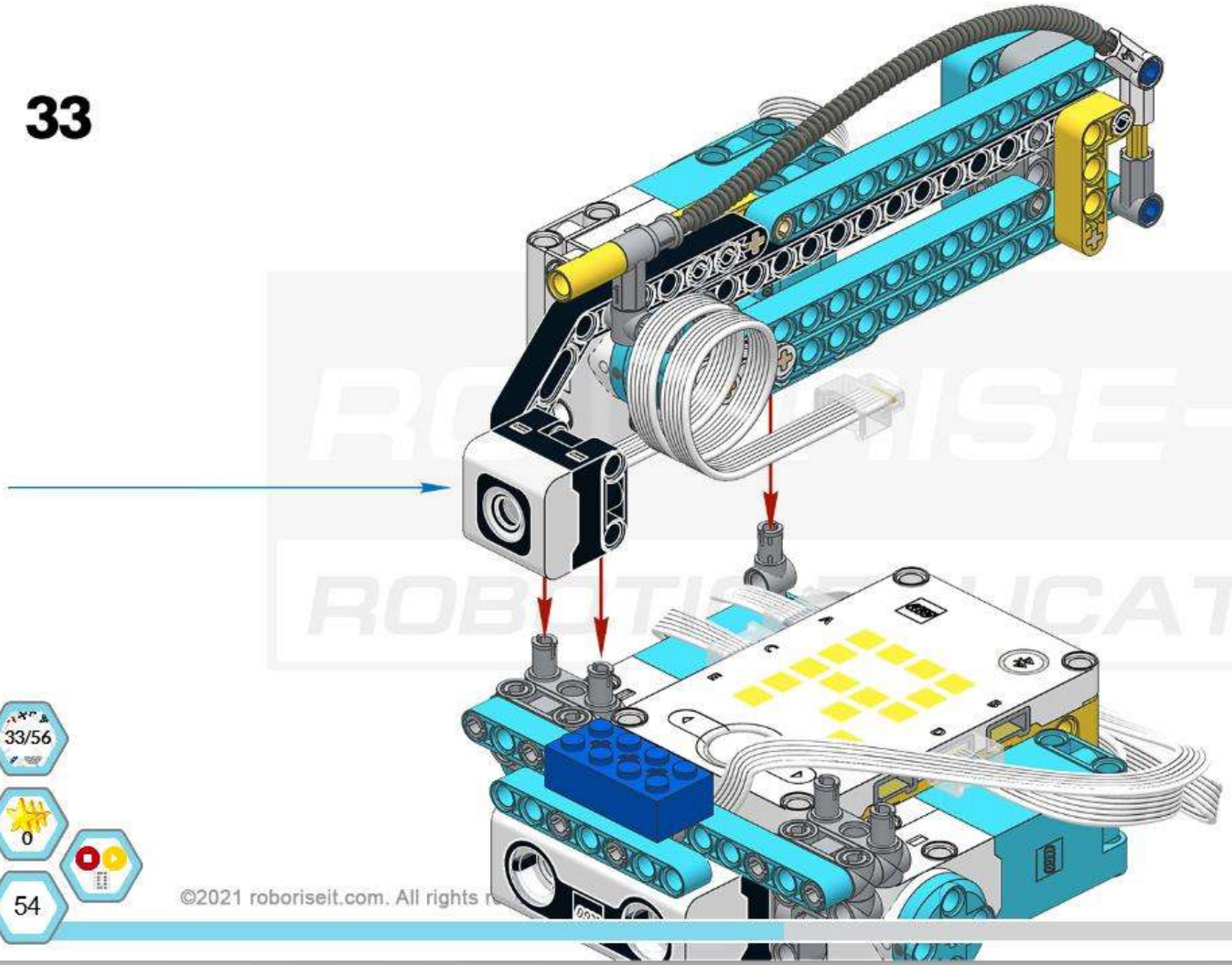
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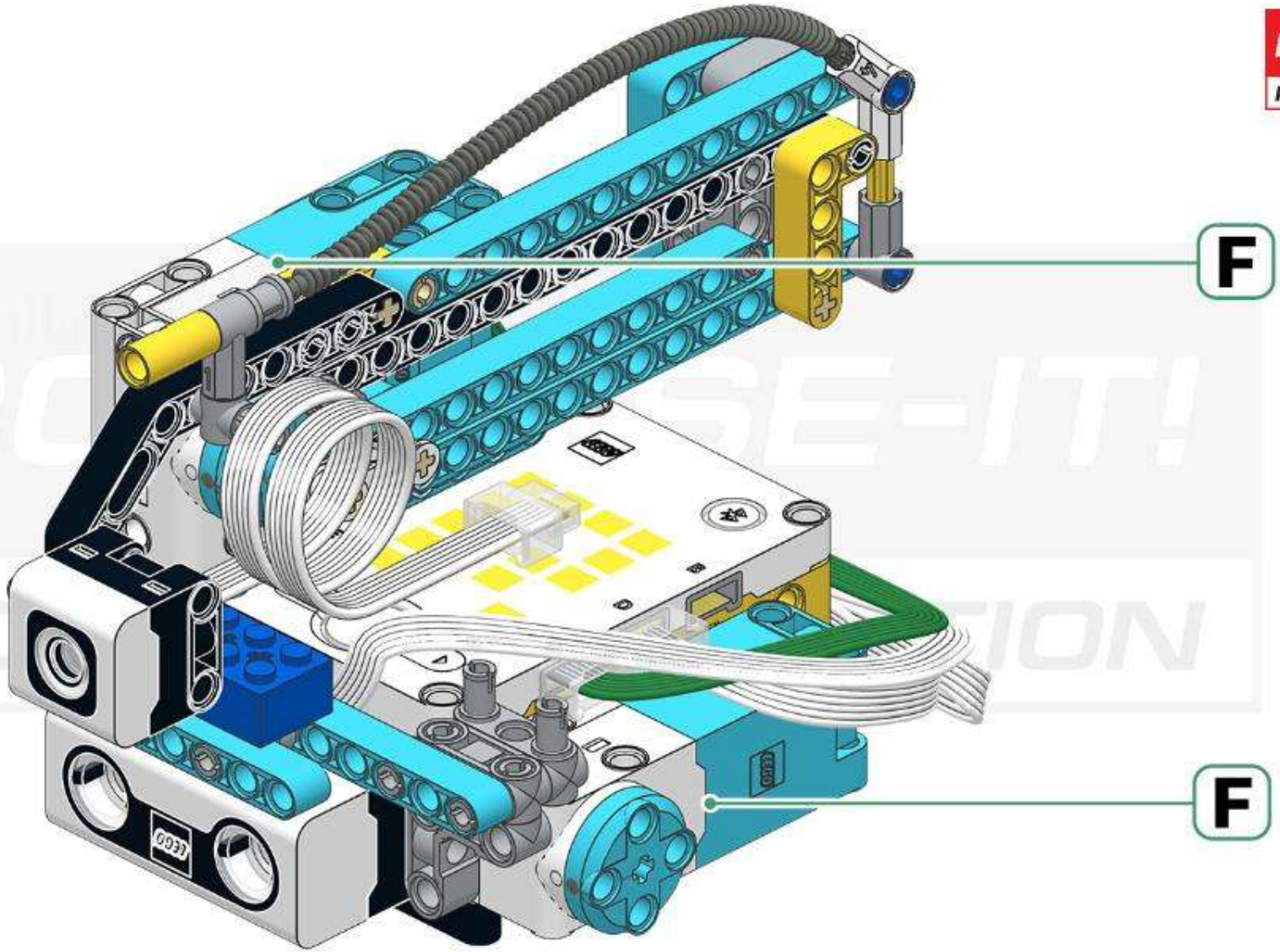
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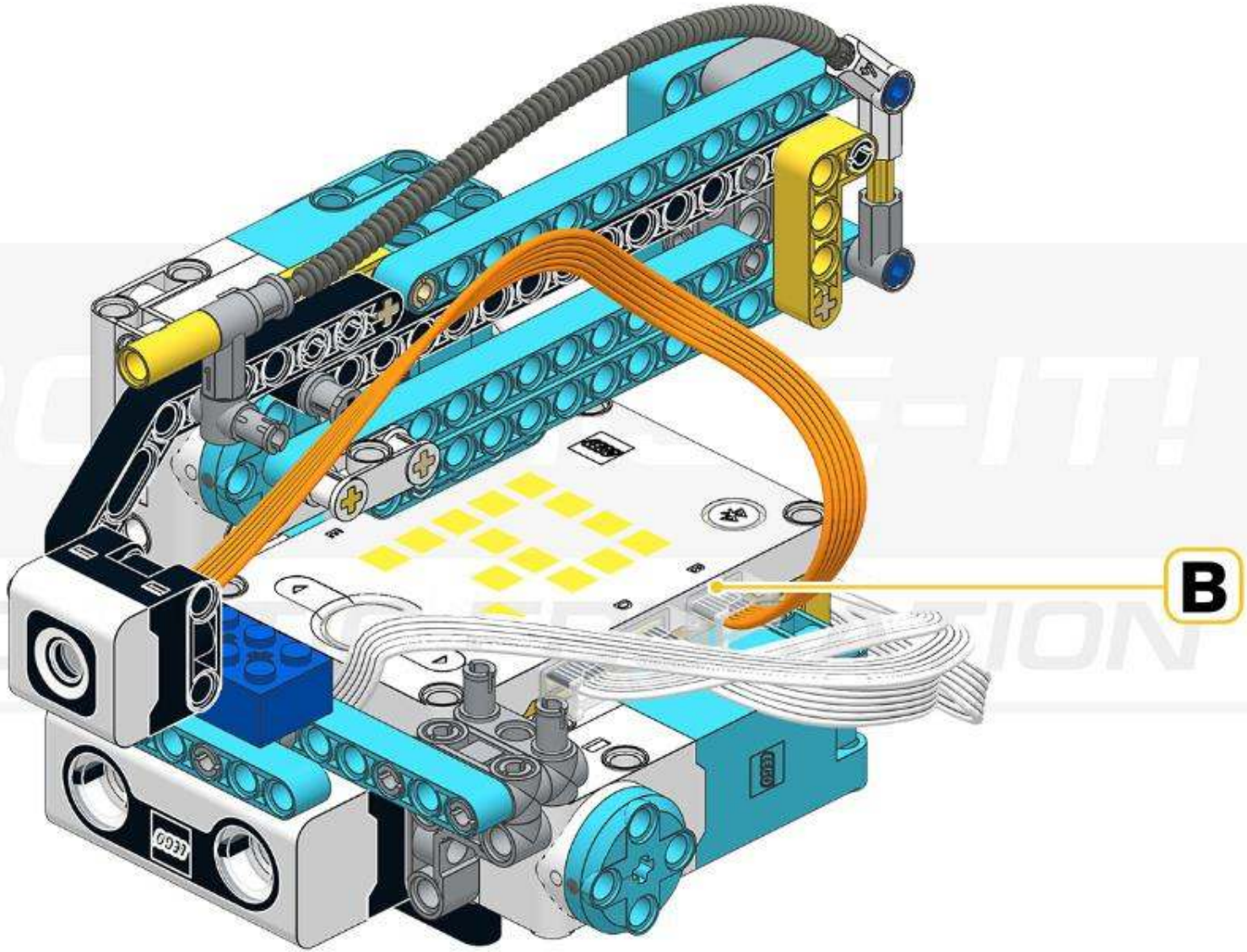
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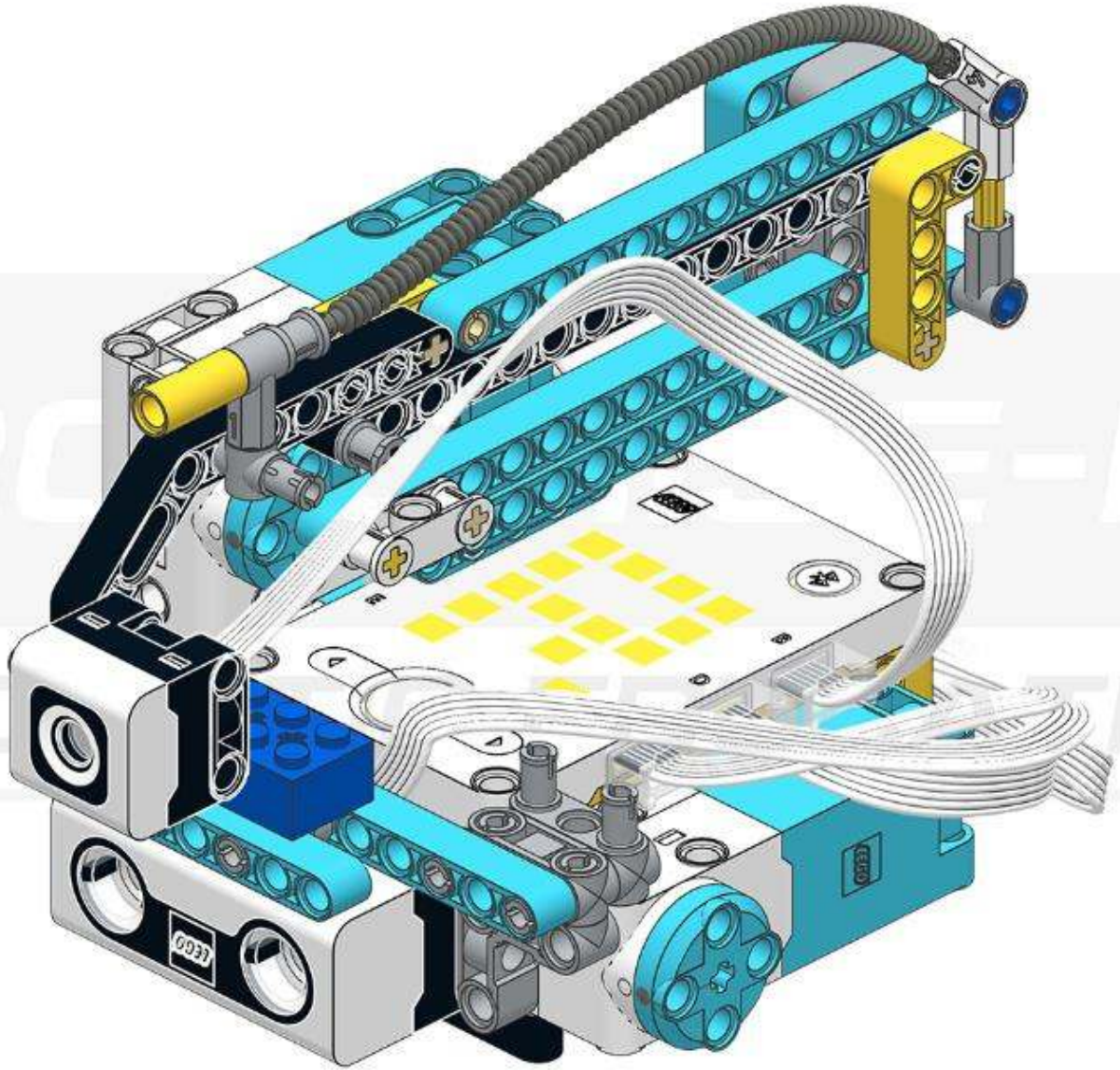
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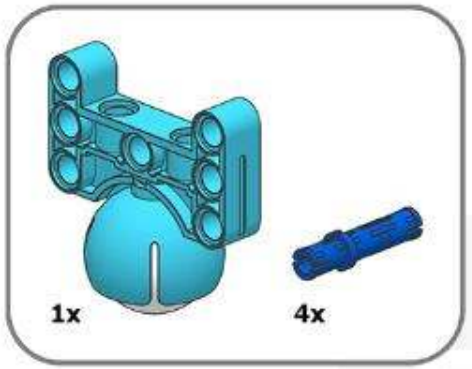
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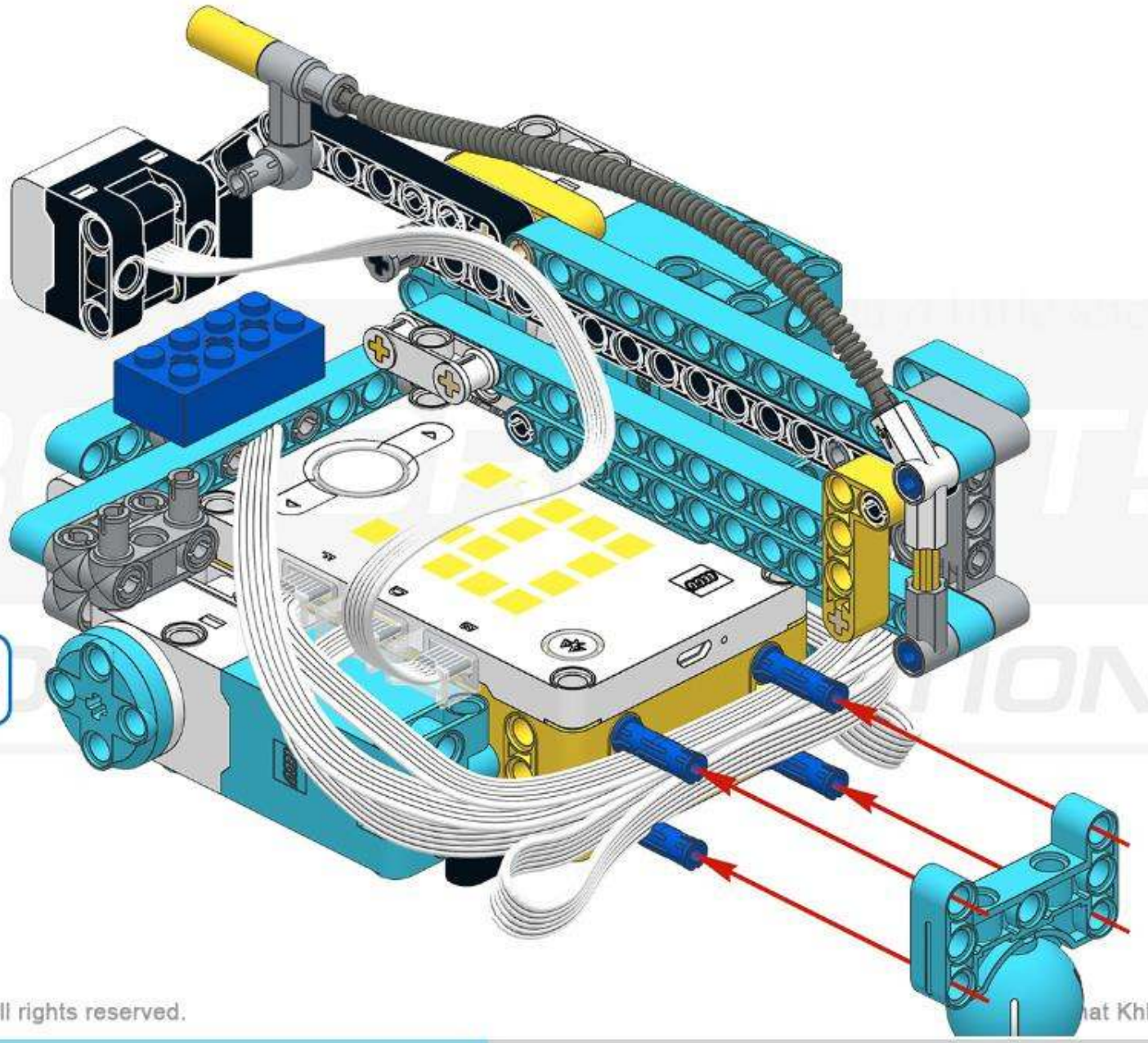
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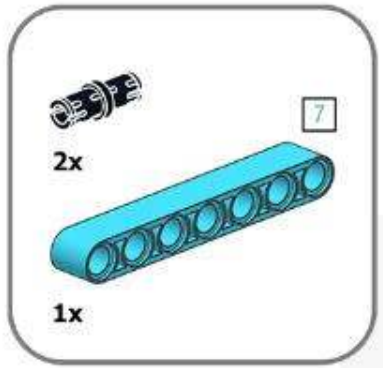
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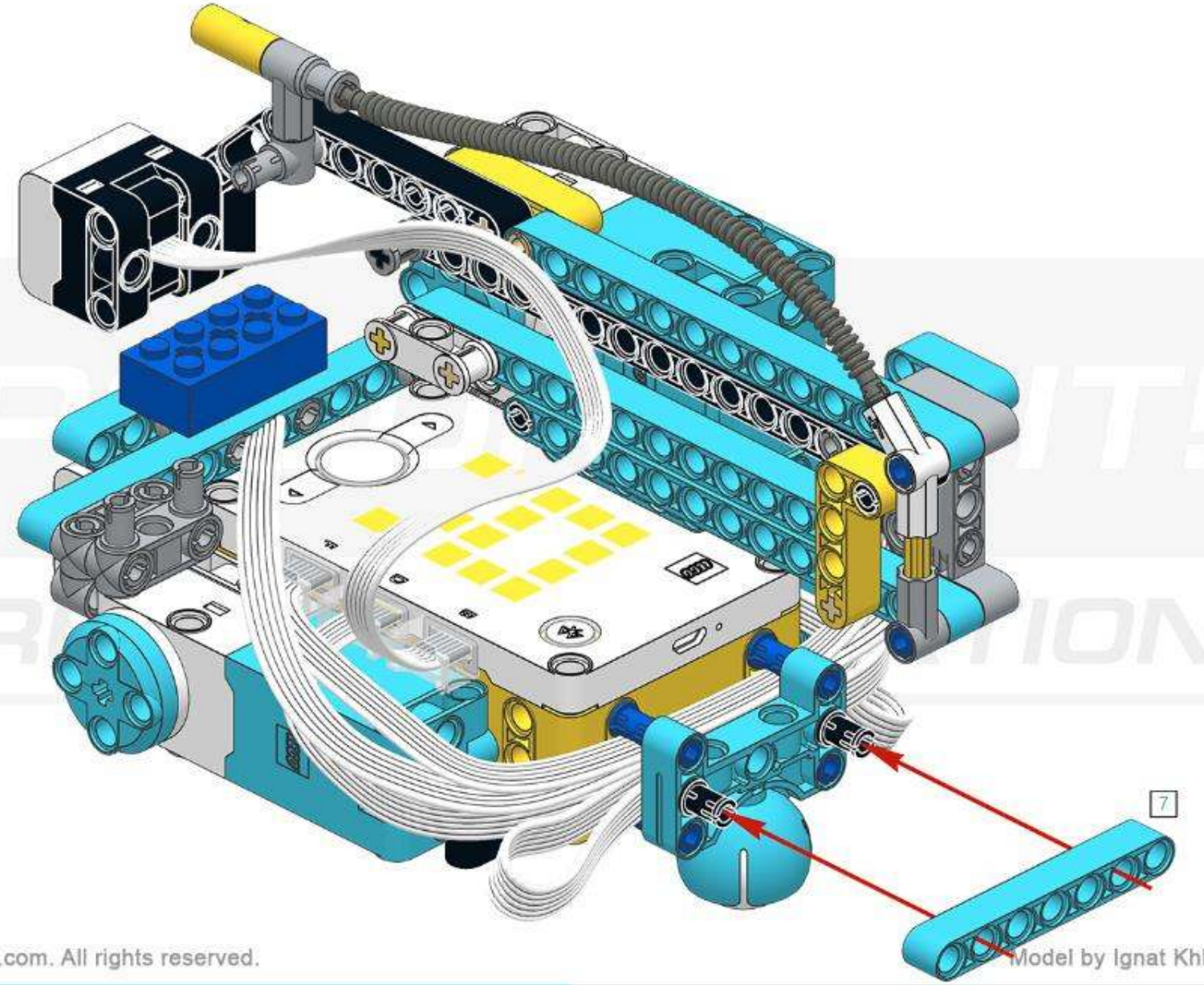
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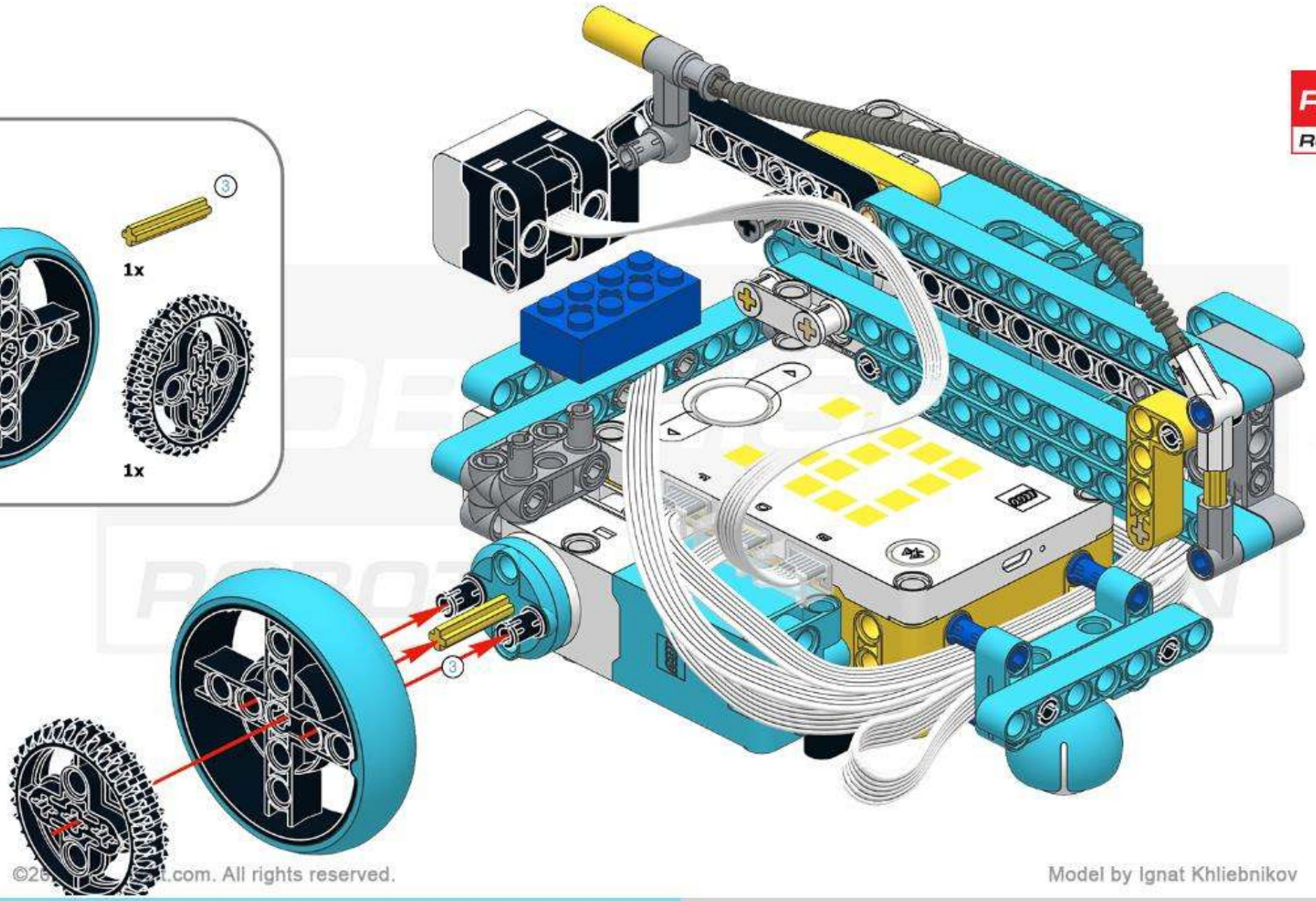
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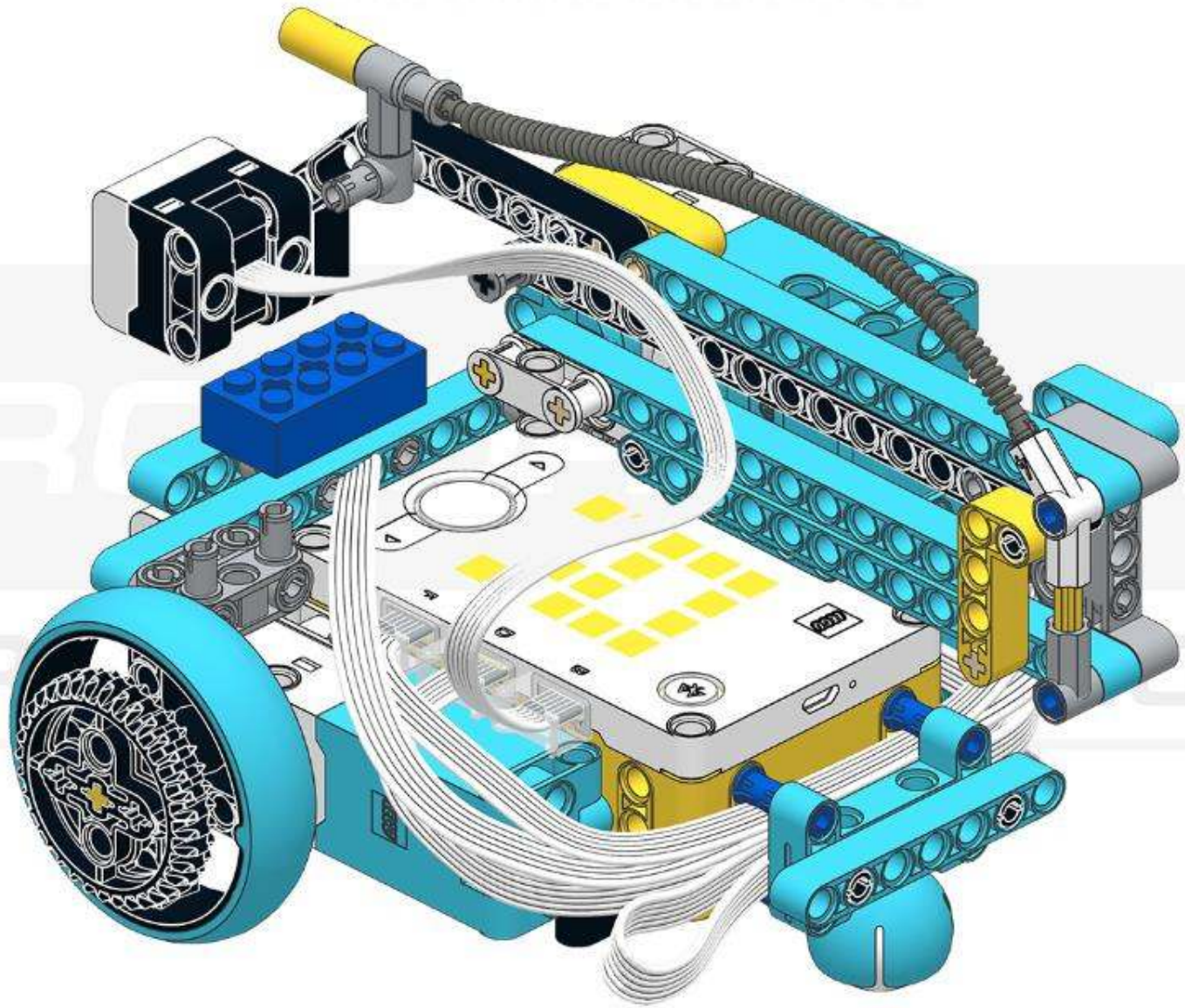
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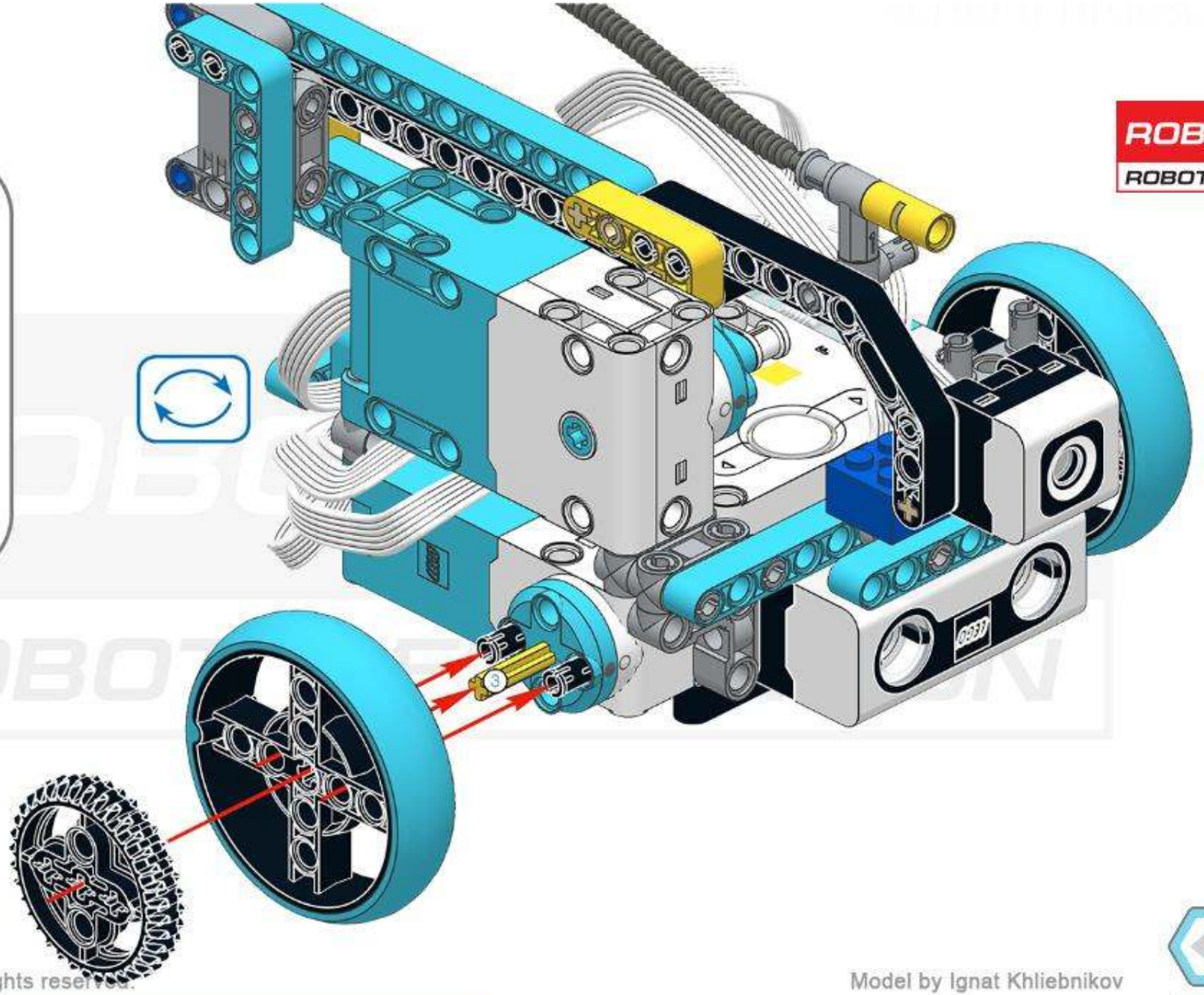
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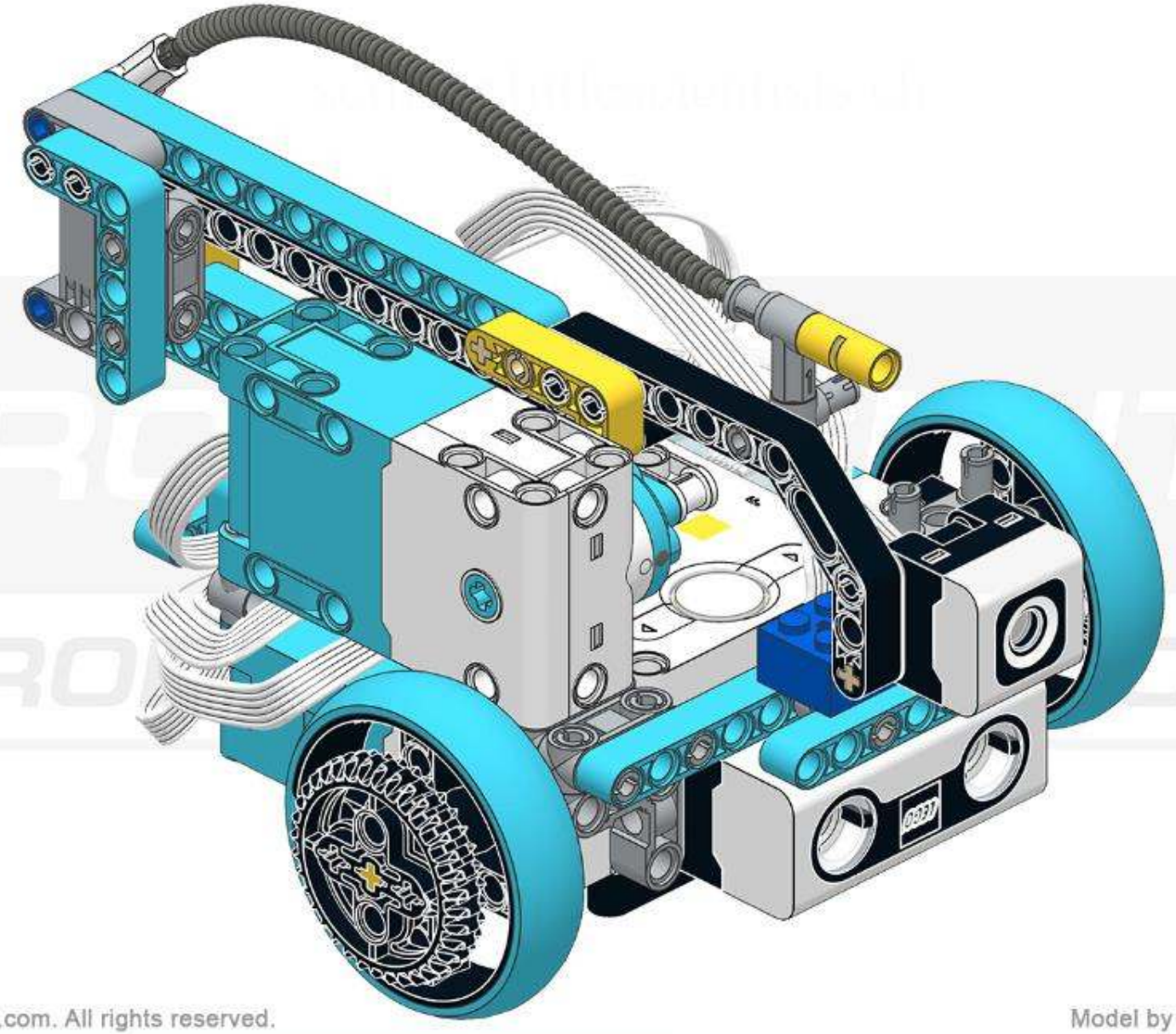
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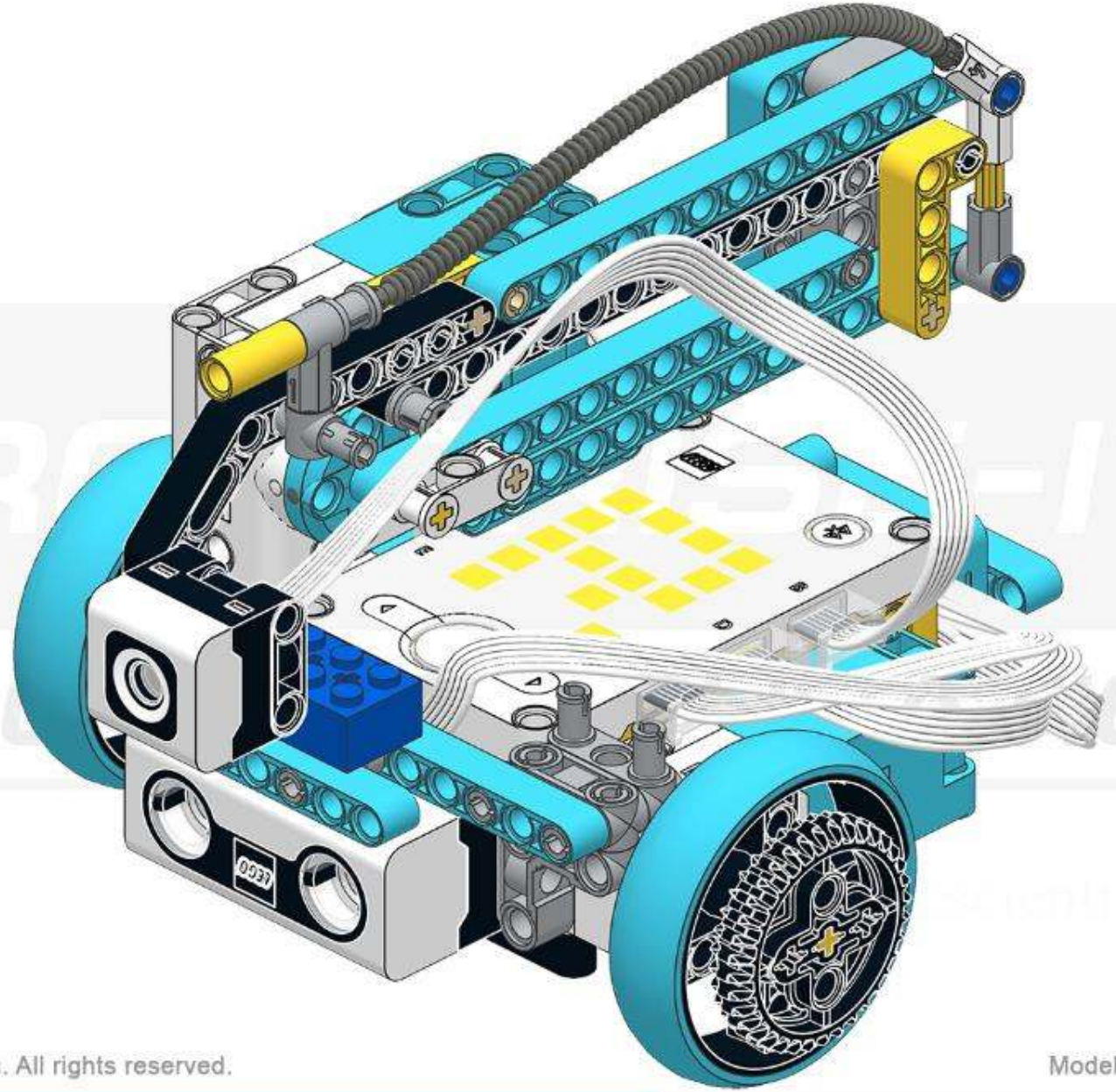
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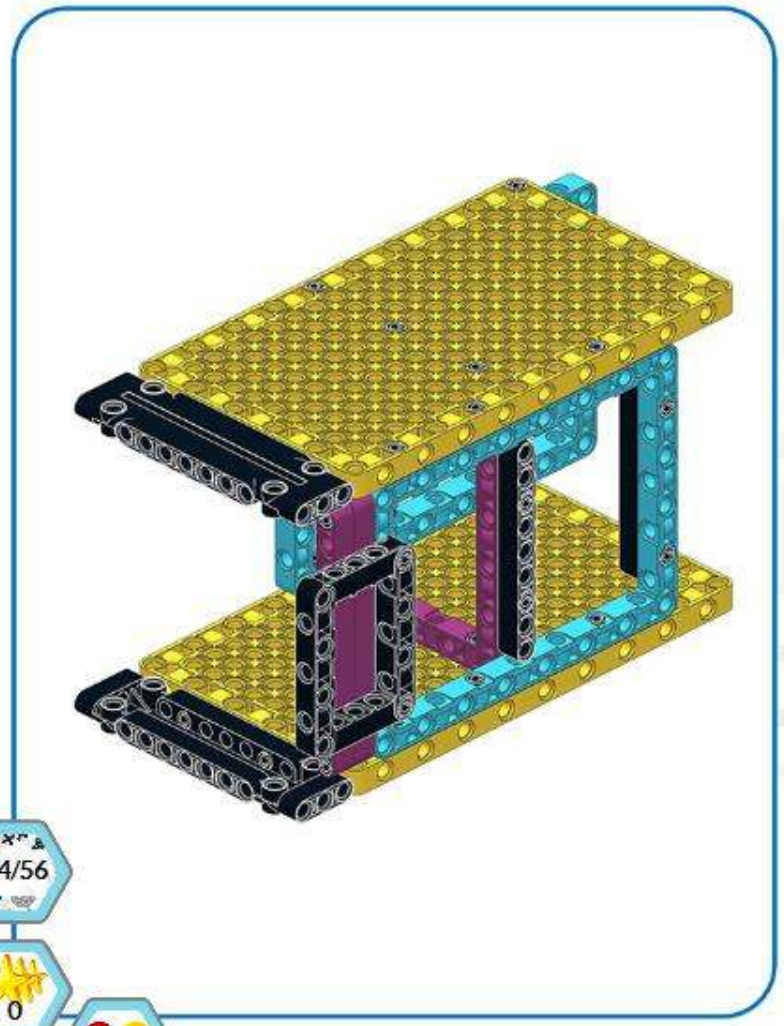
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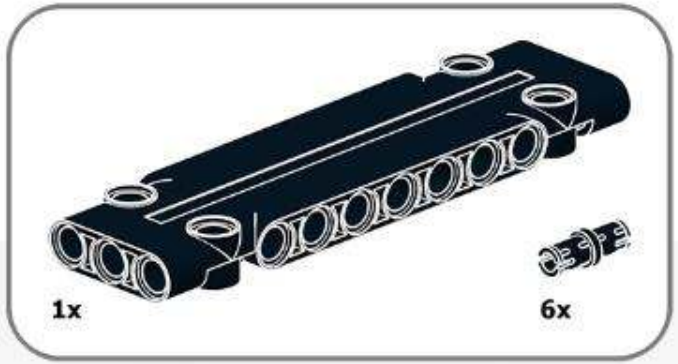
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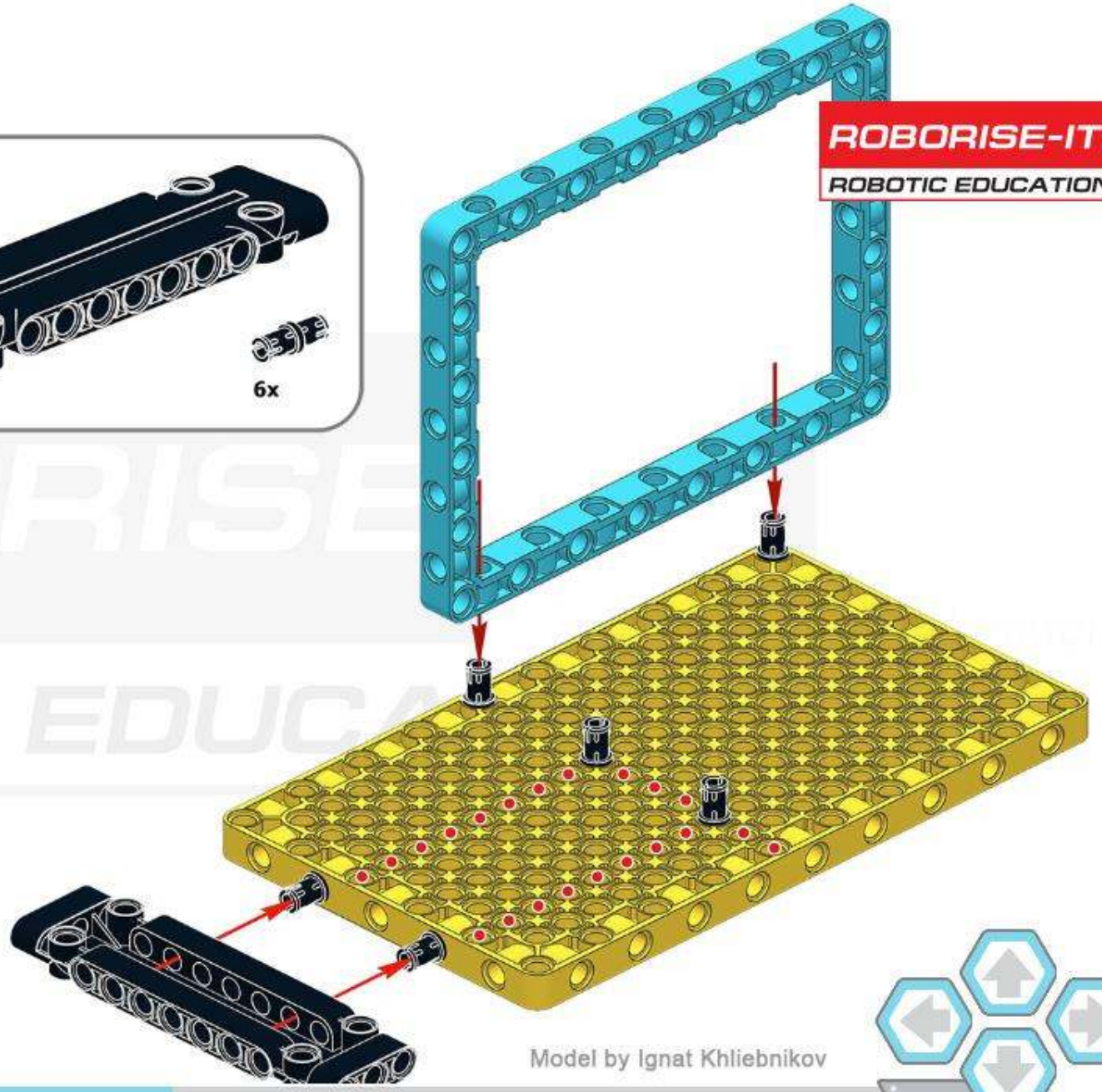
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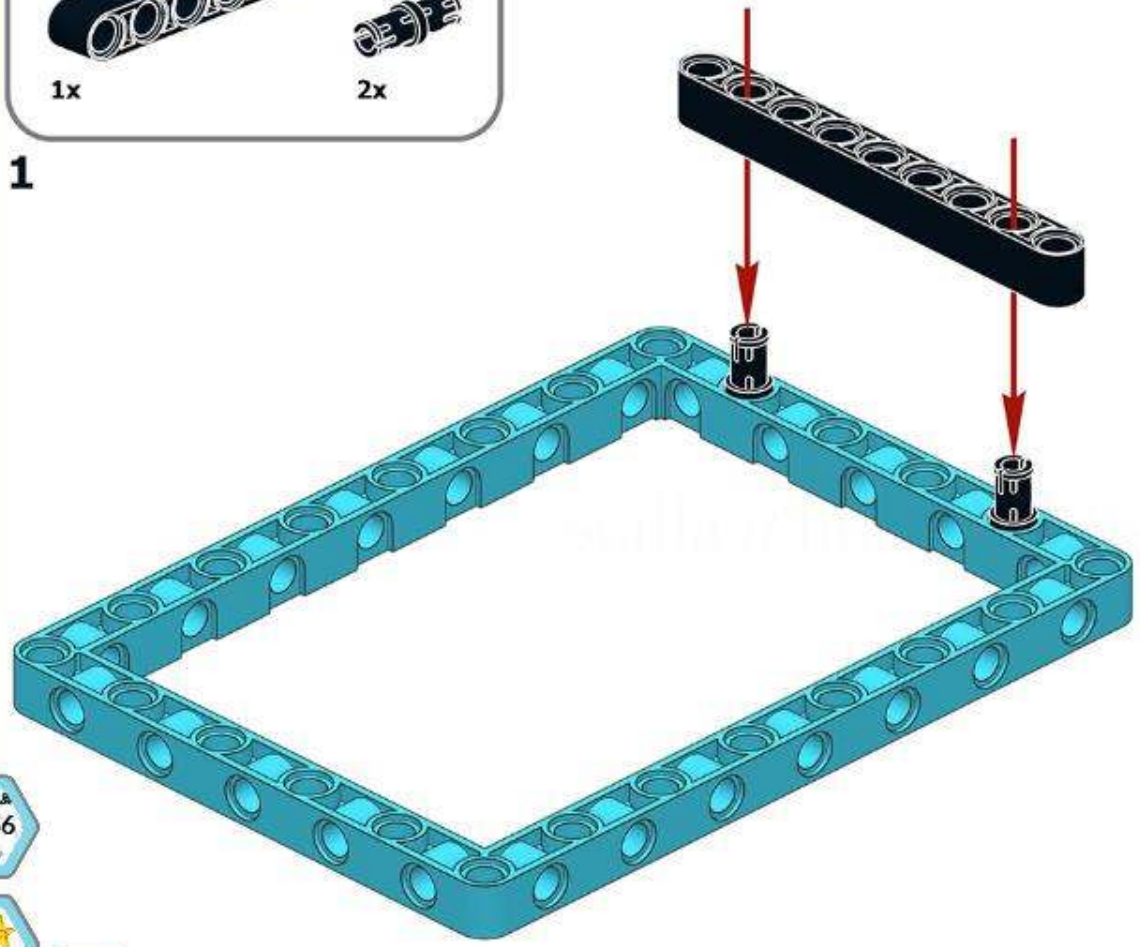
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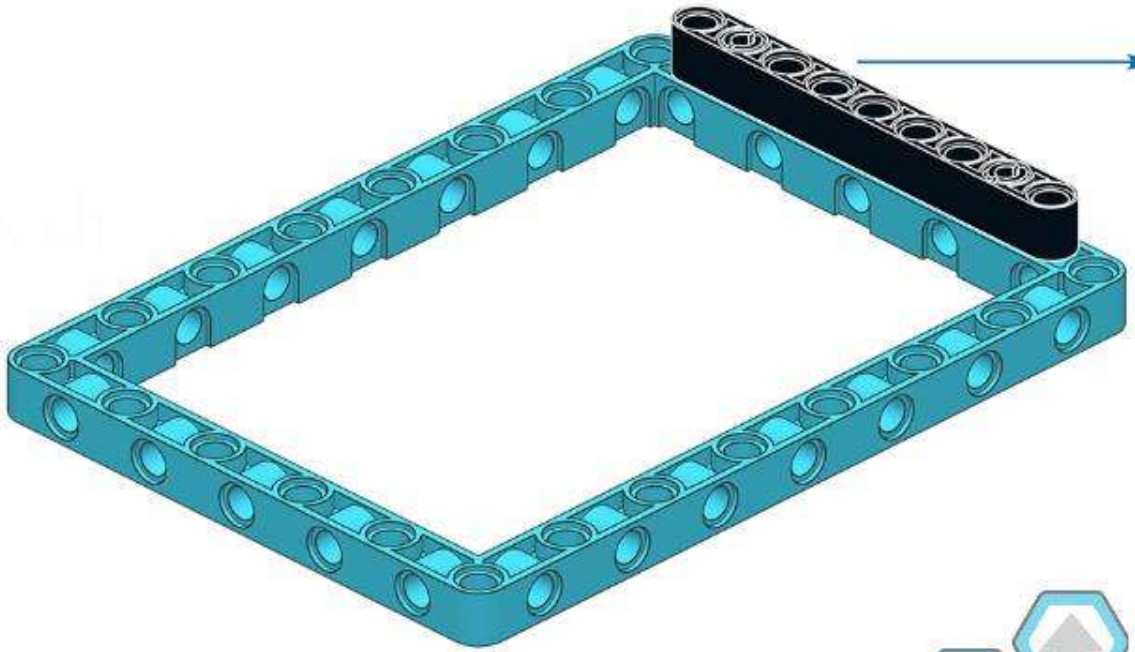
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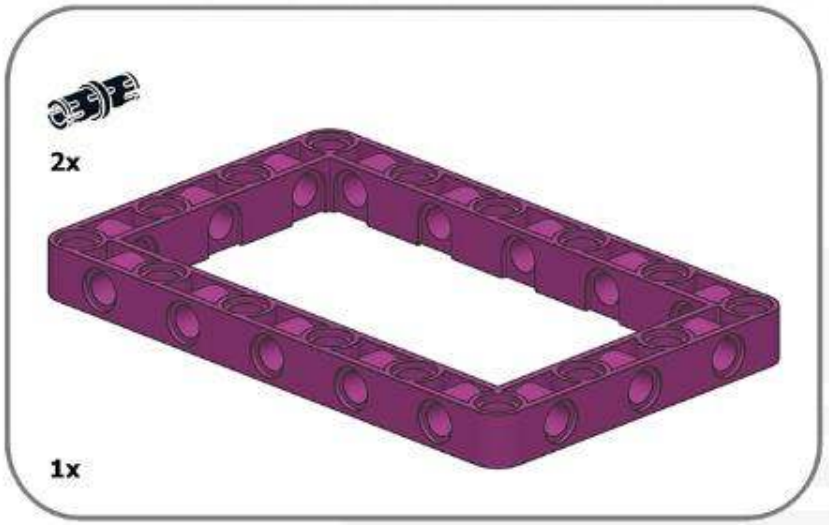


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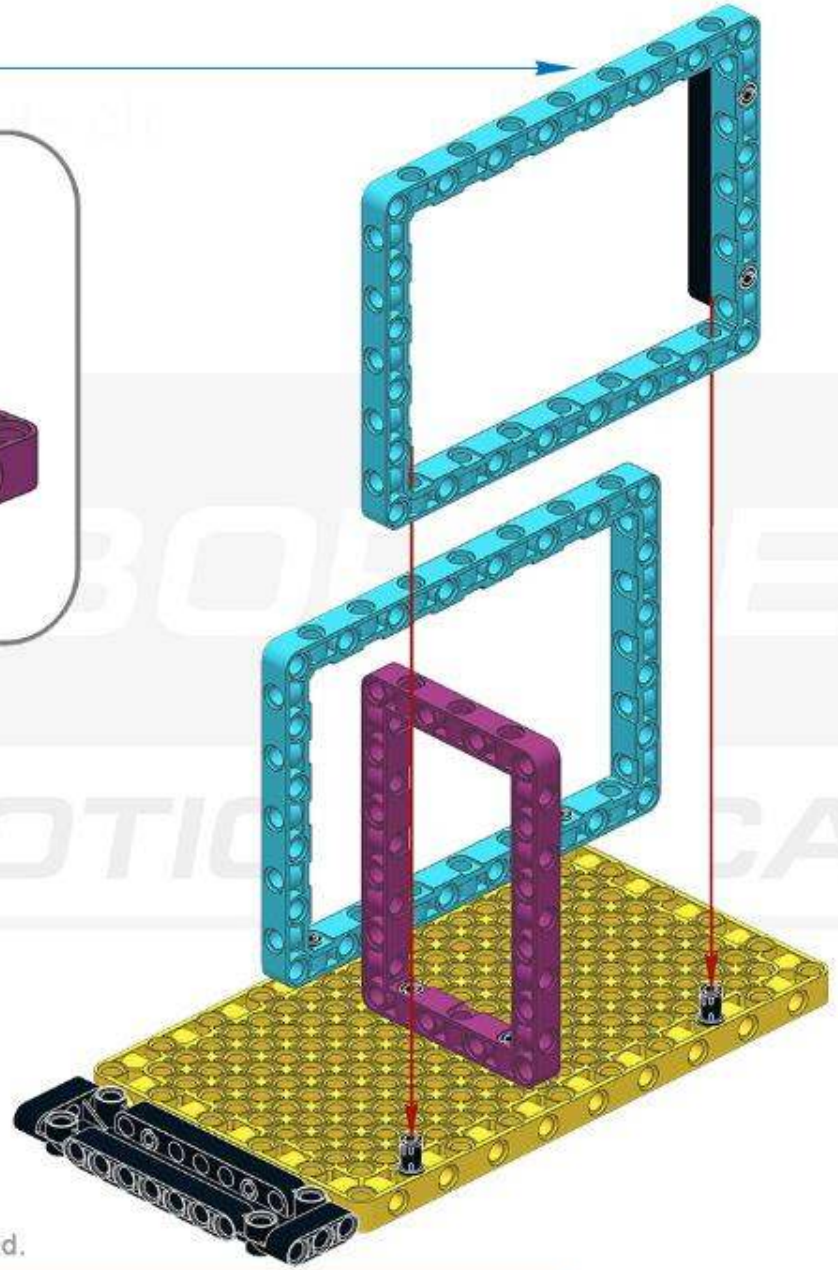
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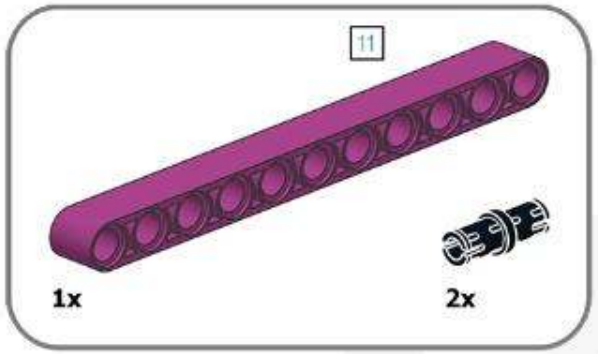
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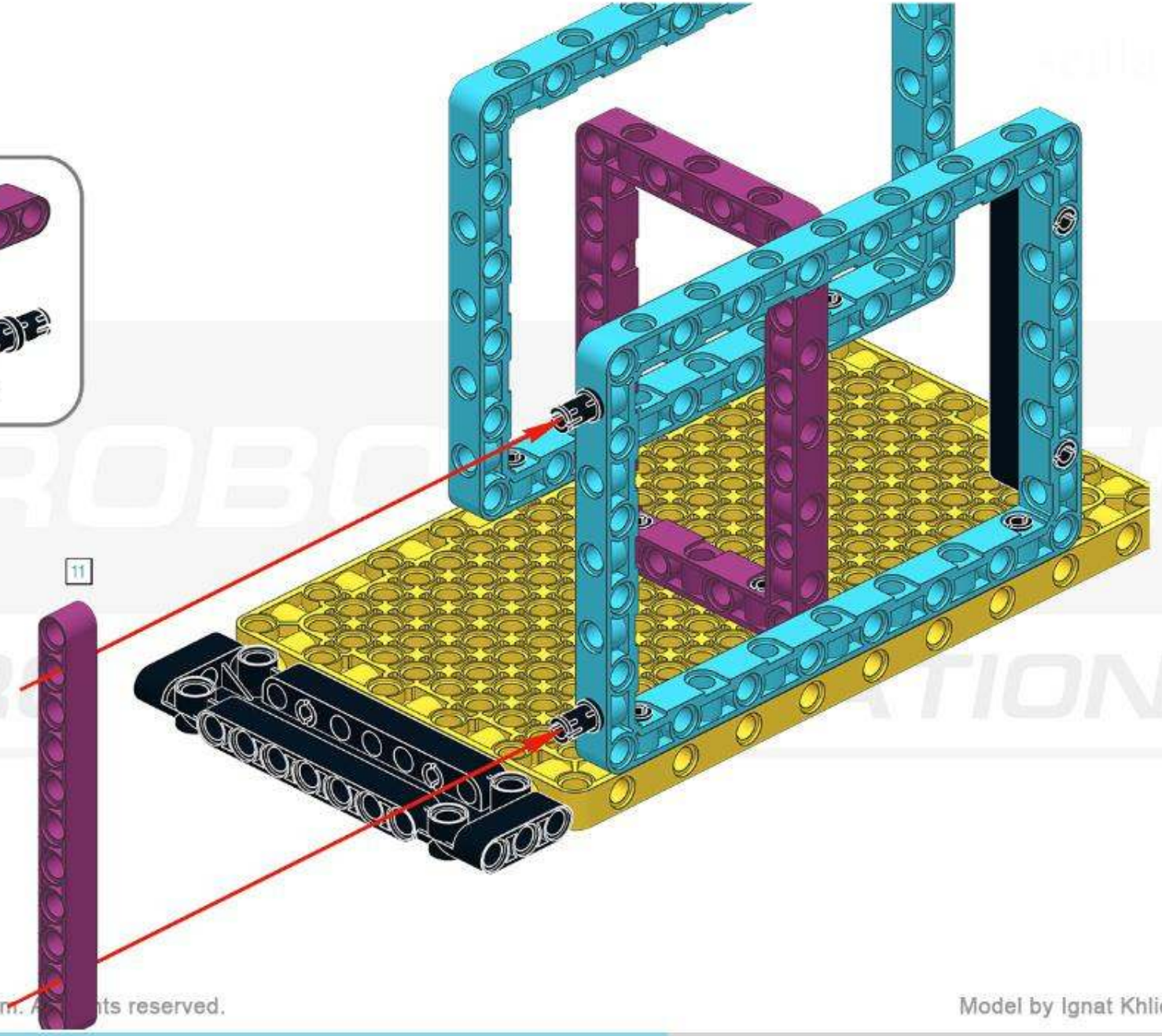
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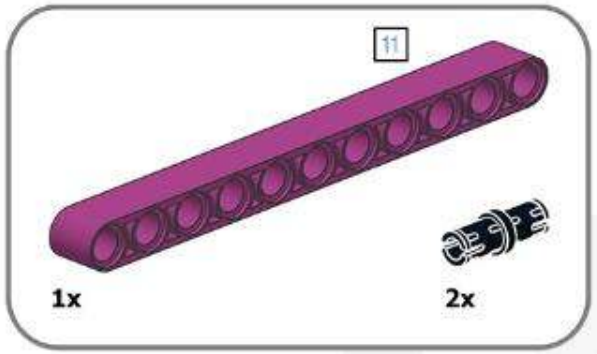
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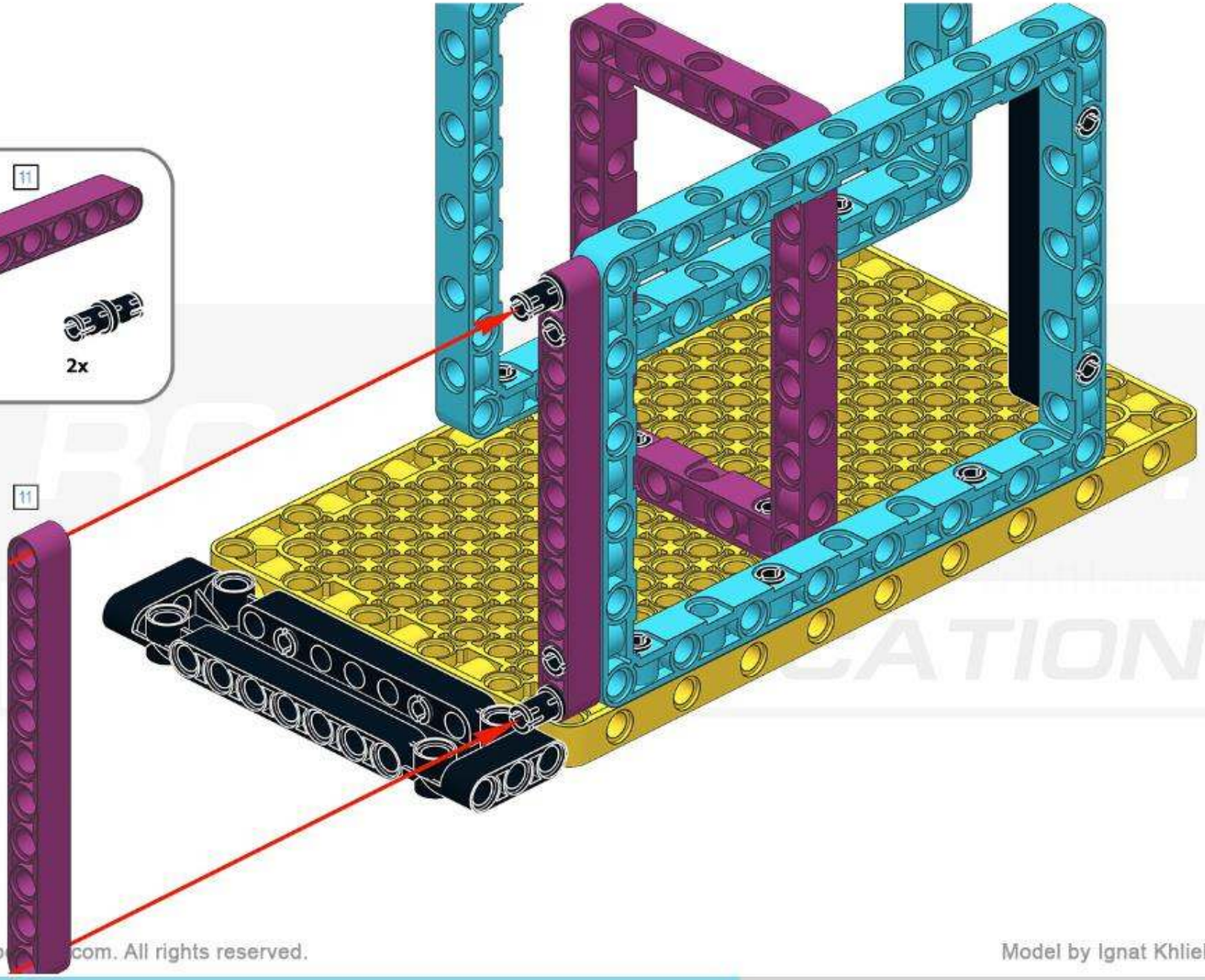
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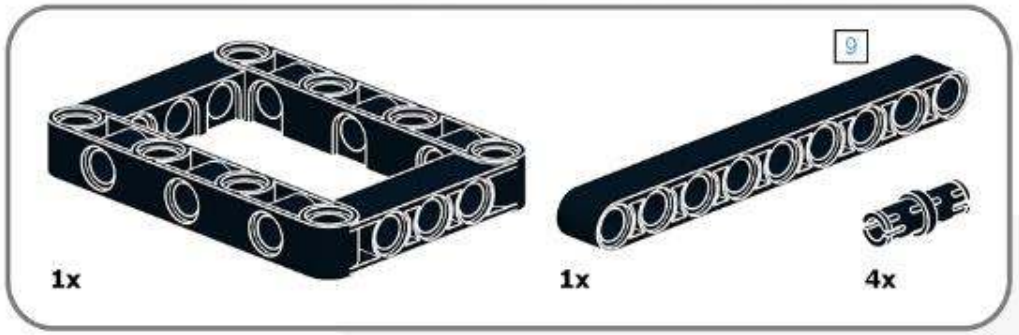
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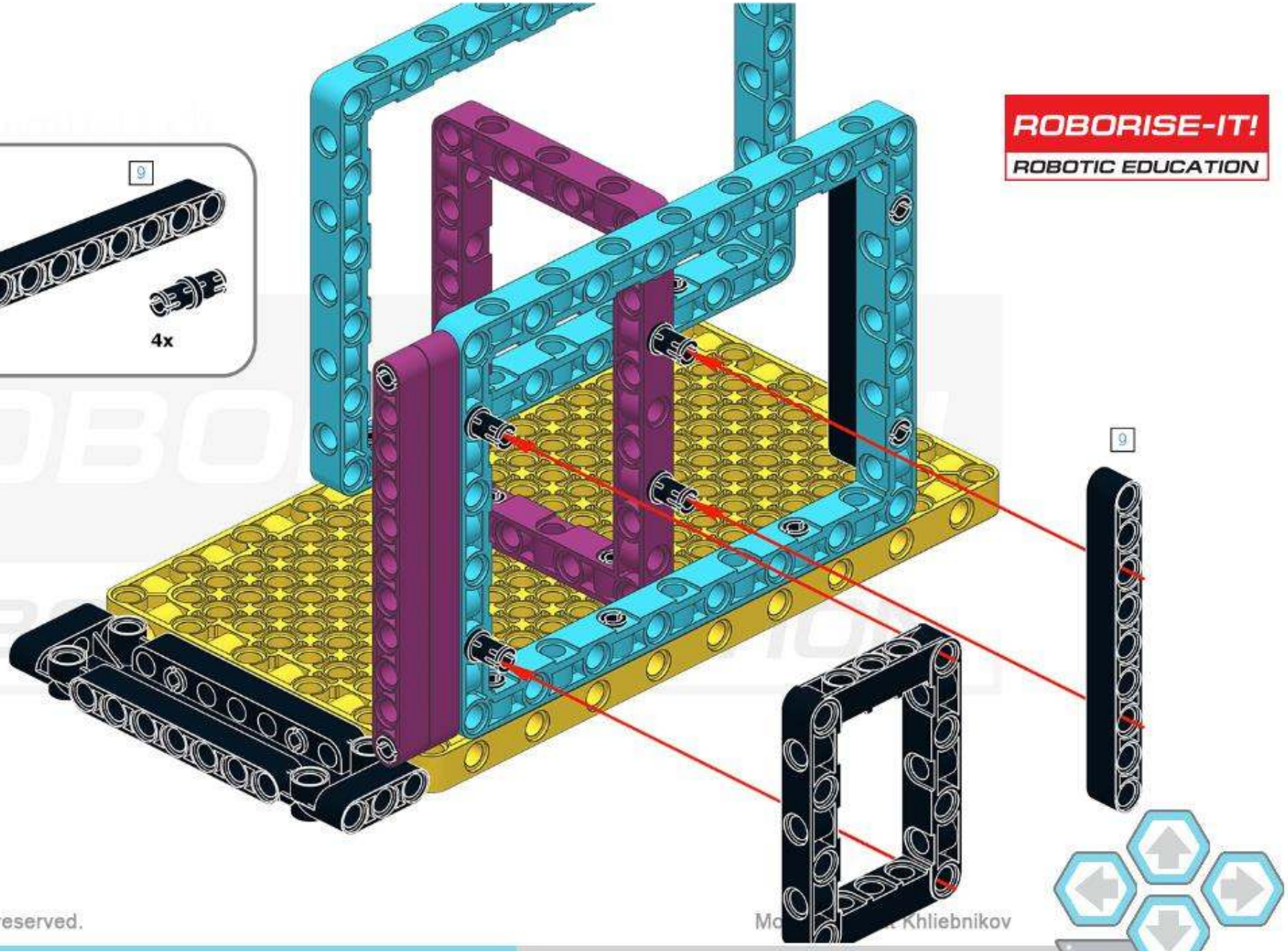
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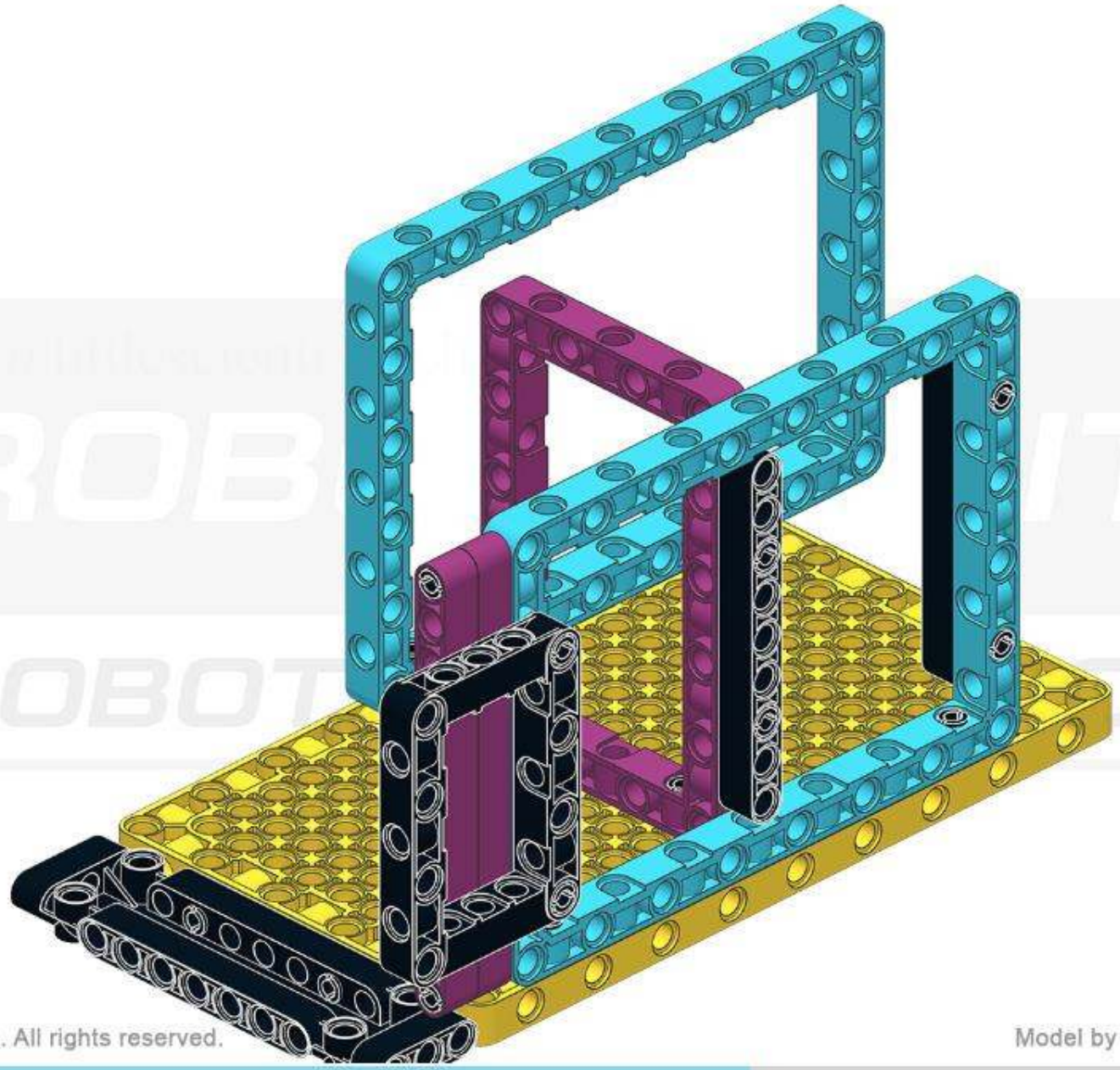
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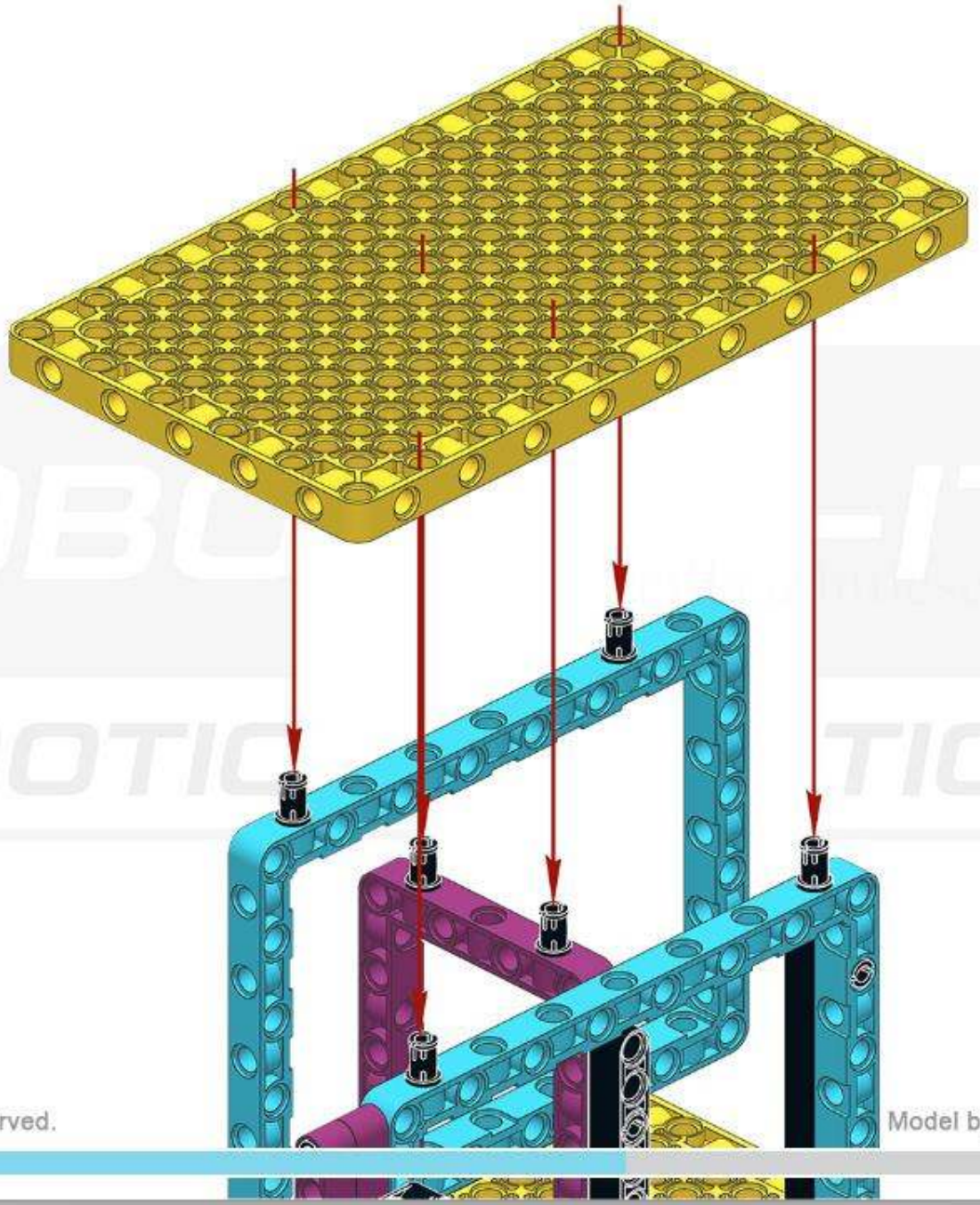
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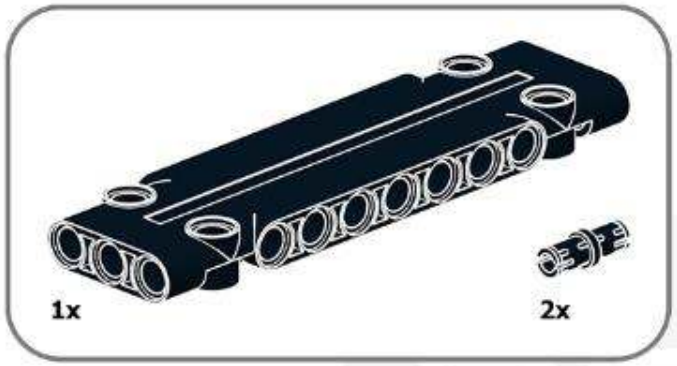
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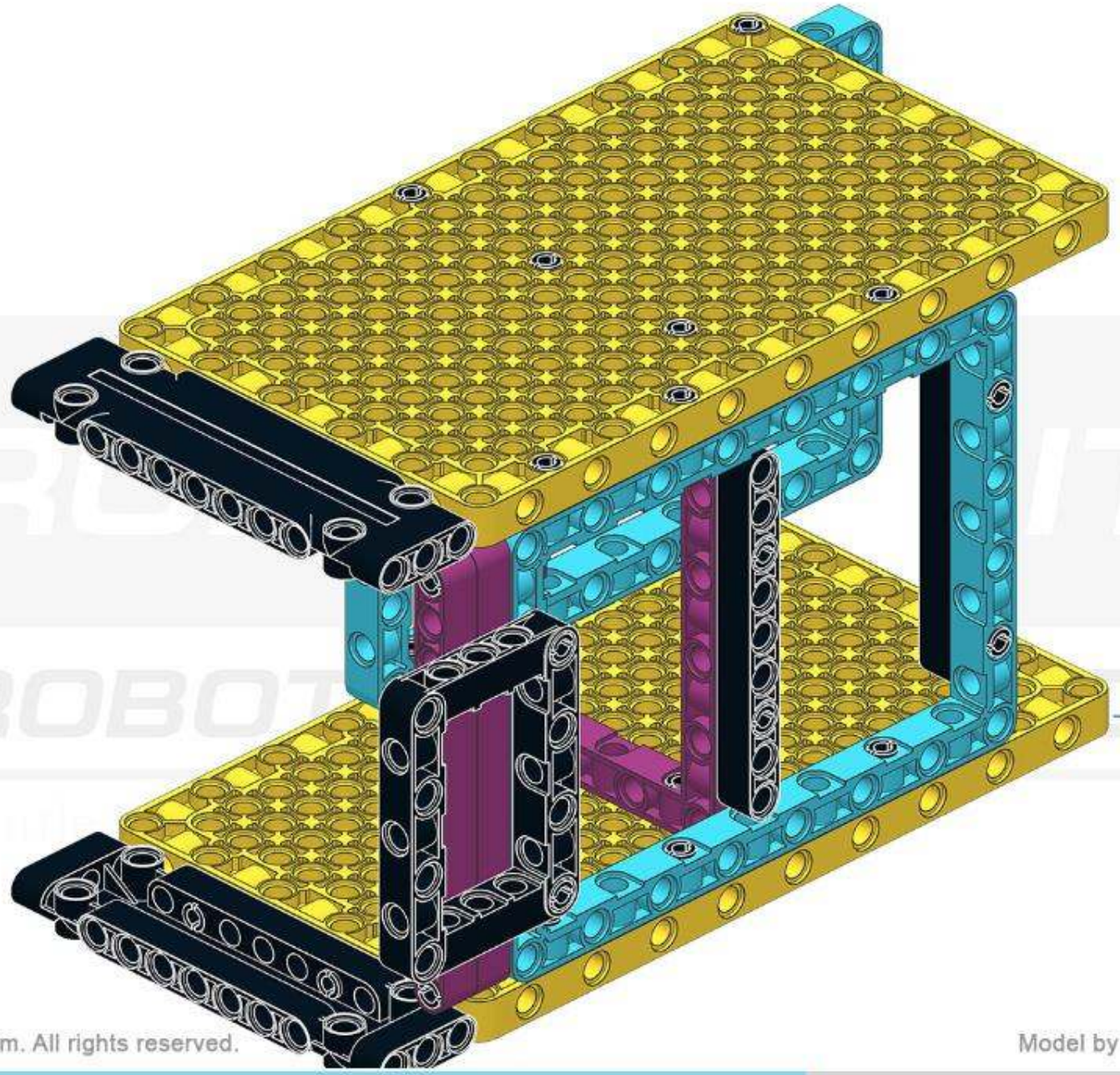
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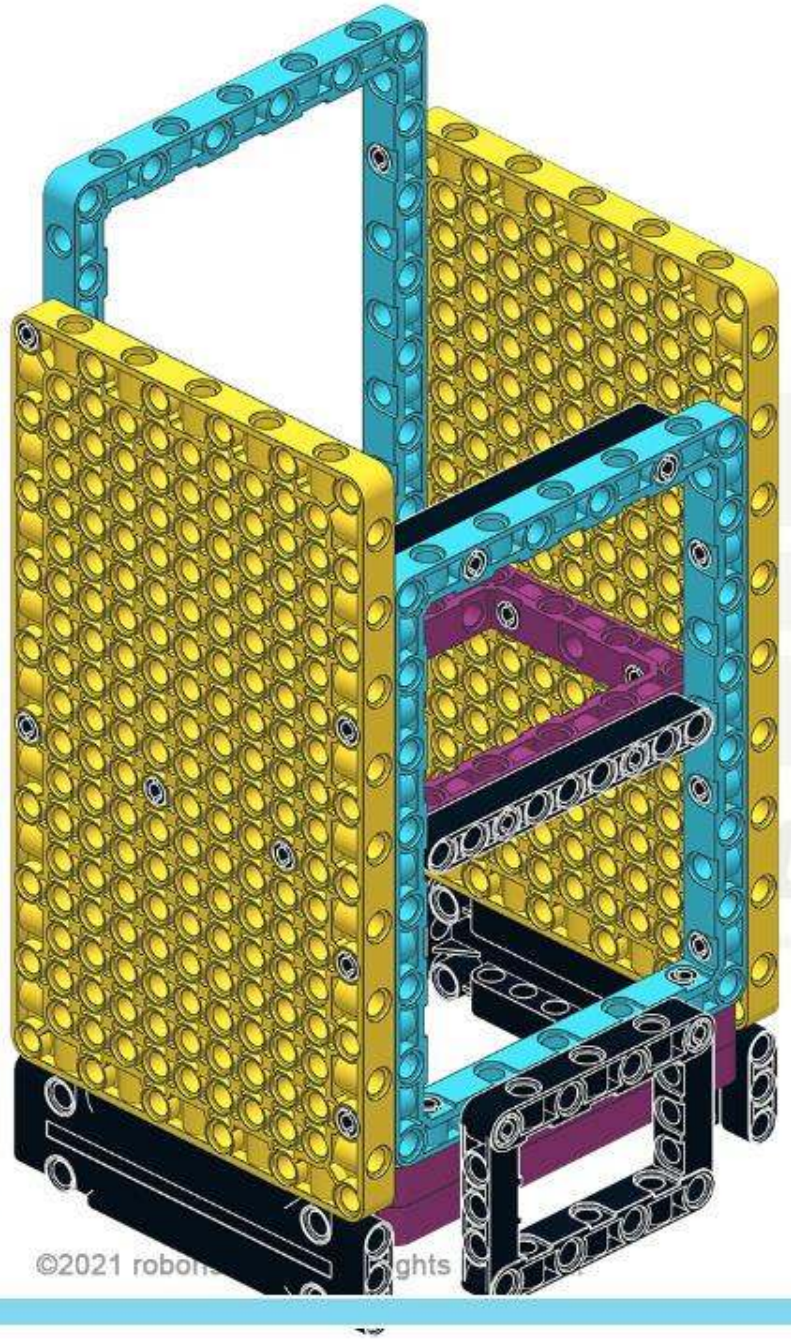
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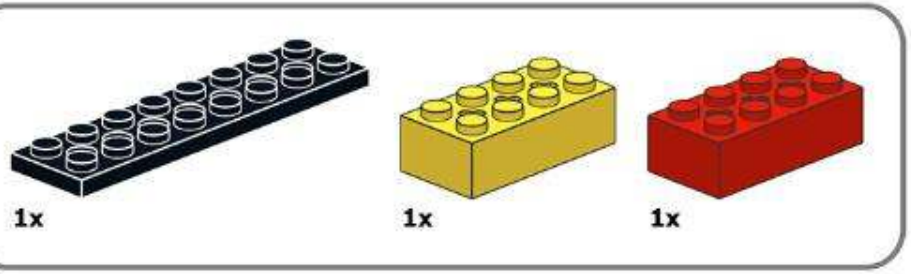
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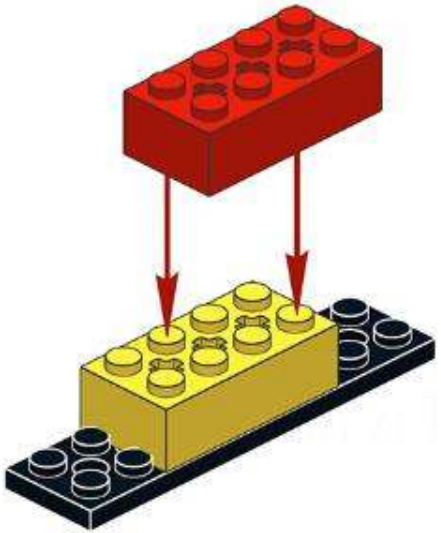
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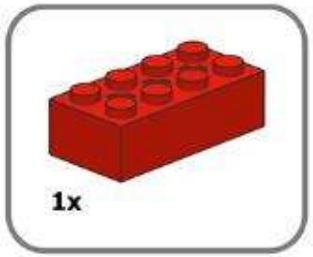
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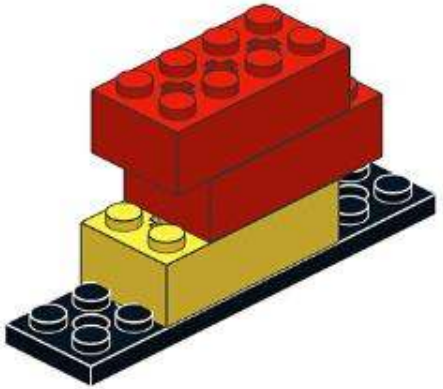


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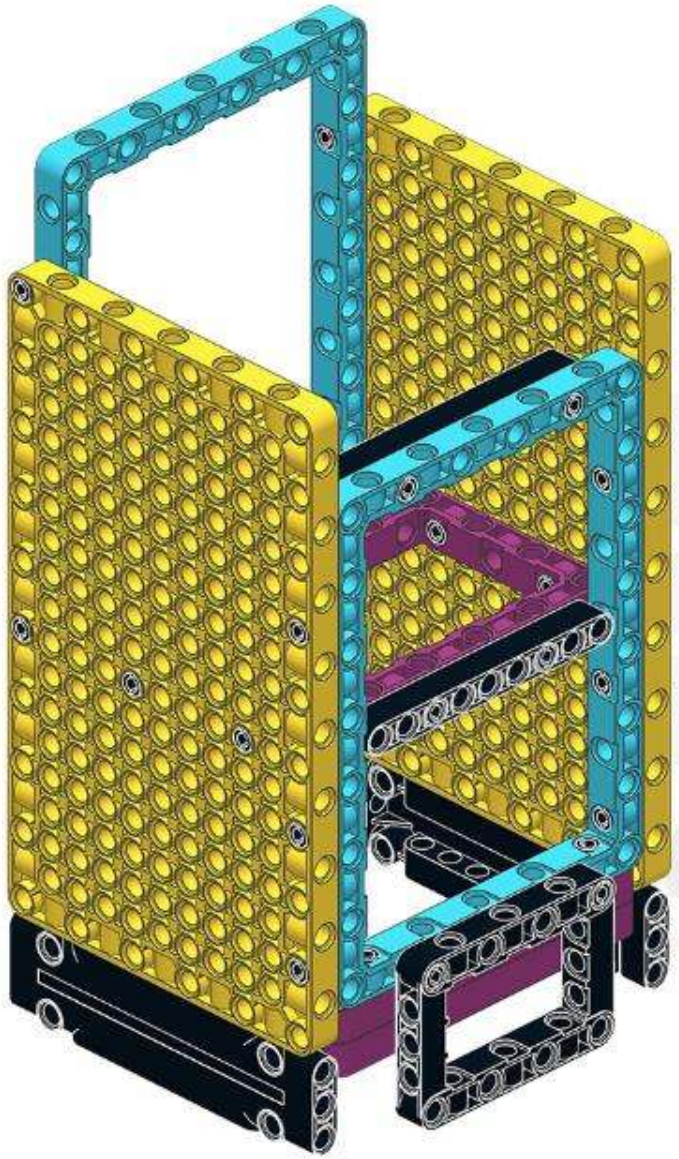
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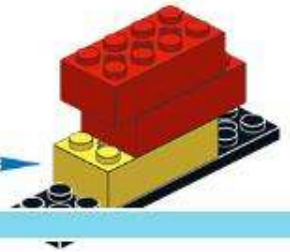
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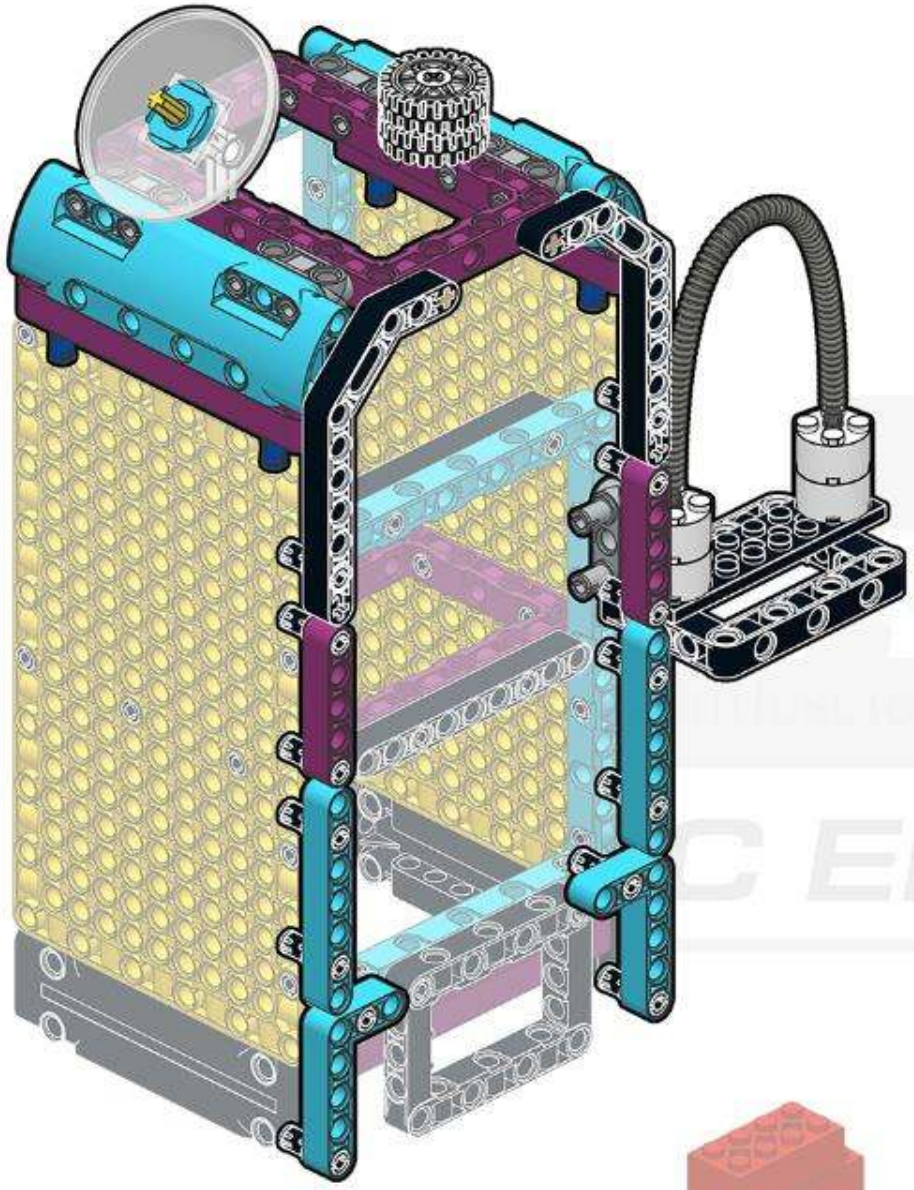
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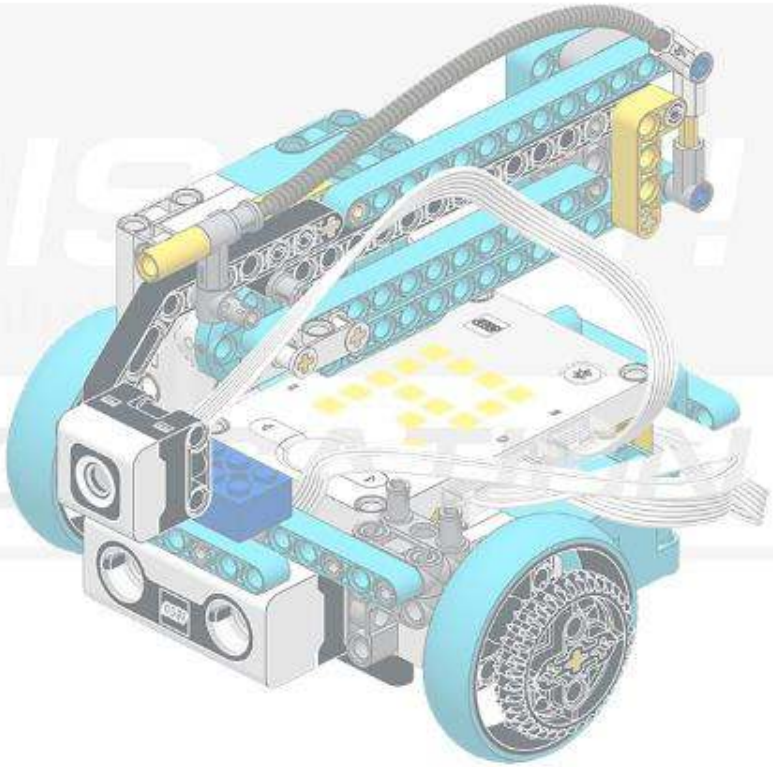
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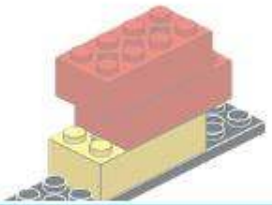
EXTENDED VERSION



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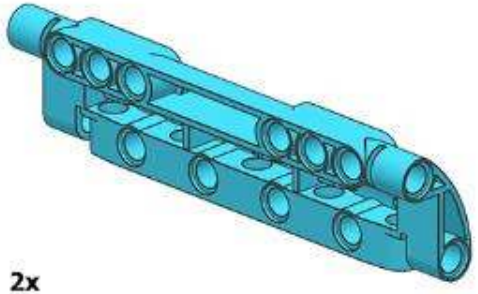
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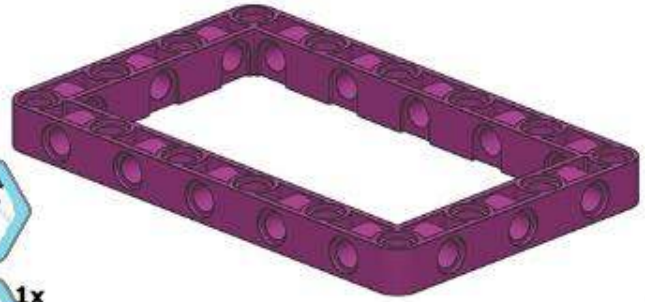
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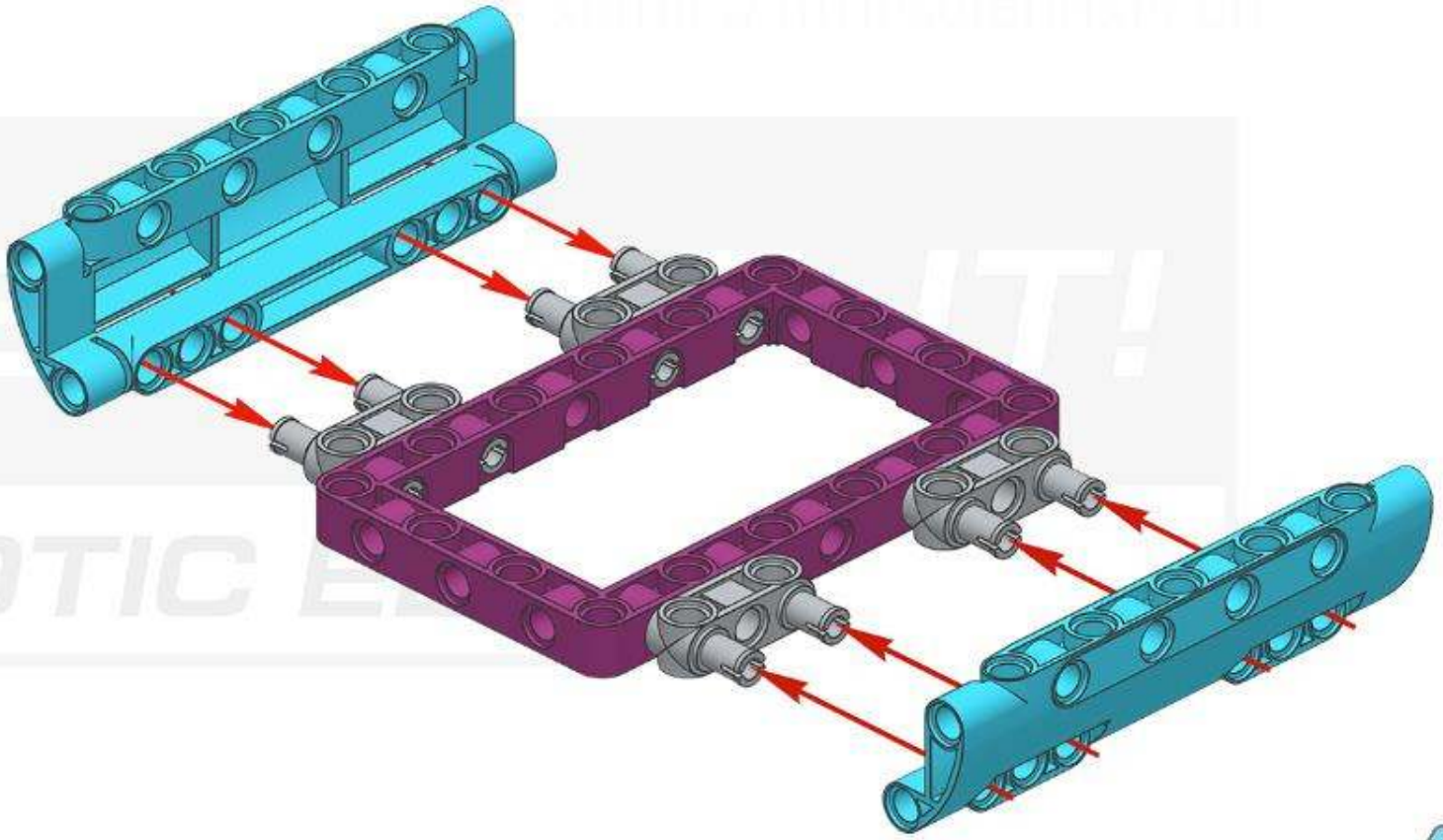
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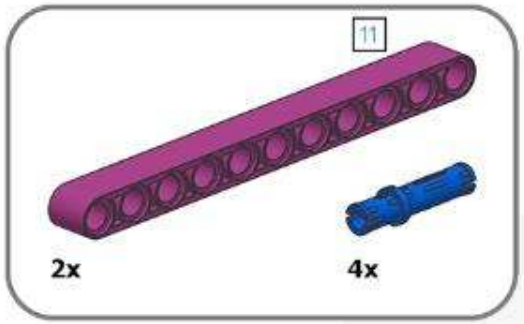
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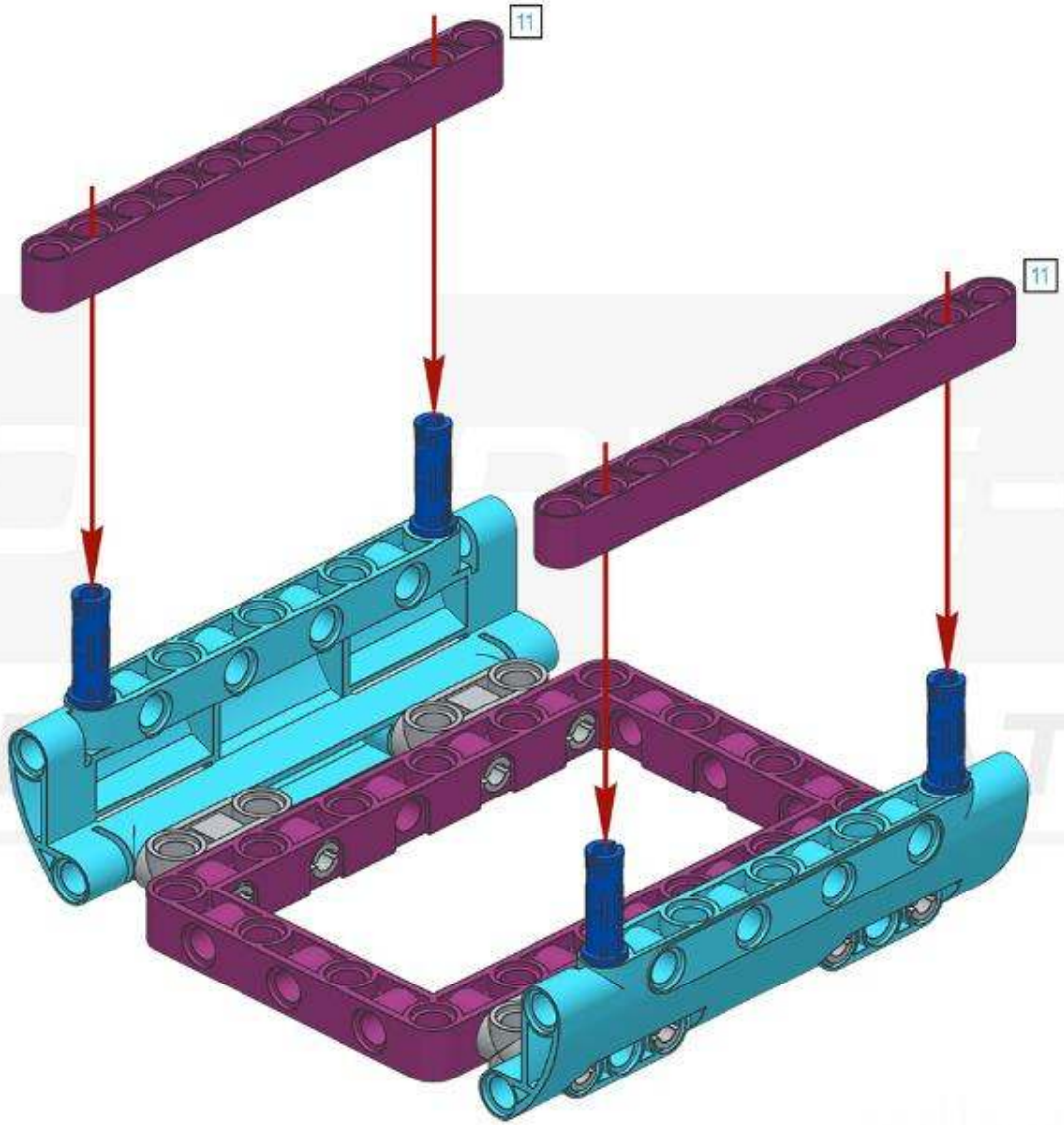
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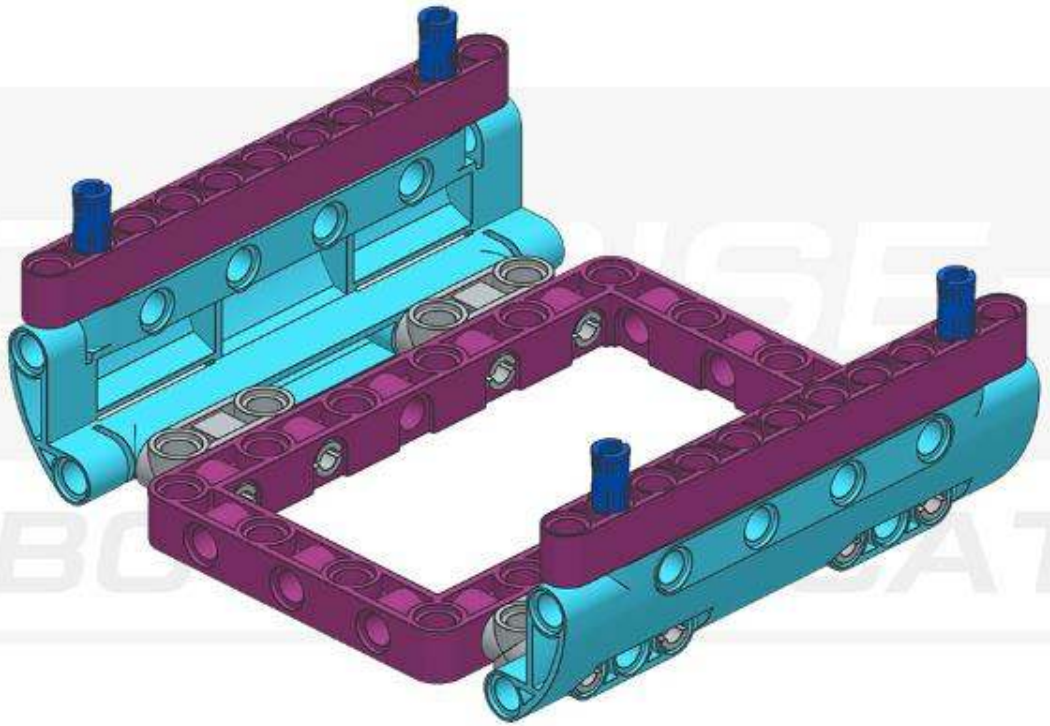
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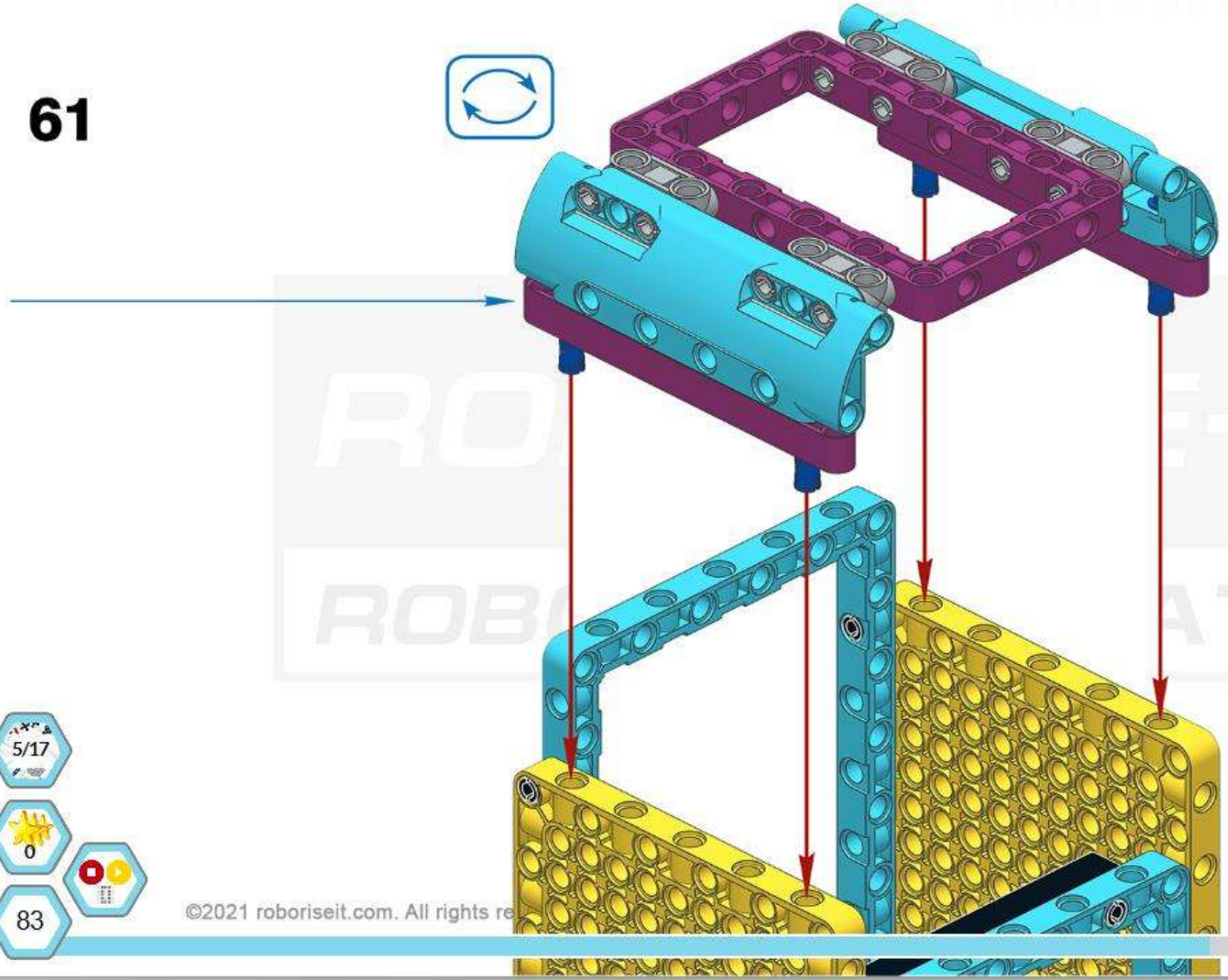
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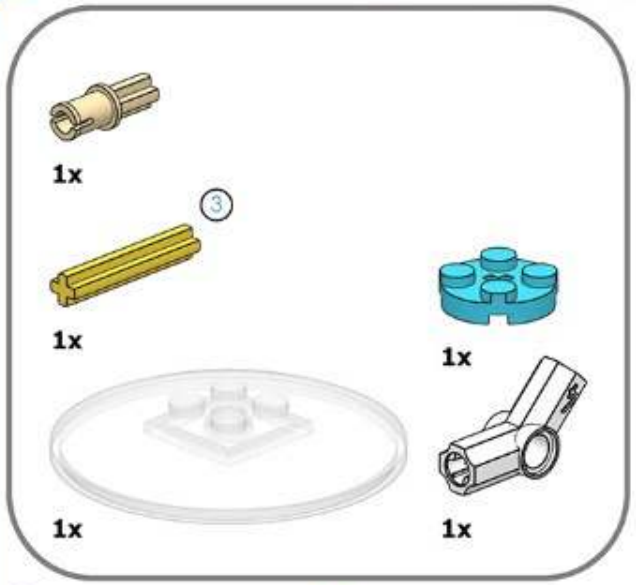
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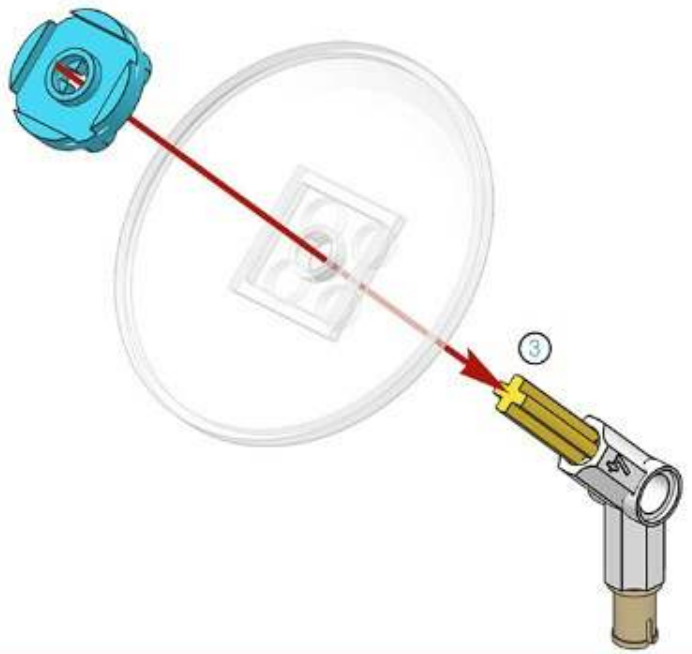
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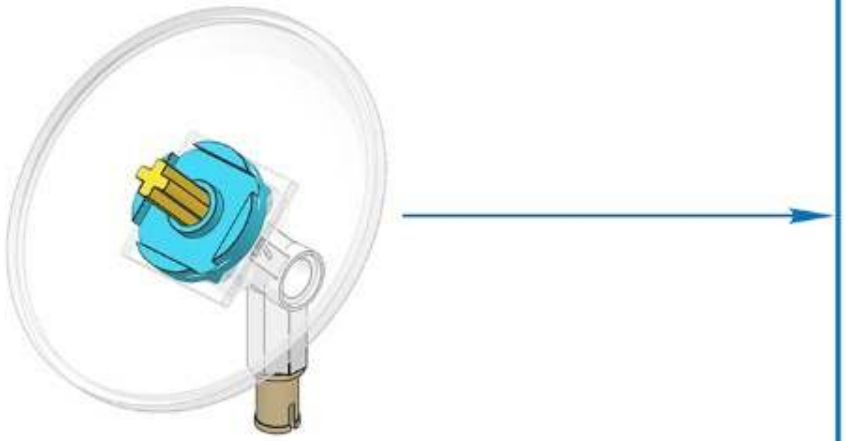


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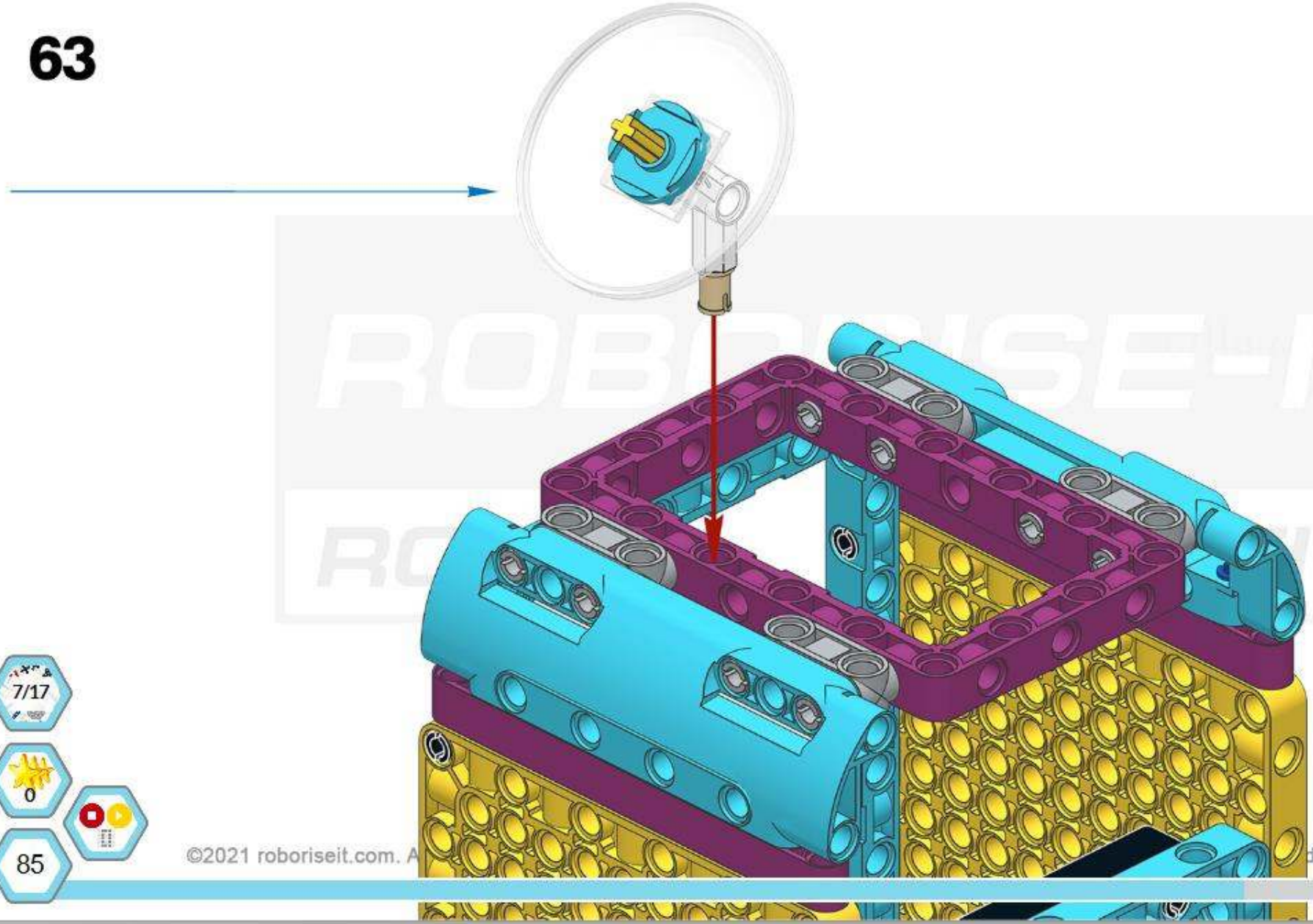


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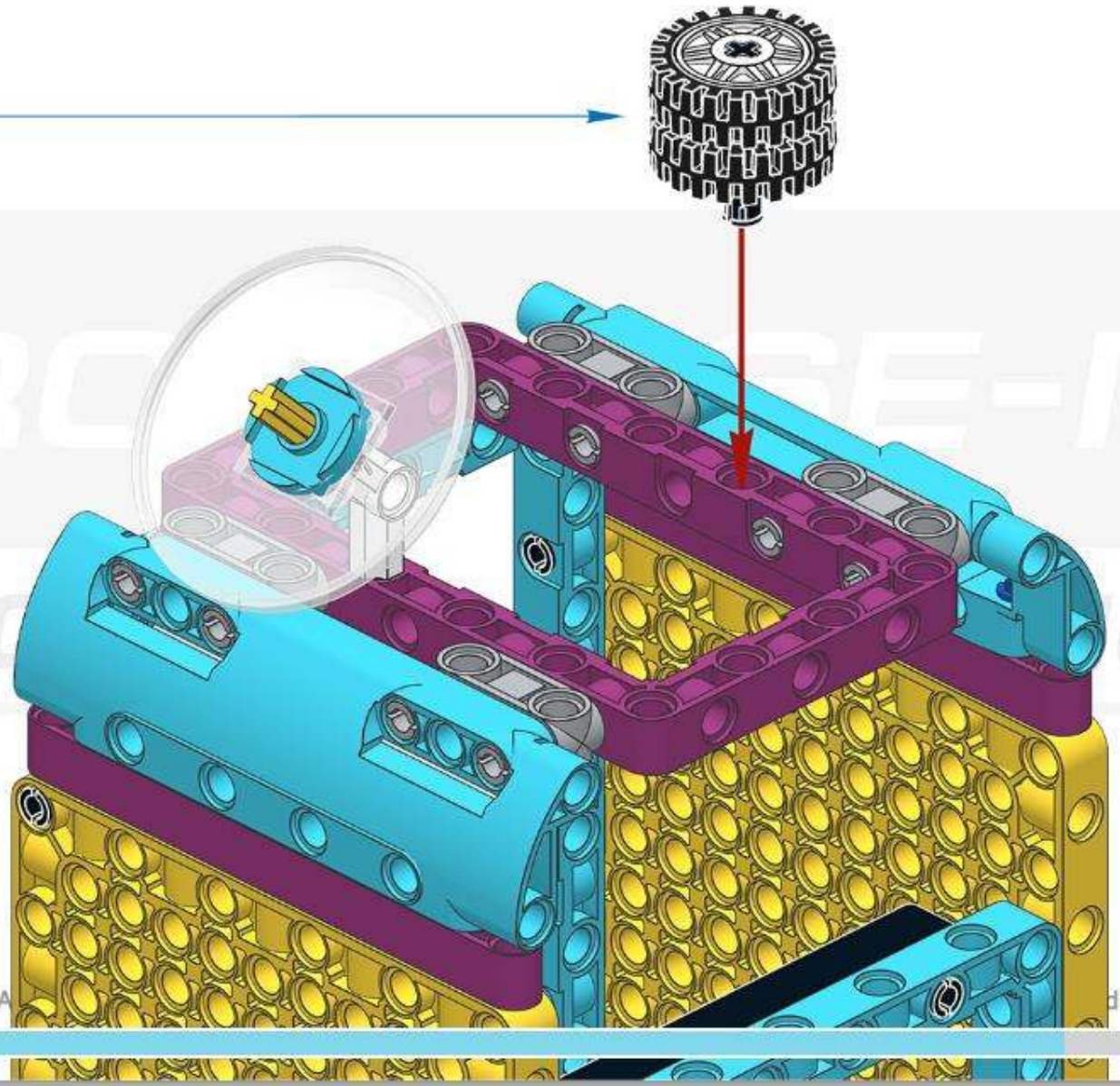
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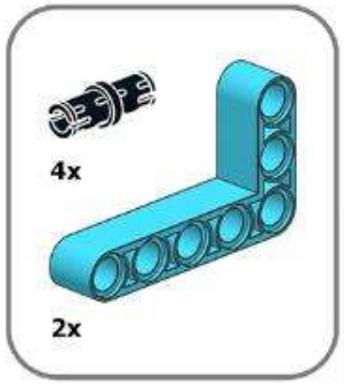
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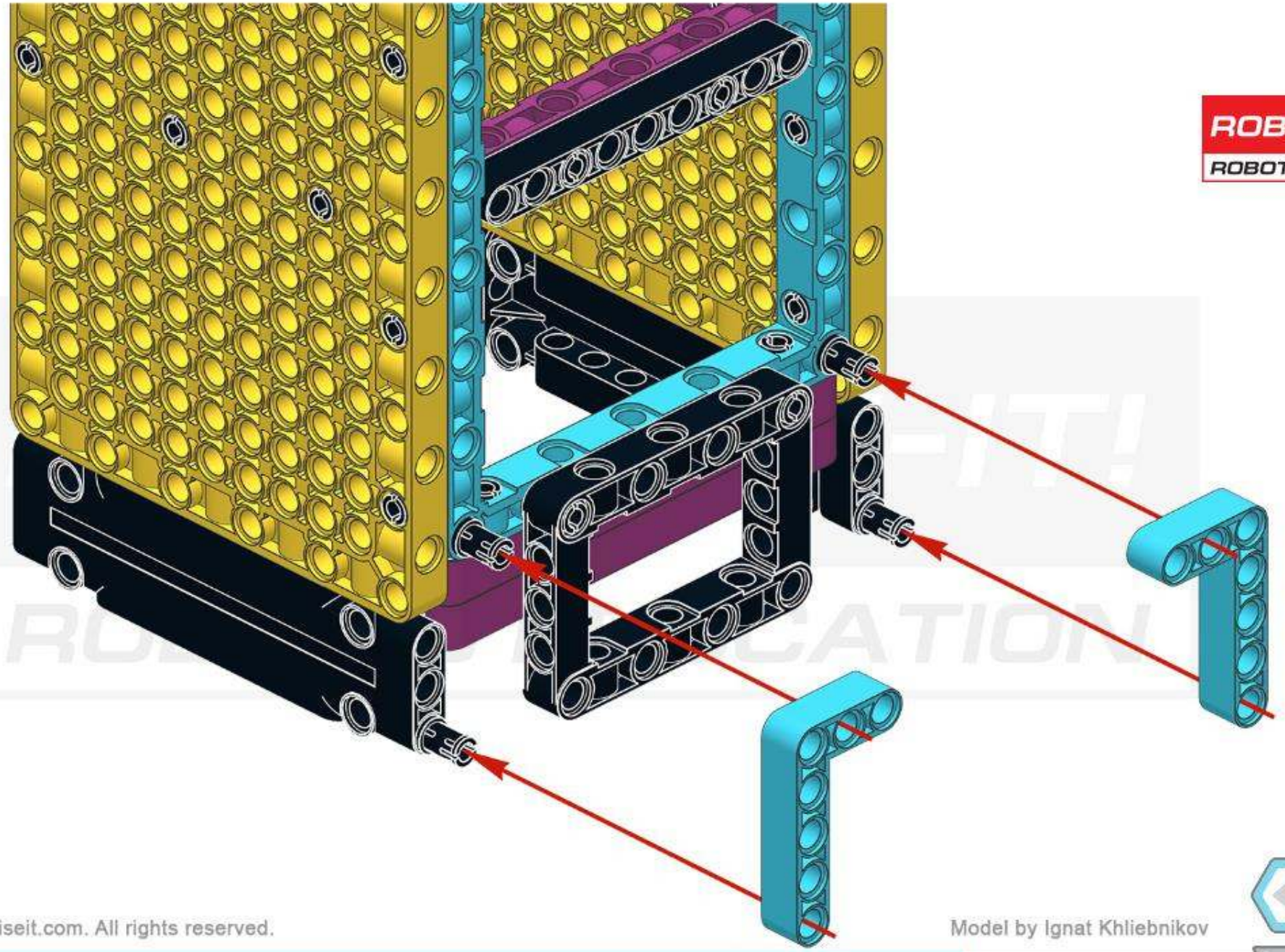
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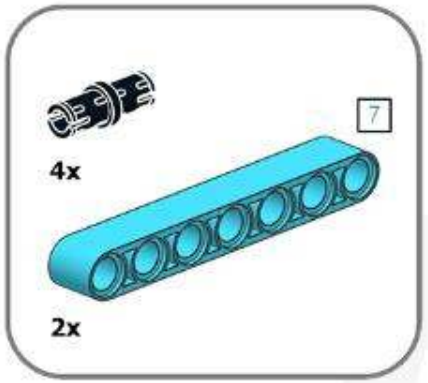
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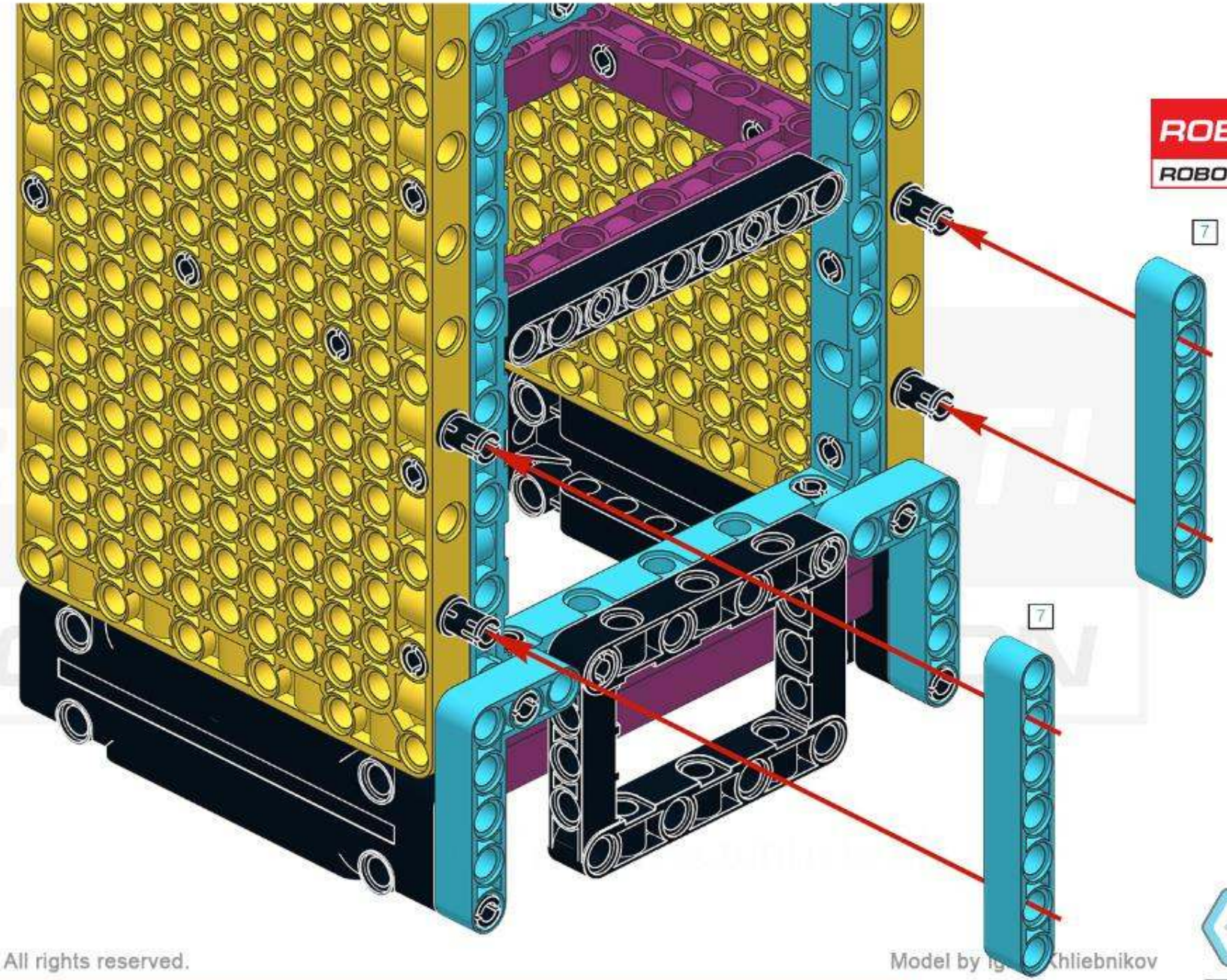
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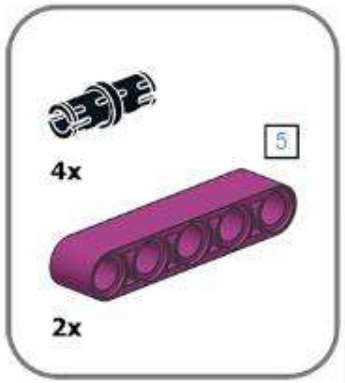
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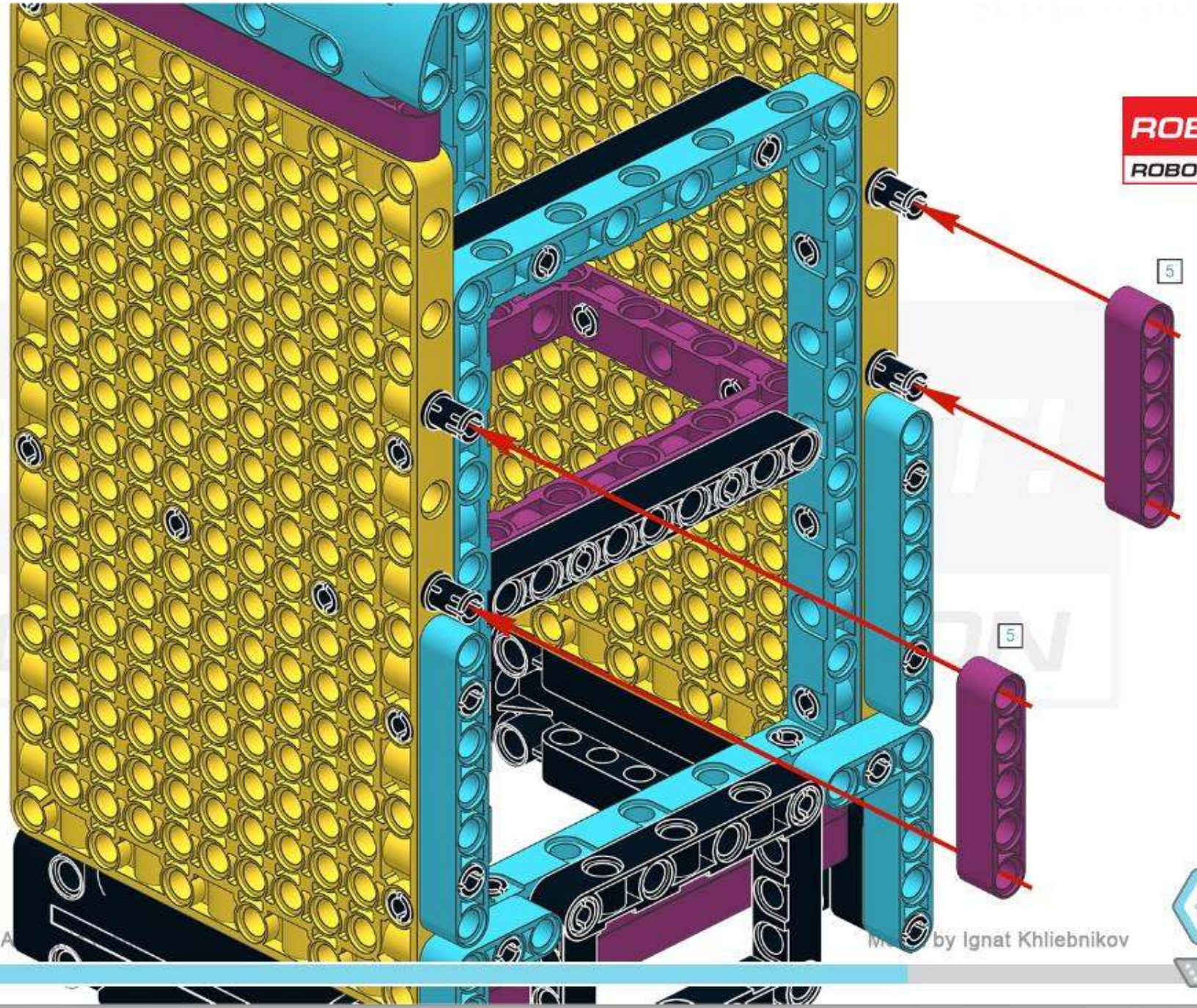
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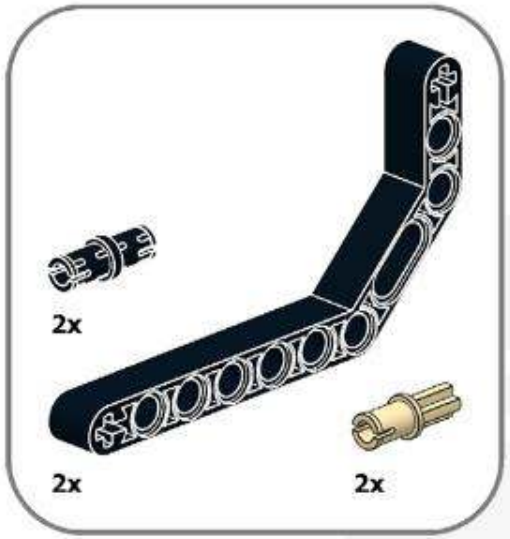
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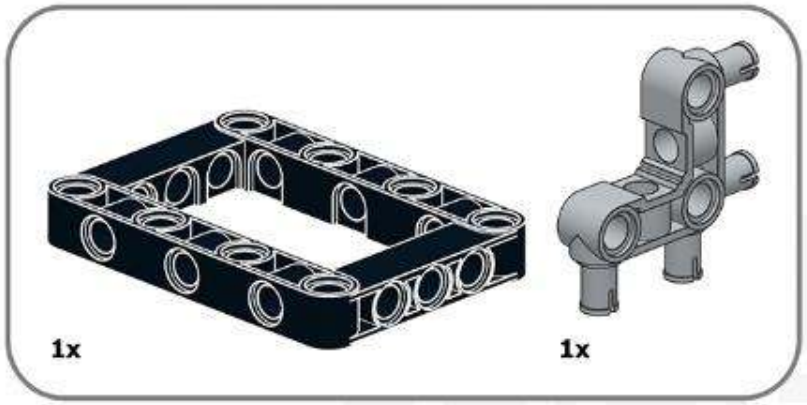
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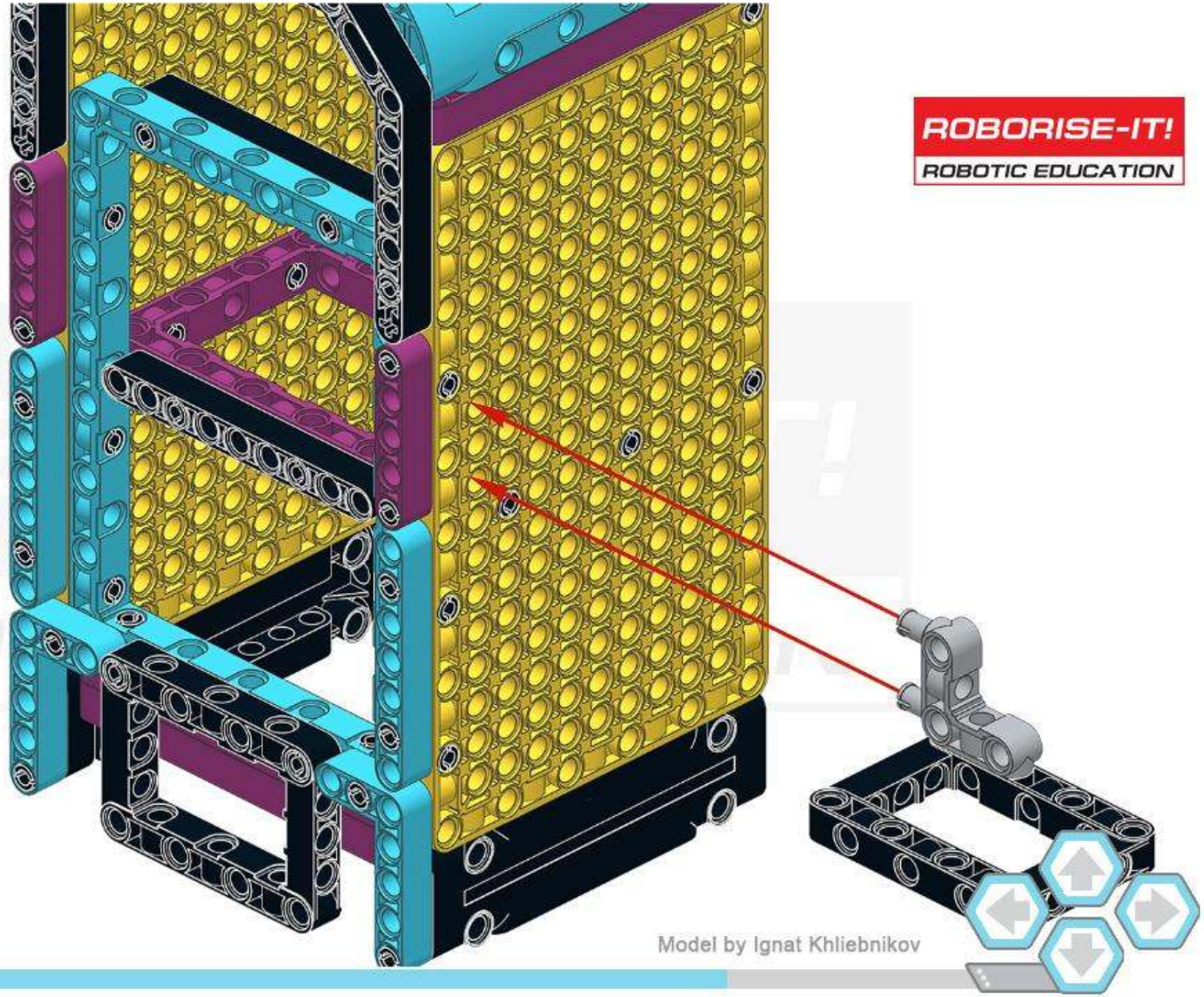
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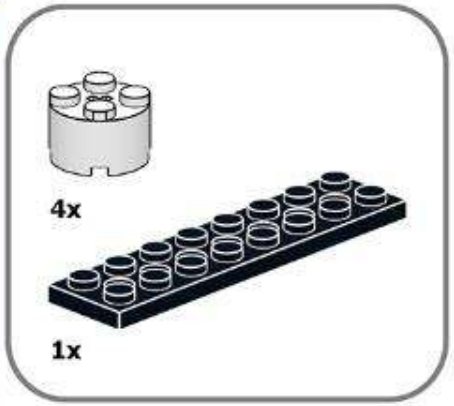
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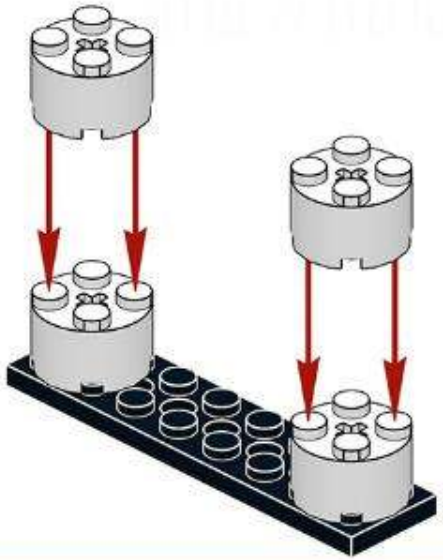
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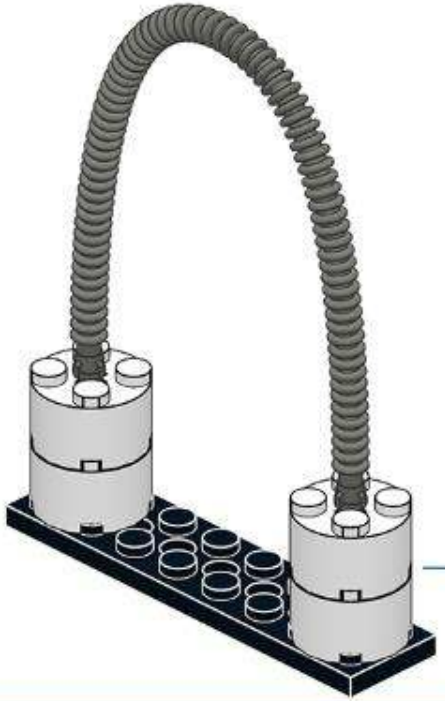


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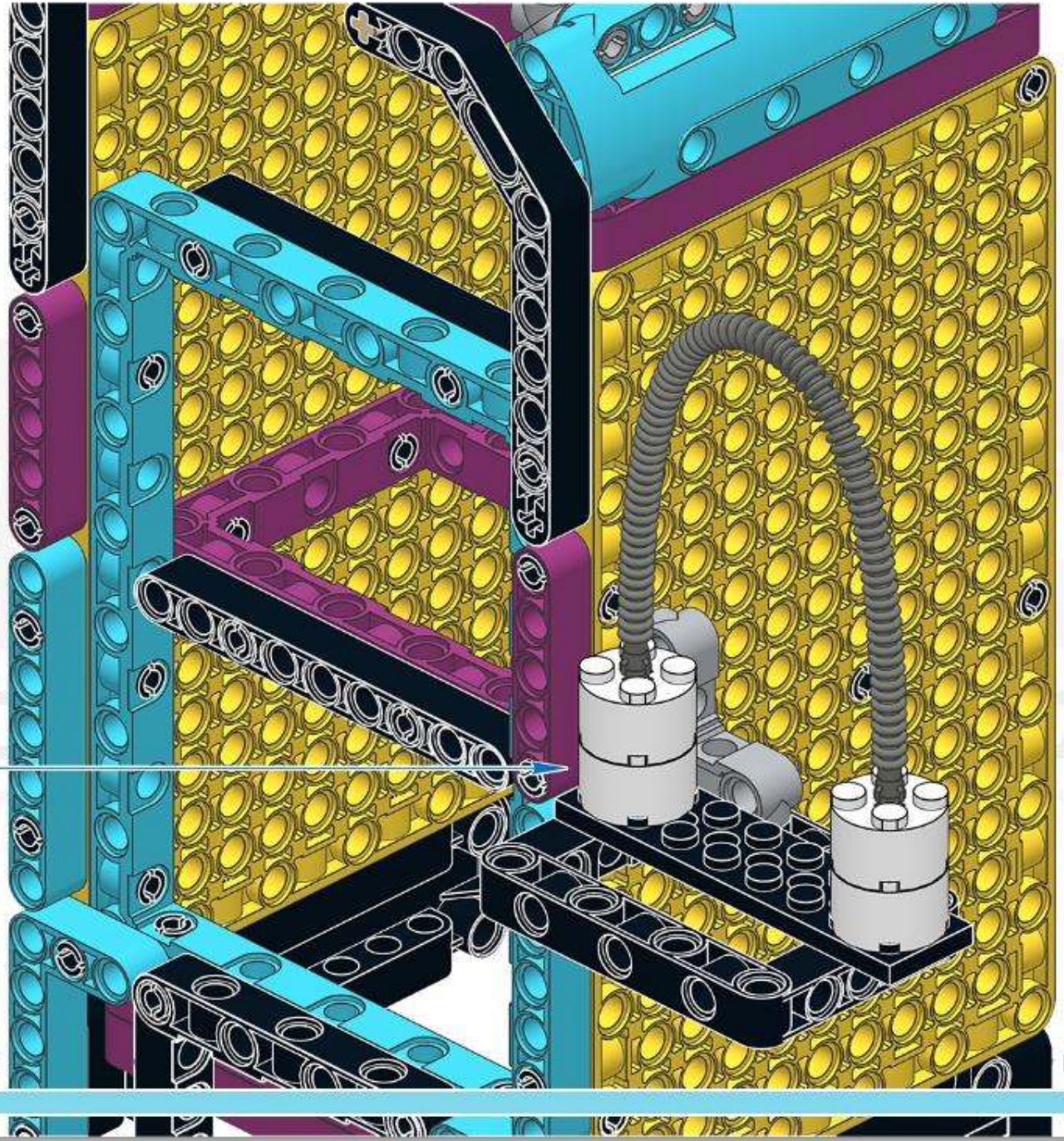
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ATTENTION

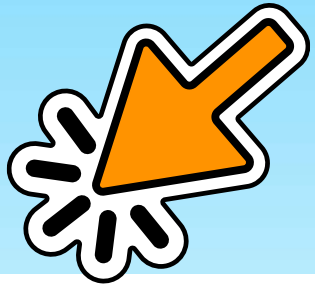
BEFORE STARTING

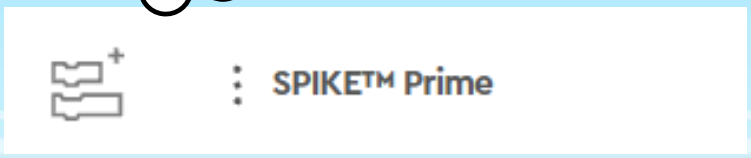


Um die Nullposition einzustellen, richte den Punkt auf dem Rotor auf den Kreis am Motorgehäuse aus.

To set the zero position, line up the dot on the rotor with the circle on the motor frame.

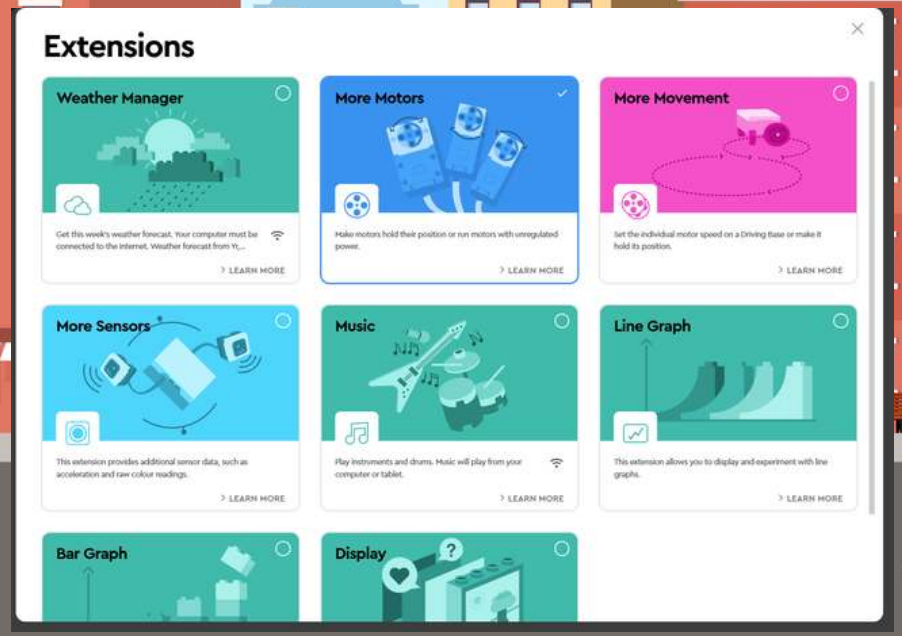
IMPORTANT





Wenn du den in dieser Anleitung verwendeten Block nicht finden kannst, klicke auf das Symbol in der unteren linken Ecke (siehe Abbildung links). Daraufhin öffnet sich ein Fenster, in dem du die für dieses Projekt benötigten Erweiterungsblöcke finden und hinzufügen kannst.

If you cannot find the specific block used in these instructions, click the icon in the bottom-left corner (shown in the figure on the left). This will open a window where you can find and add the extension blocks needed for this project.





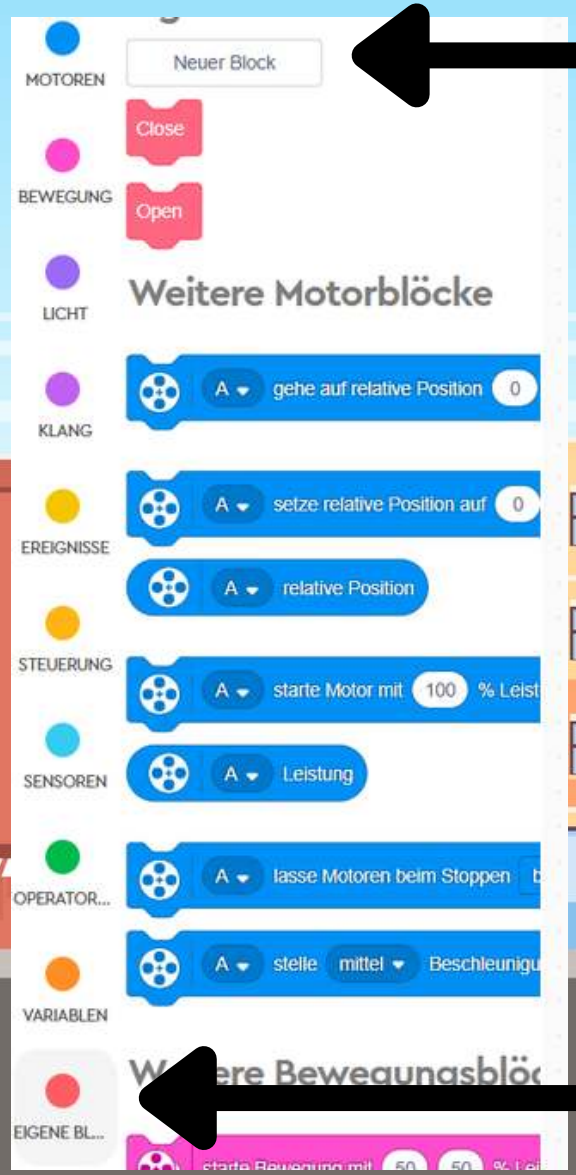
TASK 1



Programmiere dein Feuerwehrauto so, dass es selbstständig zu dem Gebäude fährt, an dem der Notfall vorliegt

In diesem Schritt teilen wir dem Computer mit, welche Motoren er verwenden soll (Anschlüsse C und D), und legen die Reifenabmessungen fest, was für genaue Entfernungsmessungen erforderlich ist. Außerdem legen wir die Geschwindigkeit und die Fahrtrichtung fest. Der Roboter fährt vorwärts, bis der Abstandssensor ein Objekt erkennt, das näher als 5 cm entfernt ist; dann stoppen die Motoren.

TASK 2

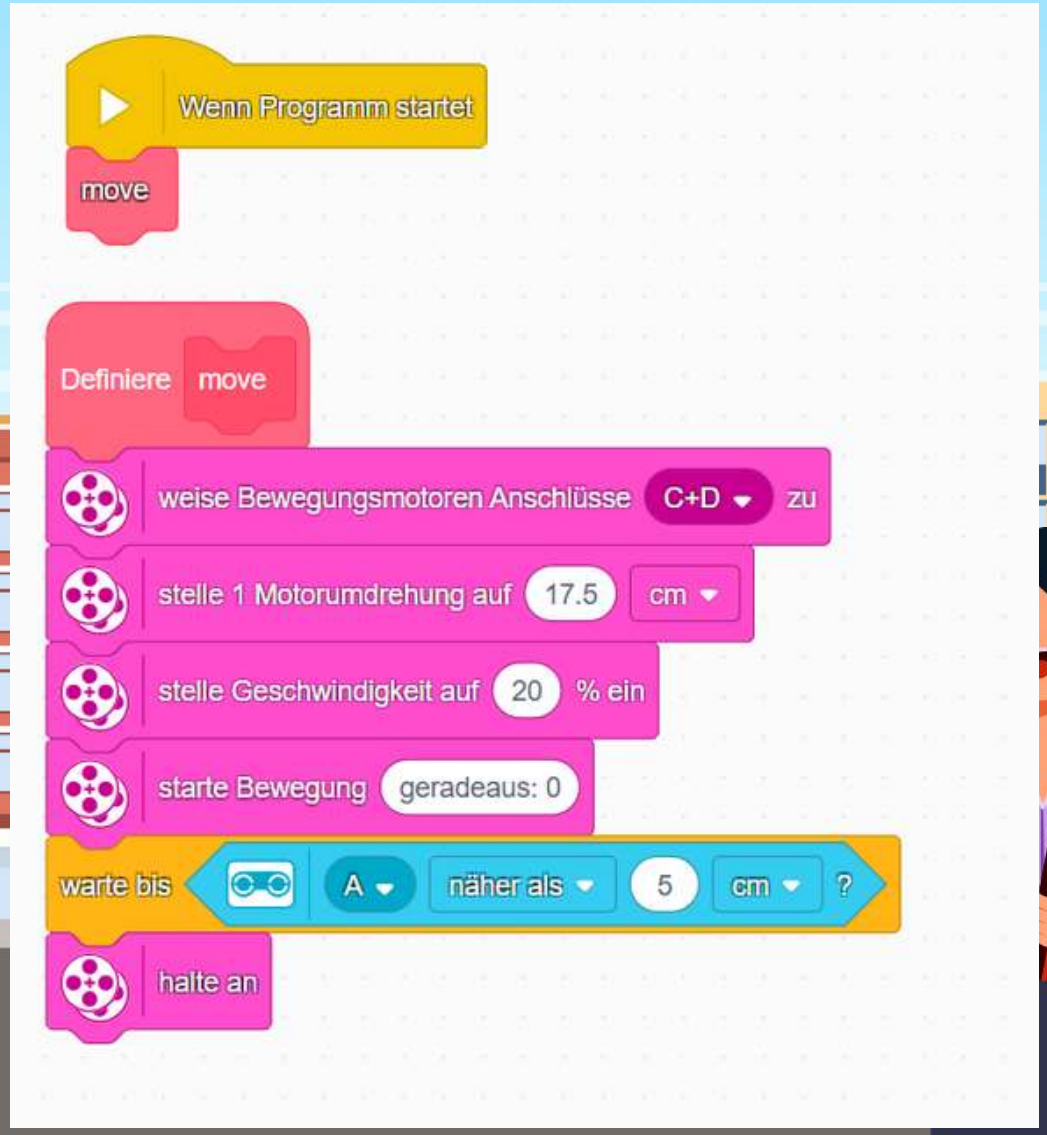


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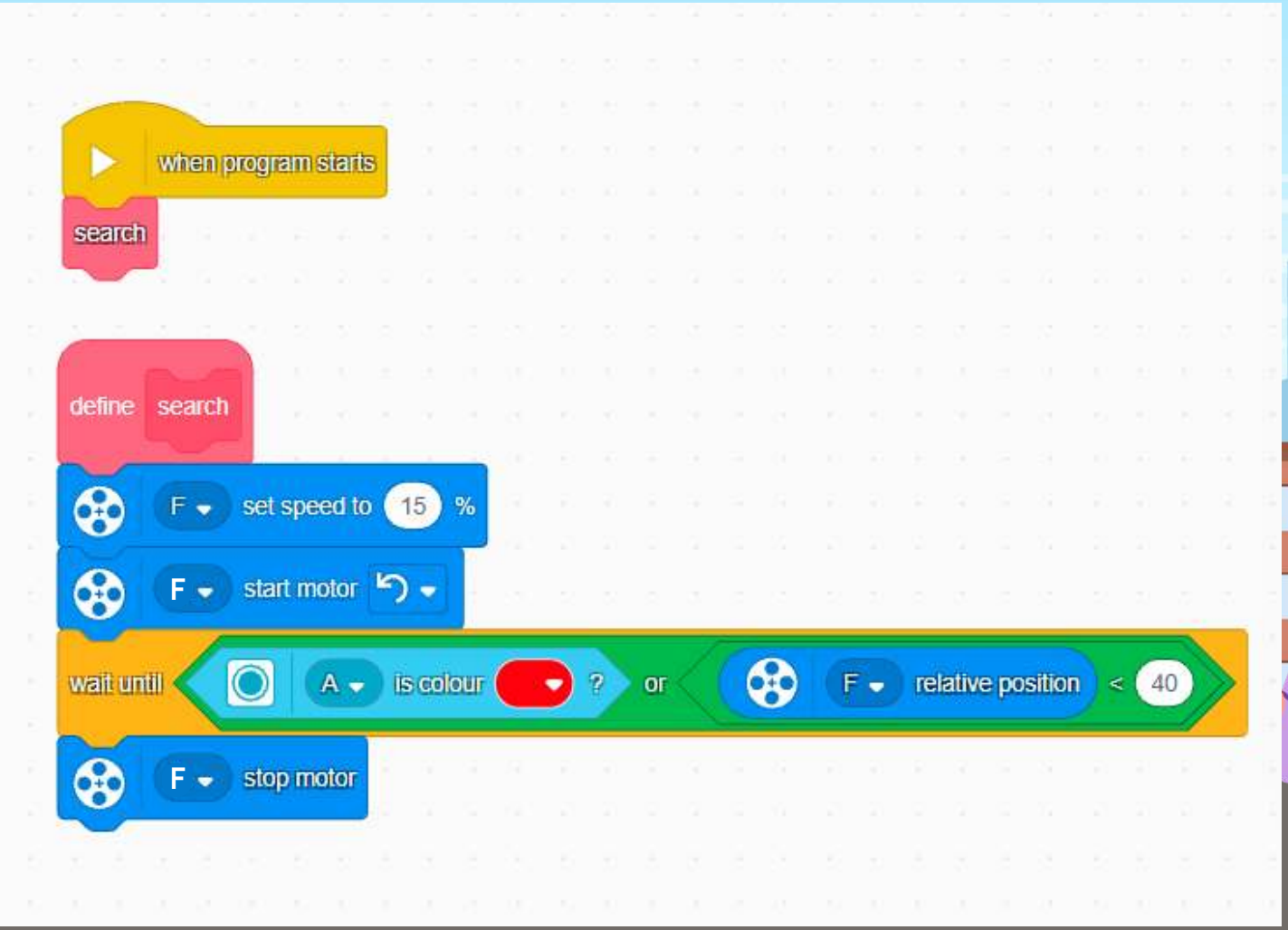
Erstelle mit der Option „Meine Blöcke“ einen neuen Block und programmiere deinen Roboter so, dass er sich bewegt

TASK 2



Um unser Programm übersichtlich zu halten, erstellen wir einen „Eigene Blöcke“. Indem wir diese Vorbereitungsschritte in den neuen Block packen, machen wir den Code viel leichter lesbar und zu debuggen. Wir können diesen Block dann einfach „aufrufen“, wann immer wir diese Anweisungen in unserem Hauptcode brauchen, ohne sie jedes Mal neu schreiben zu müssen.

TASK 3

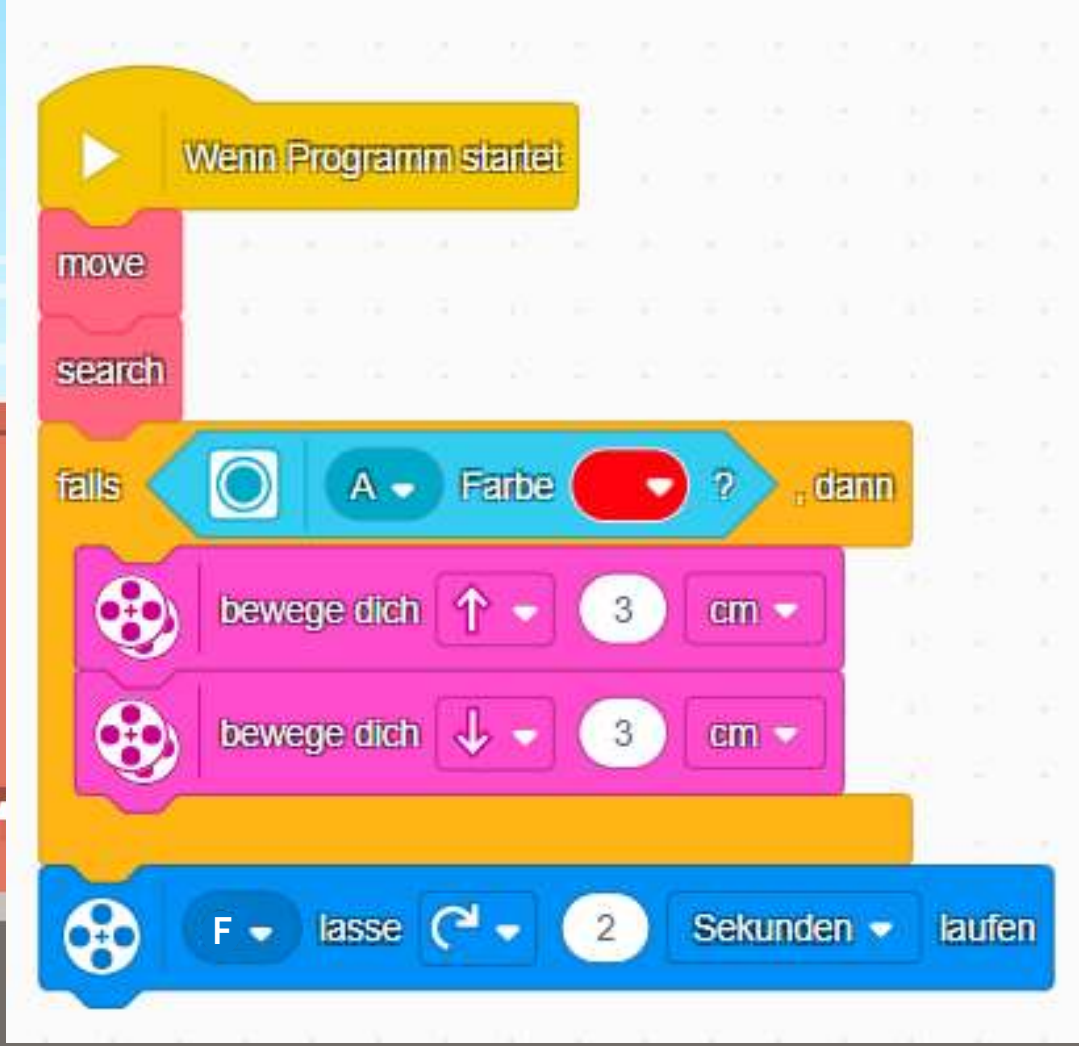


Für präzise Bewegungen ist es wichtig, die Motorpunkte auszurichten (siehe Bild).



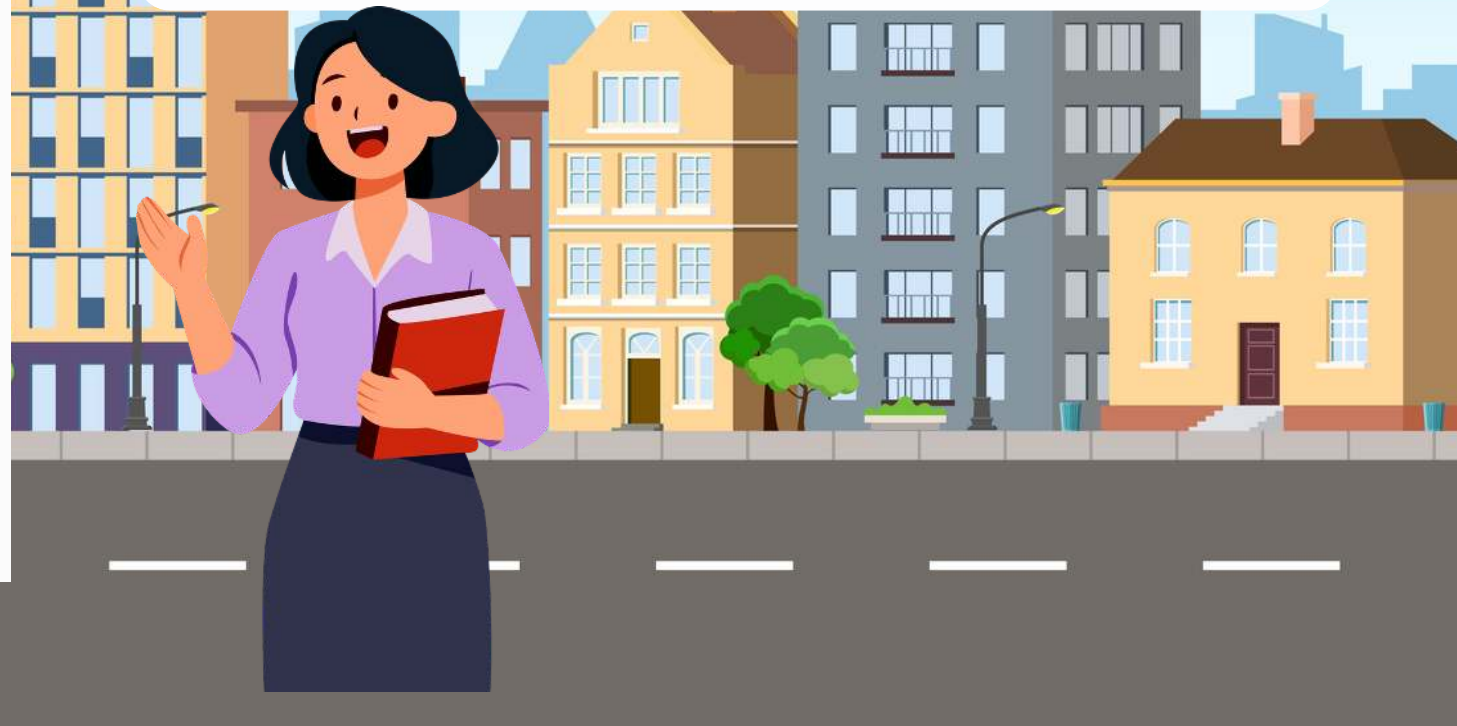
Als Nächstes programmieren wir den Roboter so, dass er nach einem „Feuer“ sucht (dargestellt durch den roten Block)

TASK 3



das Feuer löschen

Wenn der Farbsensor den roten Block erkennt, bewegt sich der Roboter hin und her, um das Feuer zu löschen, bevor er seinen Arm absenkt.





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